DIGITAL SIGNAL PROCESSING (CPE-415)

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Introduction to Digital Signal Processing

What is signal?

- Anything that conveys or carries some information is called a signal

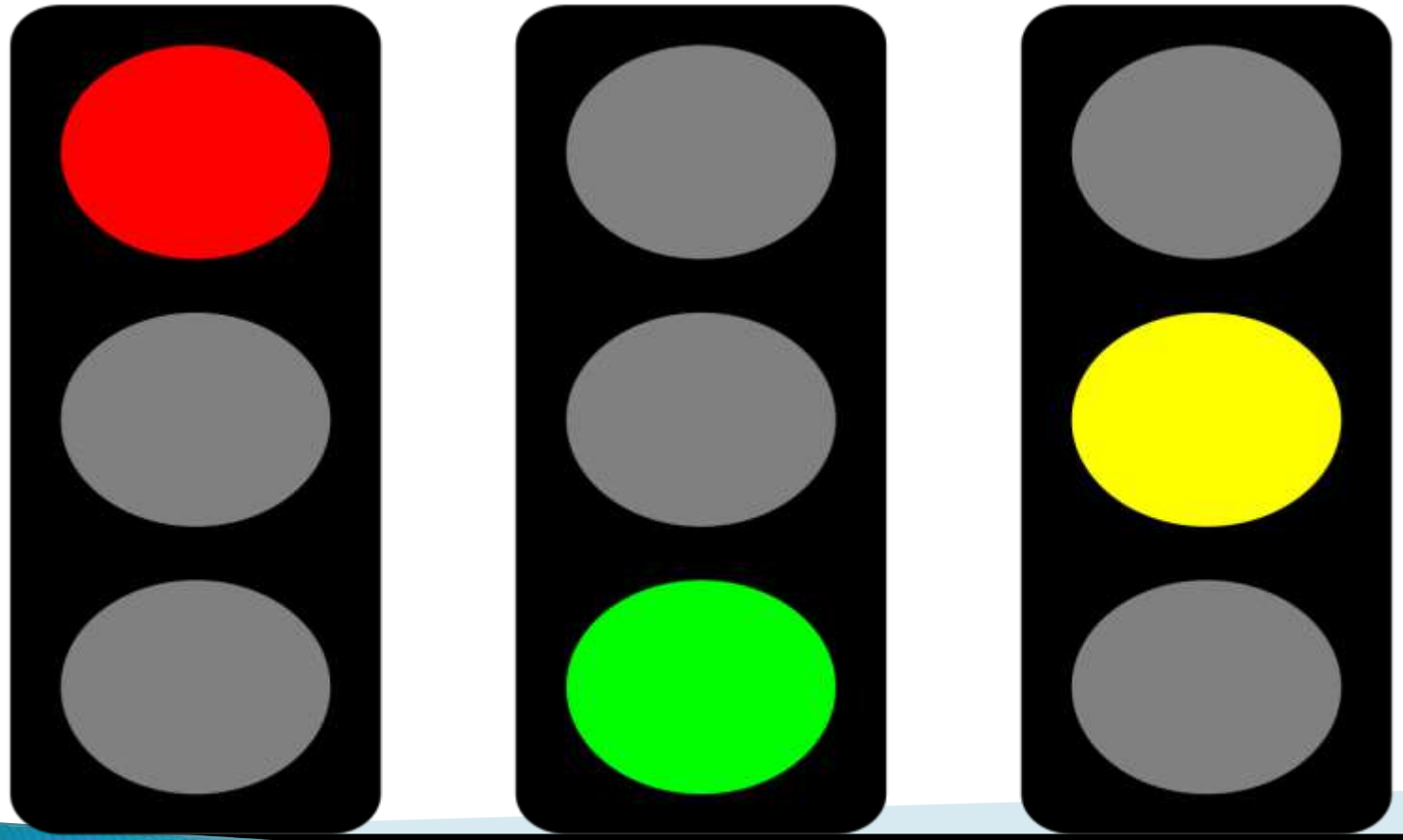
-In electrical , signal refers to any-time varying voltage, current or electromagnetic waves that caries some information.

We usually encounter signals in our day to day life.

- Music
- Speech
- Photos
- Heart beat
- Etc.

- Signal can be a function of time, distance, position, temperature, pressure etc.

Traffic Signal Light



1 2 3

Classification of Signals

- **Depending on the number of sources for the signals :**

1. One Channel
2. Multichannel

- **Depending on the number of independent variables :**

1. One Dimensional Signal
2. Multidimensional Signal

Classification of Signals

- **Depending on whether the independent variable is continuous or discrete :**

1. Continuous Signal
2. Discrete Signal

- **Depending on whether the dependent variable is continuous or discrete :**

1. Analog Signals
2. Digital Signals

Classification of Discrete Time Signals (Depending on their characteristics)

1. Deterministic and Nondeterministic Signals
2. Causal, Non-causal and Anti-causal Signals
3. Symmetric (Even) and Anti-Symmetric (Odd) Signals
4. Periodic and Aperiodic (Non-periodic) Signals
5. Energy and Power Signals

One Channel vs Multi Channel

One Channel

- Signals that are generated by single source or sensor are called one channel signals.
- Example 1: The recording of room temperature with respect to time.
- Example 2: The audio output of a mono speaker



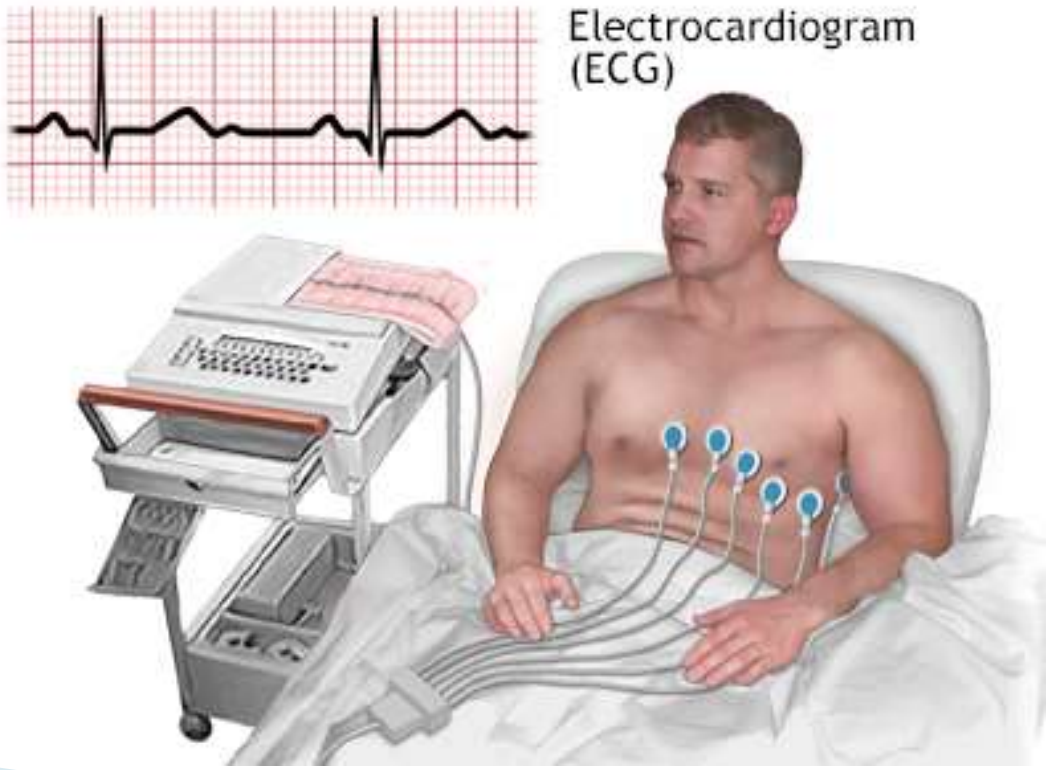
One Channel vs Multichannel

Multichannel

- Signals that are generated by multiple sources or sensors are called multichannel signals.
- Example 1: The recording of the room temperature and humidity.
- Example 2: The audio output of two stereo speakers

One Channel vs Multichannel

- Example 3: The record of ECG at eight different places in a human body. (8 channel signal)



One Dimensional vs Multidimensional

One Dimensional

- Signals which is dependent on single independent variable.

Example 1: Music and speech.

One Dimensional vs Multidimensional

Multidimensional

- Signals which is dependent on more than one independent variables.

Example 1. Photograph (The intensity and brightness at each point of photograph is a function of two spatial coordinates “X” and “Y”.

One Dimensional vs Multidimensional

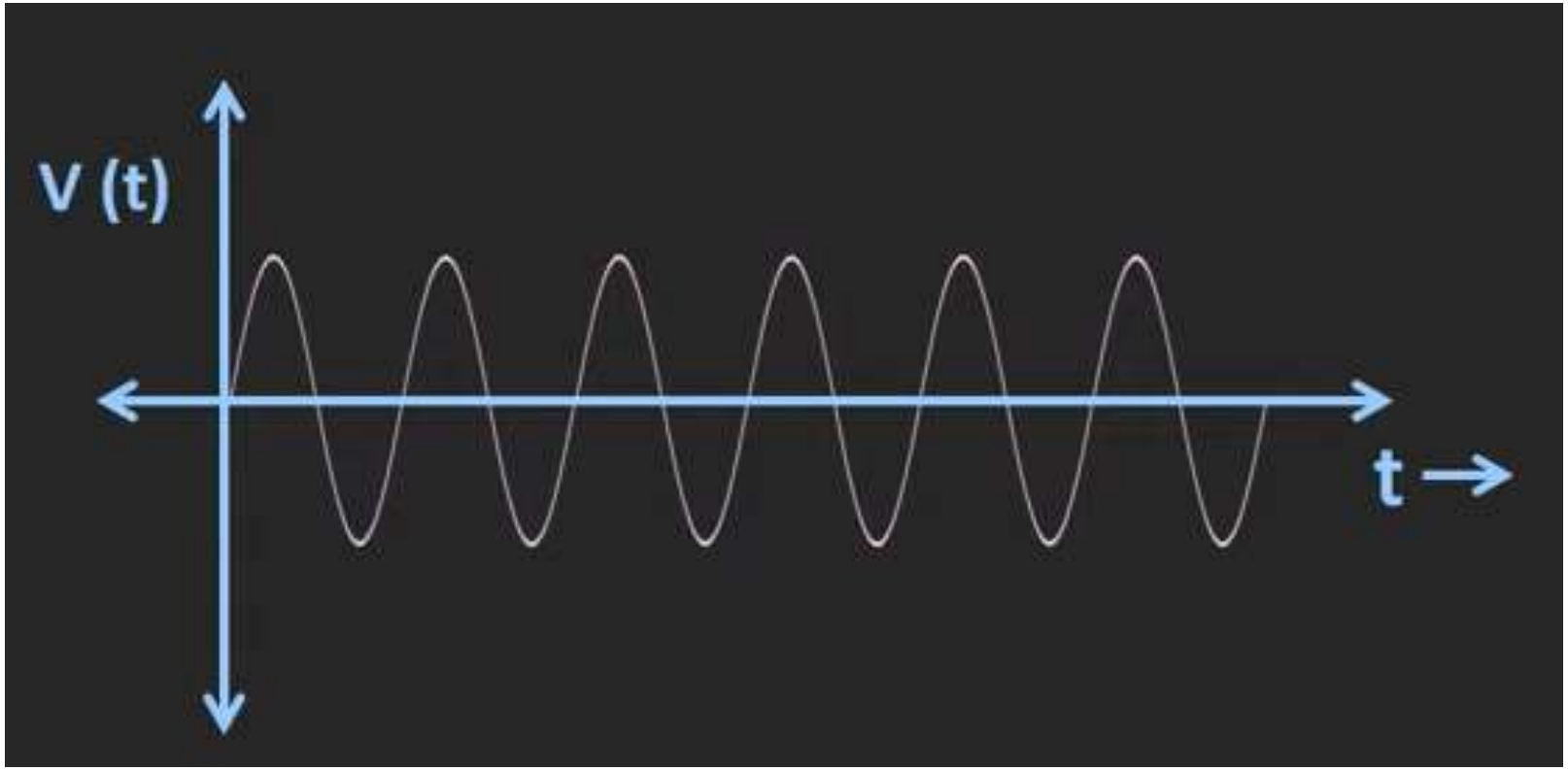
Example 2.Motion Picture (Function of 3 spatial coordinates intensity, brightness and time.

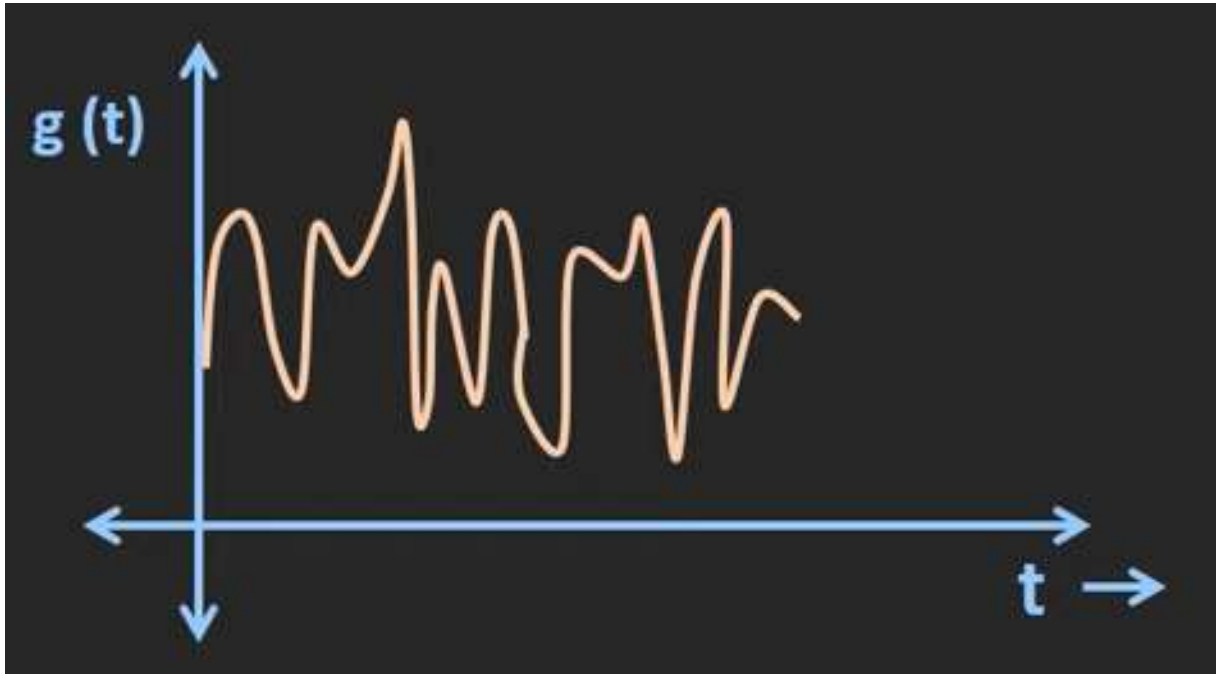
Continuous vs Discrete

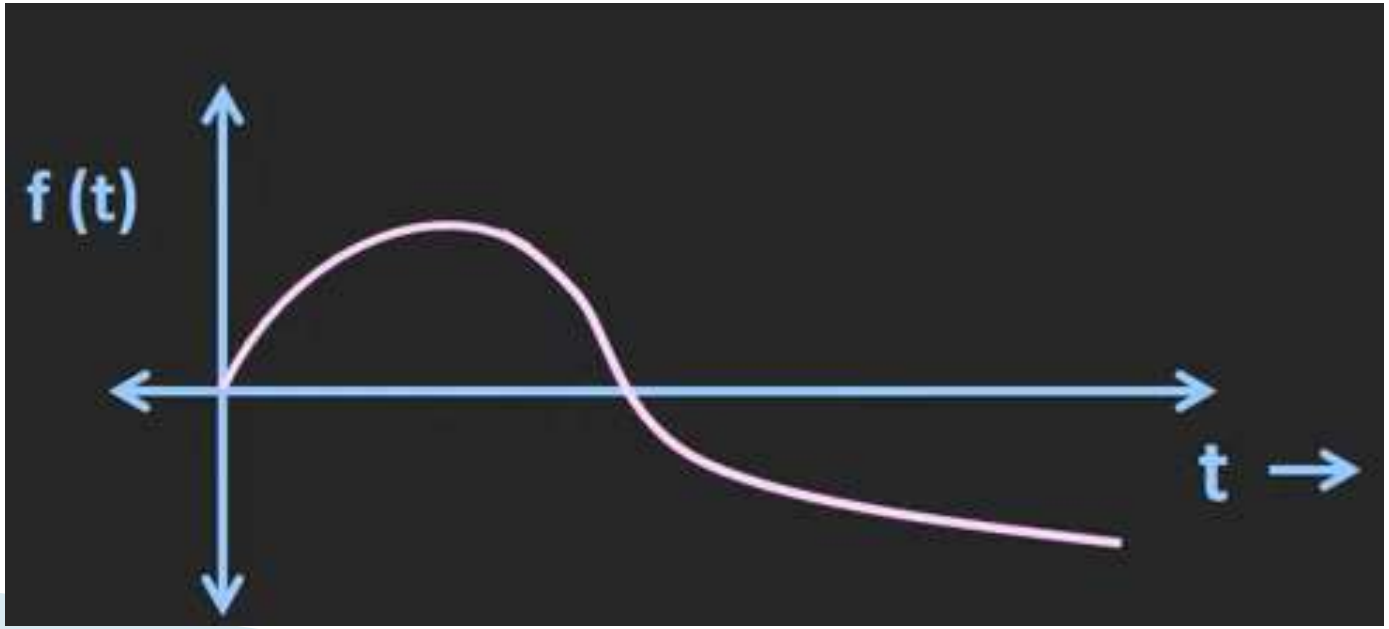
Continuous time Signal

If the signal is specified for every value in time then it is known as the Continuous time signal

Examples




g (t)
t

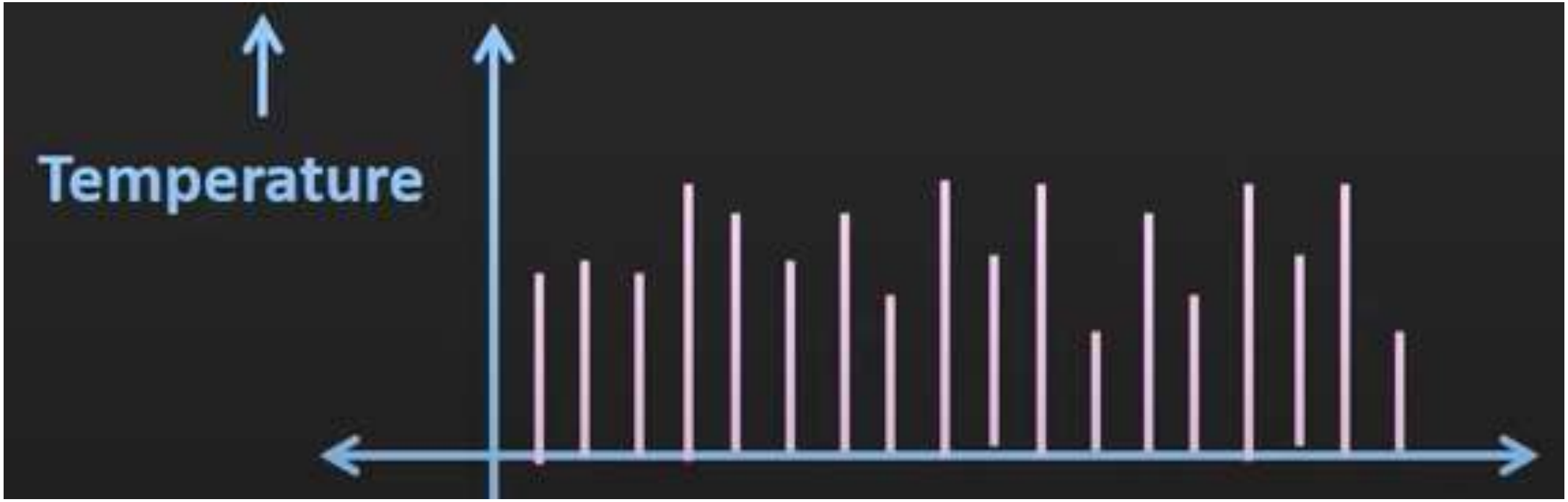

f (t)
t

Continuous vs Discrete

Discrete time Signal

If the signal is specified only for discrete time instances then it is called Discrete time signal.

Example

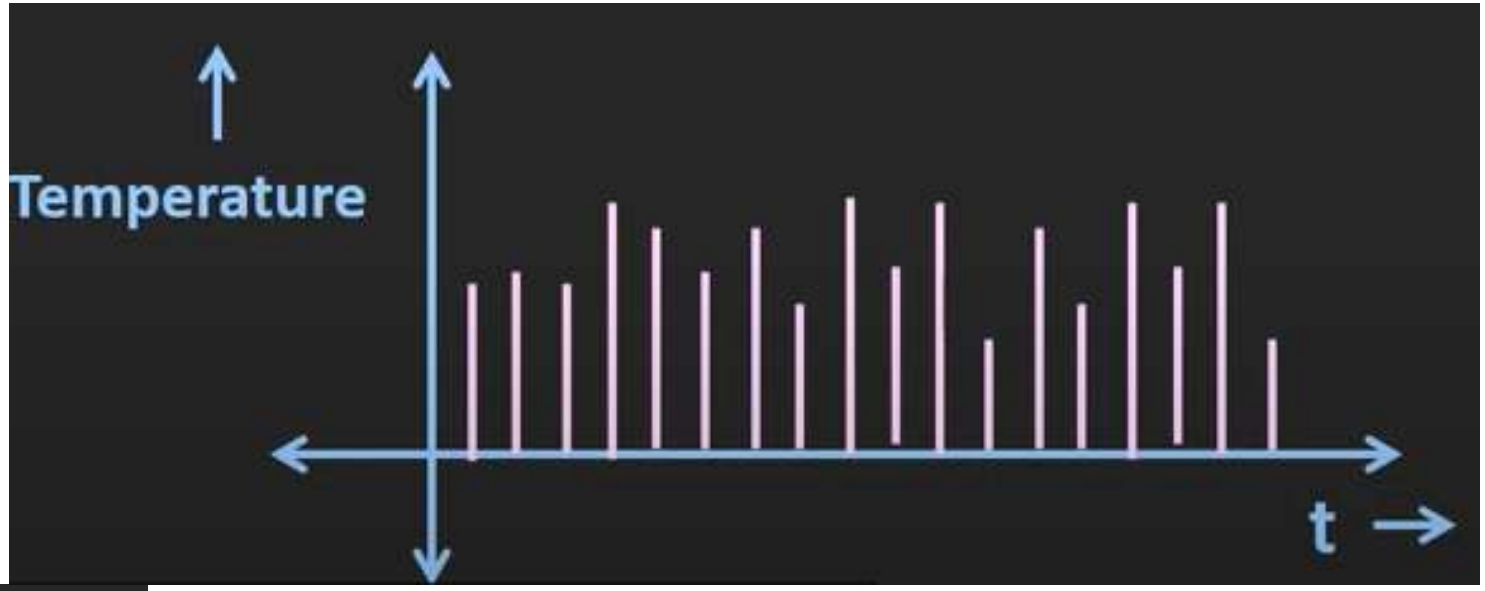


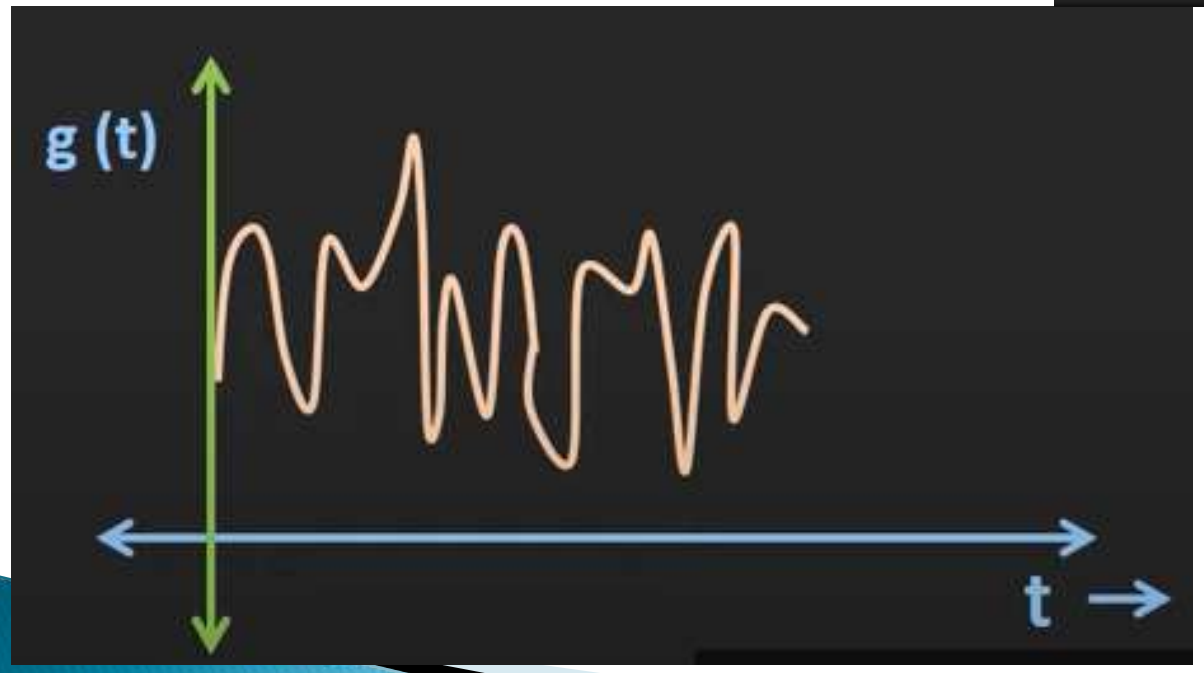
Analog vs Digital

Analog Signal

A signal whose amplitude can take any value in the continuous ranges is analog signal.

Example



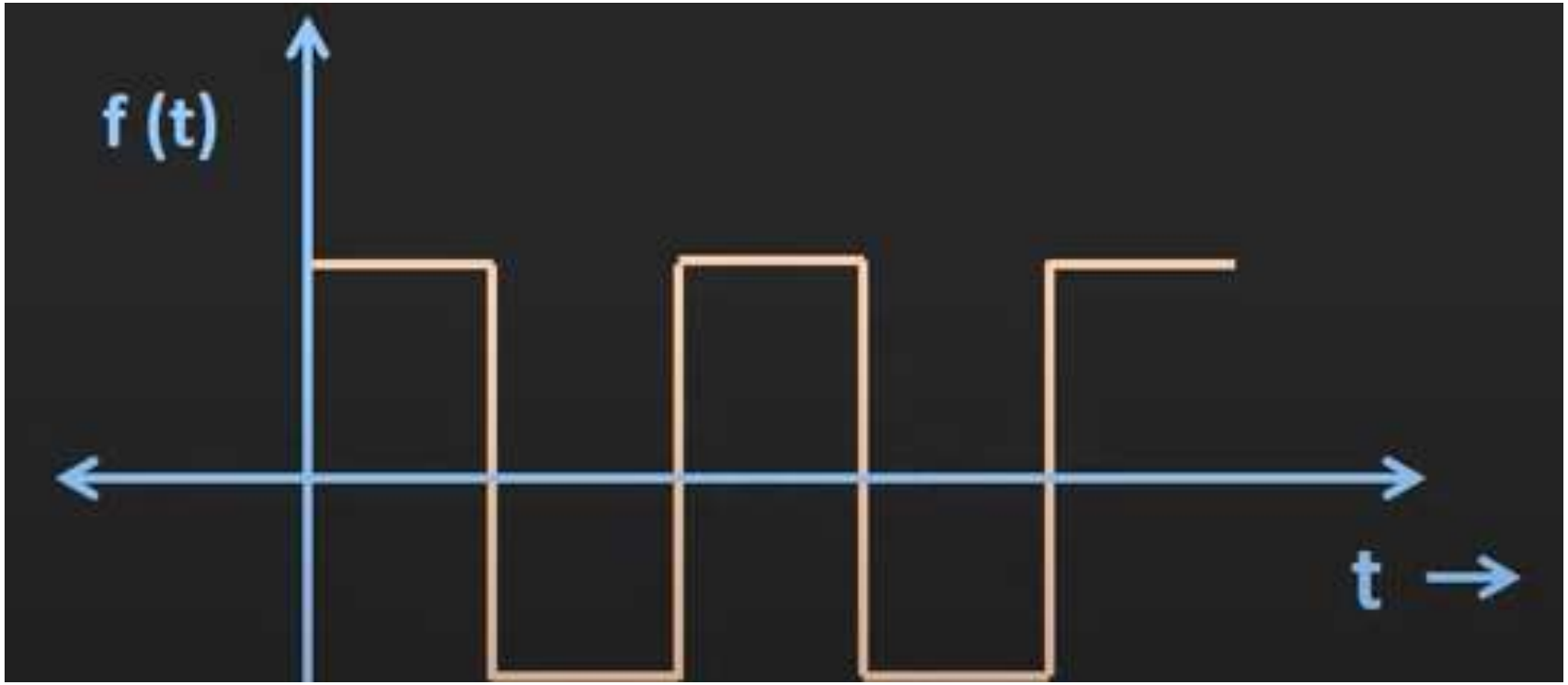


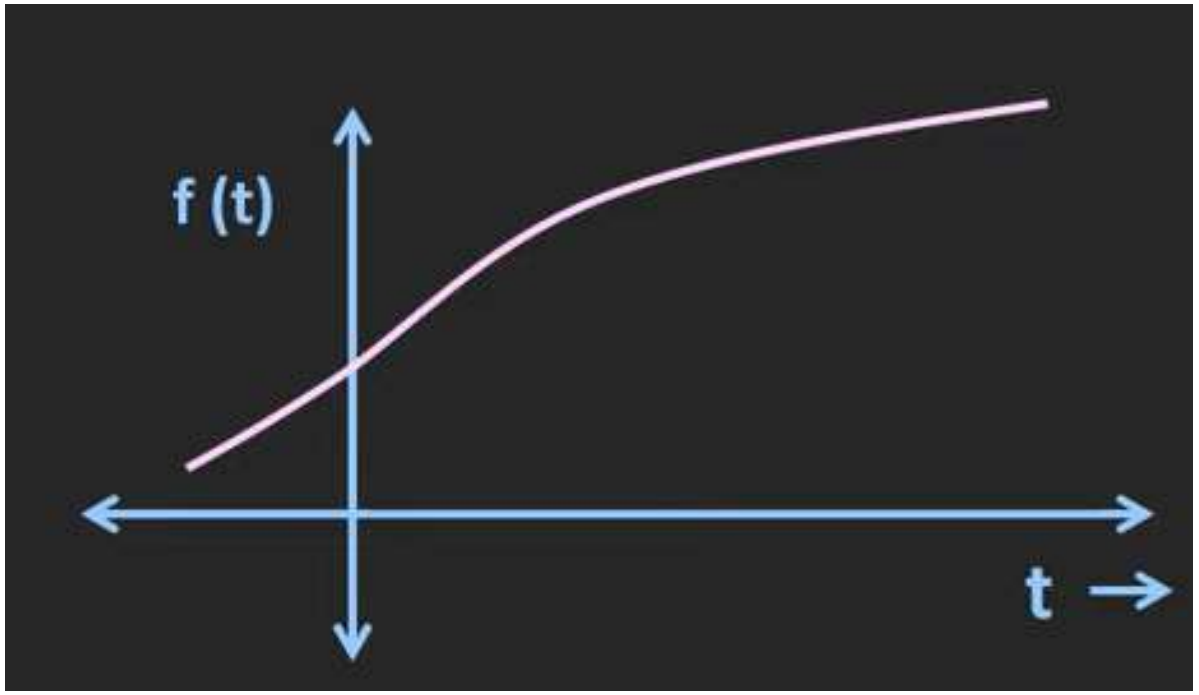
Analog vs Digital

Digital Signal

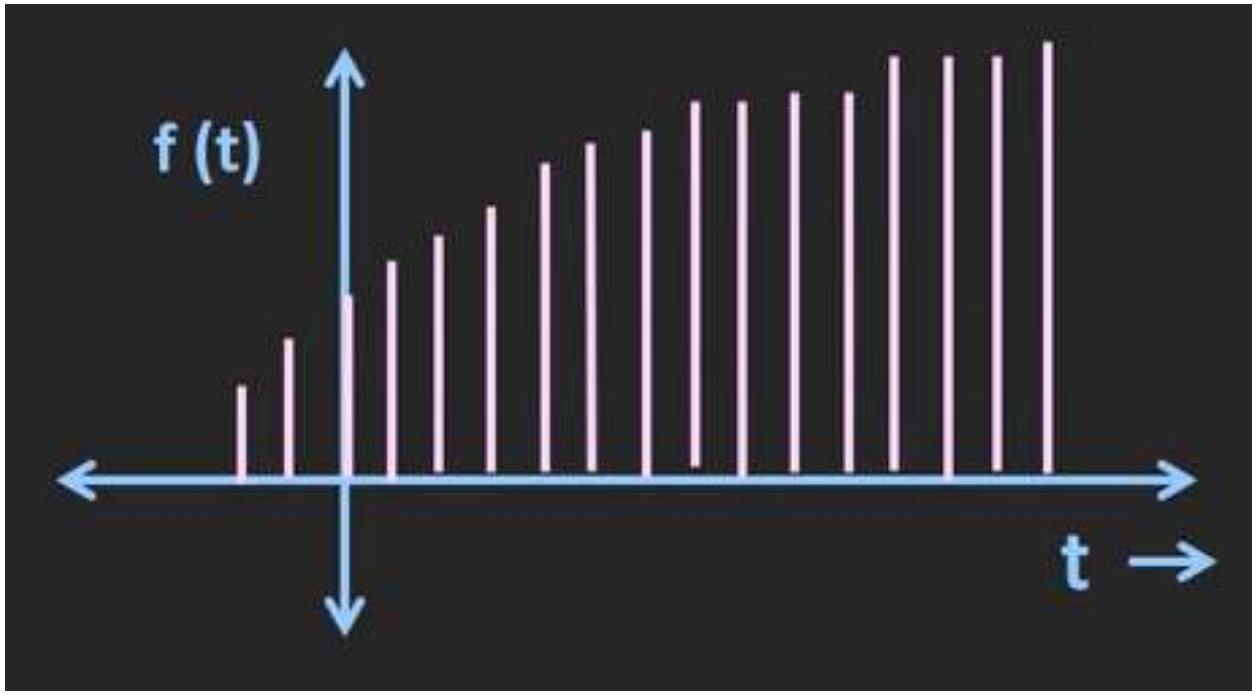
A signal whose amplitude can take only finite number of value is the digital signal.

Example

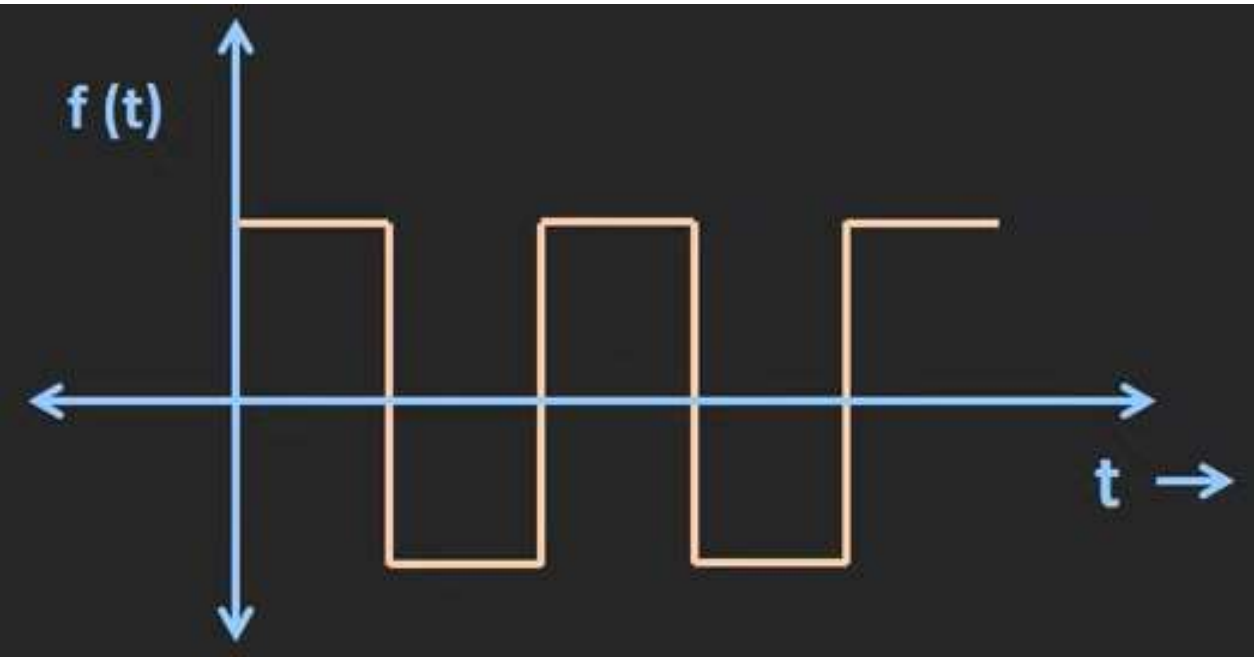




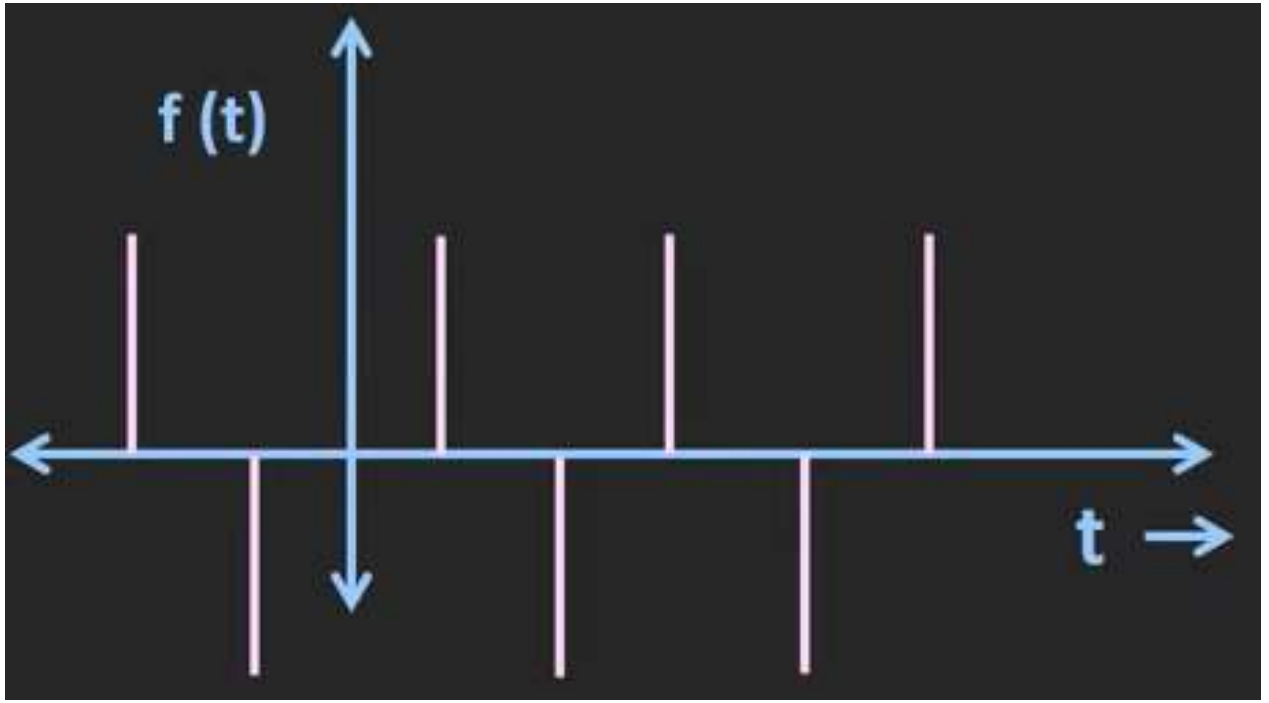
Analog and Continuous time signal



Analog and Discrete time signal



Digital and Continuous time signal

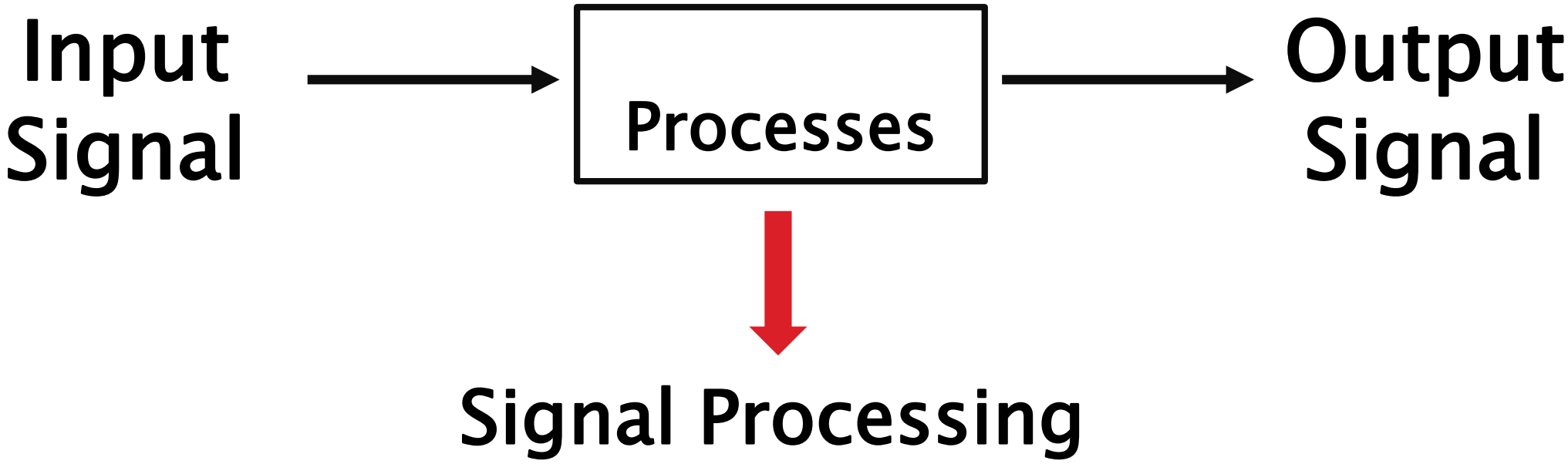


Digital and Discrete time signal

What is system?

- Basically it is a set of elements which are connected together and it takes an input signal, process it and give as an output in response to an input signal

System



Signal Processing

- refers to any kind of **modification, synthesis, and analysis** of a **signal**

- performing any operation on signals

Analog Signal Processing

- A type of signal processing conducted on continuous analog signals by analog circuits whose major components are resistors, capacitor and inductors, etc.

Digital Signal Processing

- Is the use of digital processing, such as by computers or more specialized digital signal processors, to perform a wide variety of signal processing operations

- It involves applying various mathematical and computational algorithms to analog and digital signals to produce a signal that’s of higher quality than the original signal.

Advantages of Digital system over an analog system

- Reproductivity of the results.
- Accuracy of results.
- Flexibility and functionality.
- Ease of design: No specific math skill needed to visualize the behavior of small digital circuits.
- Programmability.
- Due to the integration of millions of digital logic elements on a single miniature chip forming for very low cost integrated circuits.

Advantages of Digital system over an analog system

- More reliable than an analog system due to better immunity to noise.

- Speed: A digital logic integration of millions of digital logic elements on a single miniature chip forming low cost integrated circuits.

Advantages of Digital circuits over an analog circuits

- The Digital circuit is very accurate.

- The Digital circuit generates a digital signal which is more reliable than analog signals.

- Compare to the analog circuits is that signal represented digitally can be transmitted without degradation due to noise.

- Digital signal can be used by digital circuit components, which are very cheap and easy to produce in many components on a single chip.

Advantages of Digital circuits over an Analog circuits

- Enables transmission of signal over a long distance.

- Digital signal maintains its quality over long distances better than analog signals.

- Digital signals carry more information per second than analog signals circuit.

- Digital signal is used to construct a digital circuit since the cost of digital components is very cheap.

Advantages of Digital circuits over an Analog circuits

- Digital circuits are easy to design and cheaper than analog circuits

- The hardware implementation in a digital circuit is more flexible than analog.

- The signal processing function such as encryption and compression are employed in digital circuits to maintain the secrecy of the information.

Advantages of Digital signal processing over Analog signal processing

Advantages of Digital signal processing

• DSP offers high accuracy. Hence, filters designed in DSP have tighter control over the output accuracy.

•Comparatively cheaper than an analog counterpart.

•Reconfiguration is very easy and only code or DSP program needs to be flashed after changes as per requirement. Reconfiguration of an analog signal is difficult.

•DSP offers various interface types such as UART, I2C helps in interfacing other ICs with the DSP.

•DSP can be interfaced with FPGA helps in the design of a protocol stack of the entire wireless system such as WiMAX and LTE.

Advantages of Digital signal processing

- Digital system can be cascaded without any loading problem.

- The digital circuits can be reproduced easily in large quantities at a comparatively lower cost.

- The digital circuits are less sensitive to a tolerance of component values.

- DSP are easily transported because digital signals can be processed offline.

- Current computers can't process analog data so digitalization is necessary.

Advantages of Digital signal processing

- Enables transmission of signals over a long distance.

- DSP use less bandwidth so that you can cram more information into the same space.

- DSP enables multi-directional transmission very easily.

- In DSP easier to translate human audio and video signals and other messages into machine language.

- Universal compatibility is possible in DSP compared to ASP.
- Mathematical processing algorithms can be easily implemented in DSP.

Advantages of Digital signal processing

- Mathematical processing algorithms can be easily implemented in DSP.

- Generally, all application needs standard hardware thus the operation of DSP is mainly dependent on software.

Disadvantages of Digital signal processing

- DSP requires an anti-aliasing filter before ADC and re-construction filter after DAC so it requires ADC and DAC modules.

- DSP dissipates higher power compared to ASP because DSP processed signals at high speed and moreover it consists of higher internal hardware resources.

- DSP chips are very costly and hence one needs to use the appropriate ICs as per requirement.

Disadvantages of Digital signal processing

- The hardware architectures and software instructions of each DSP are different so training on DSP in order to program for various applications. Hence, Highly skilled engineers can only work on the device.

- A typical DSP chip contains more than 400,000 transistors thus power dissipation is more in DSP.

- DSP is costly for small applications as compared to ASP.

- DSP requires greater bandwidth to transmit the same information of ASP.

Disadvantages of Digital signal processing

- Downgrading the quantity of data is possible but increasing the bit rate is also possible beyond the certain level.

- The detection of digital signals requires the communication system to be synchronized whereas there is no case in ASP.

Digital Signal Processor Significance

Due to their unique microprocessor design, digital signal processors (DSPs) efficiently complete digital signal processing tasks. Additionally, they are extremely important in a number of fields, such as telecommunications, audio processing, image and video processing, radar systems, control systems, and more.

- **Efficient signal processing-** DSPs are like super-fast calculators that process sounds, images, and other signals quickly and accurately.

- **Specialized architecture-** DSPs are designed specifically for tasks like making sounds clearer, removing noise, and improving picture quality.

- **Real-time processing-** DSPs work really fast, which is important for things like live video chats and streaming music or videos without delays.

- **Power efficiency-** DSPs are designed to use less energy while still doing their job well, which is useful for devices that run on batteries.

- **Versatility-** DSPs can be programmed to do different tasks, so they can adapt and work with new technologies and improve over time.

- **System integration-** DSPs can do many things on their own, like converting signals from analogue to digital and vice versa, which helps make devices smaller and cheaper.

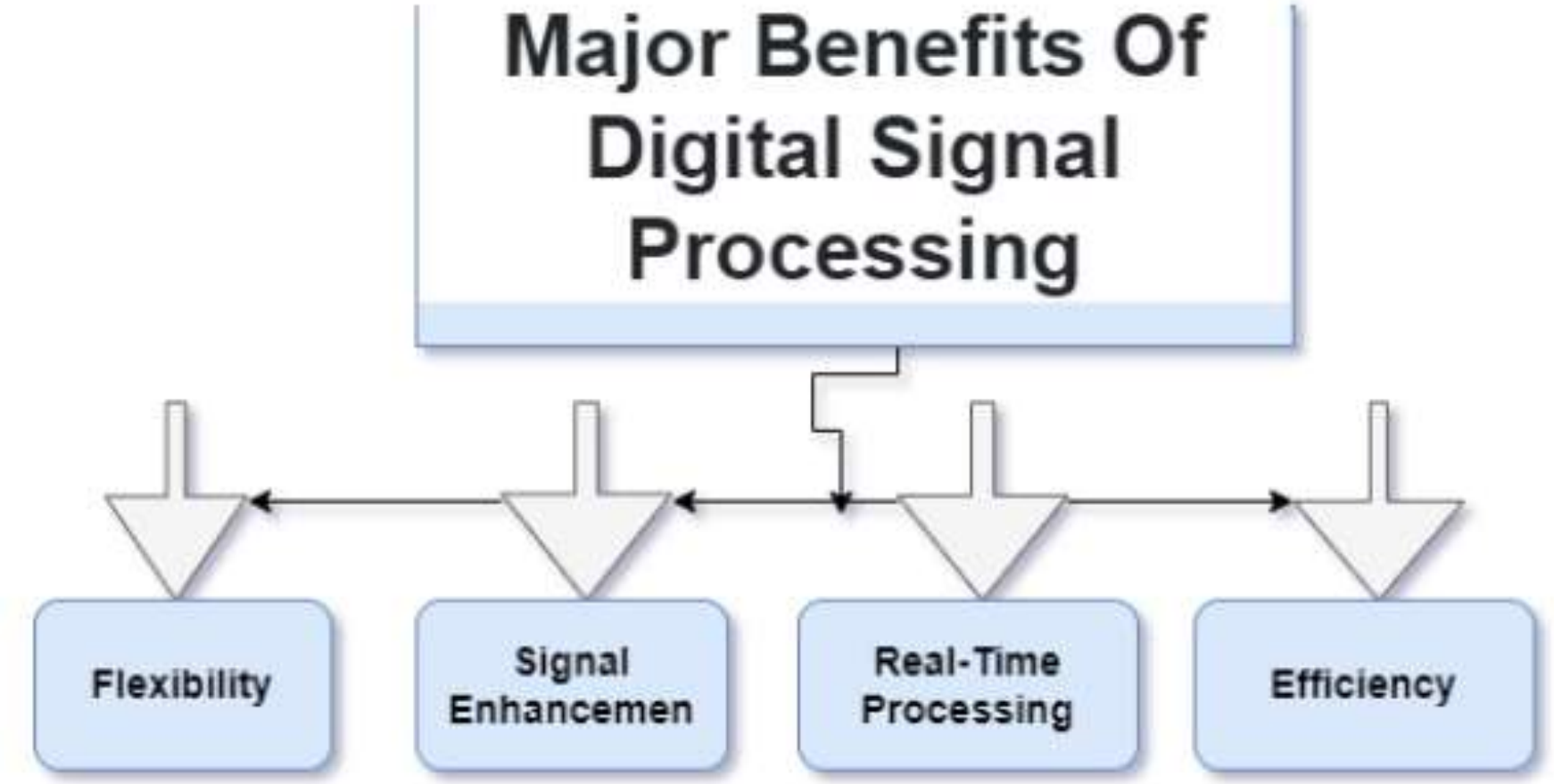
- **Performance optimization-** DSPs have tricks and shortcuts built into them to make signal processing faster and more efficient.

• **Wide application range-** DSPs are used in things like cell phones, Wi-Fi, music players, video games, medical devices, cars, and more, making them important in many different areas of technology.

• **AI processing-** DSPs play a crucial role in accelerating AI algorithms. Moreover, they can efficiently handle the computational demands of tasks like speech recognition, natural language processing, computer vision, and deep learning. Similarly, DSPs enable AI-powered applications and devices to perform tasks quickly and accurately, enhancing user experiences and enabling new capabilities like robotics, virtual assistants, and autonomous systems.

What are the Major Benefits of Digital Signal Processing?

Digital Signal Processing (DSP) offers several major benefits that contribute to its widespread use.



Major Benefits of Digital Signal Processing:

• **Flexibility:** DSP allows us to easily change and adjust algorithms to fit different signal processing needs. Moreover, it’s like having a toolbox with various tools that we can use for different tasks.

•**Signal Enhancement-** DSP algorithms can make signals better. They can reduce unwanted noise, enhance important details, and improve the overall quality of signals. Also, it’s like using a special filter to make a picture look clearer.

Major Benefits of Digital Signal Processing:

- **Real-Time Processing:** DSP can quickly process signals in real-time, meaning it can analyze and manipulate signals as they happen. In addition, this is important for applications that require immediate responses, like video games or live audio processing.

- **Efficiency:** DSP techniques are designed to make the most efficient use of computing resources. In addition, they can perform complex calculations using less memory and processing power, making them efficient and effective.

Application of Digital Signal Processing

Digital Signal Processing is used in the fields of :

- RADAR (**Ra**dio **D**etection **A**nd **R**anging)
- SONAR (**So**und **N**avigation **A**nd **R**anging)
- Voice Recognition
- Speech Processing
- Image Processing
- Consumer Electronics

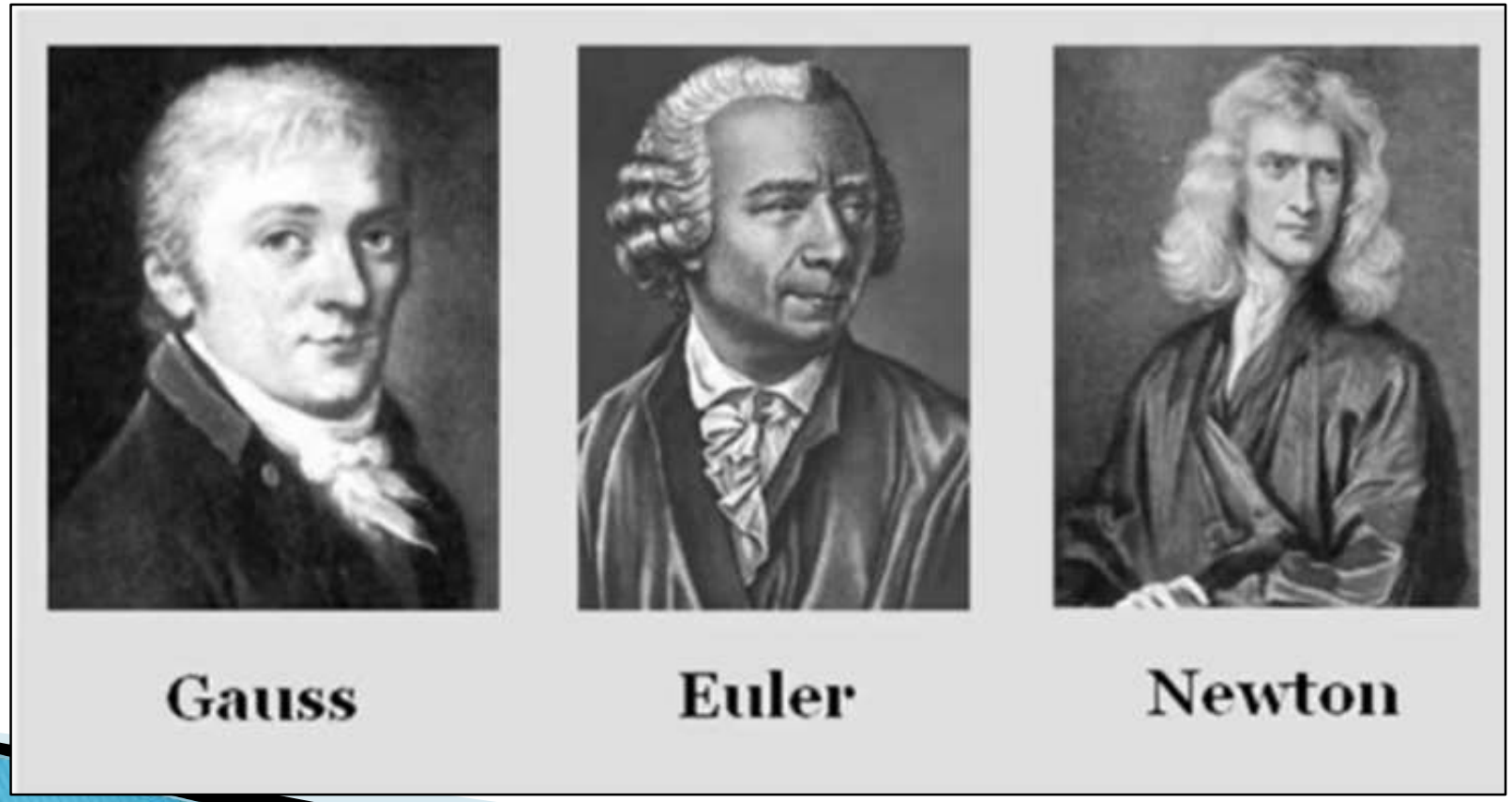
Digital Signal Processing is used in the fields of :

- Telecommunications systems
- Biomedical engineering
- Seismology

History of Signal Processing

- Signal processing is a technology that spans an immense set of disciplines including, entertainment, communications, space, exploration, medicine, hardware.

The roots of signal processing started with solving numerical techniques for complex equations. (Newton, Gauss, Euler, Fourier)



Year 1930-1940

- Radar is created and it was used for spread spectrum communications during World War II.





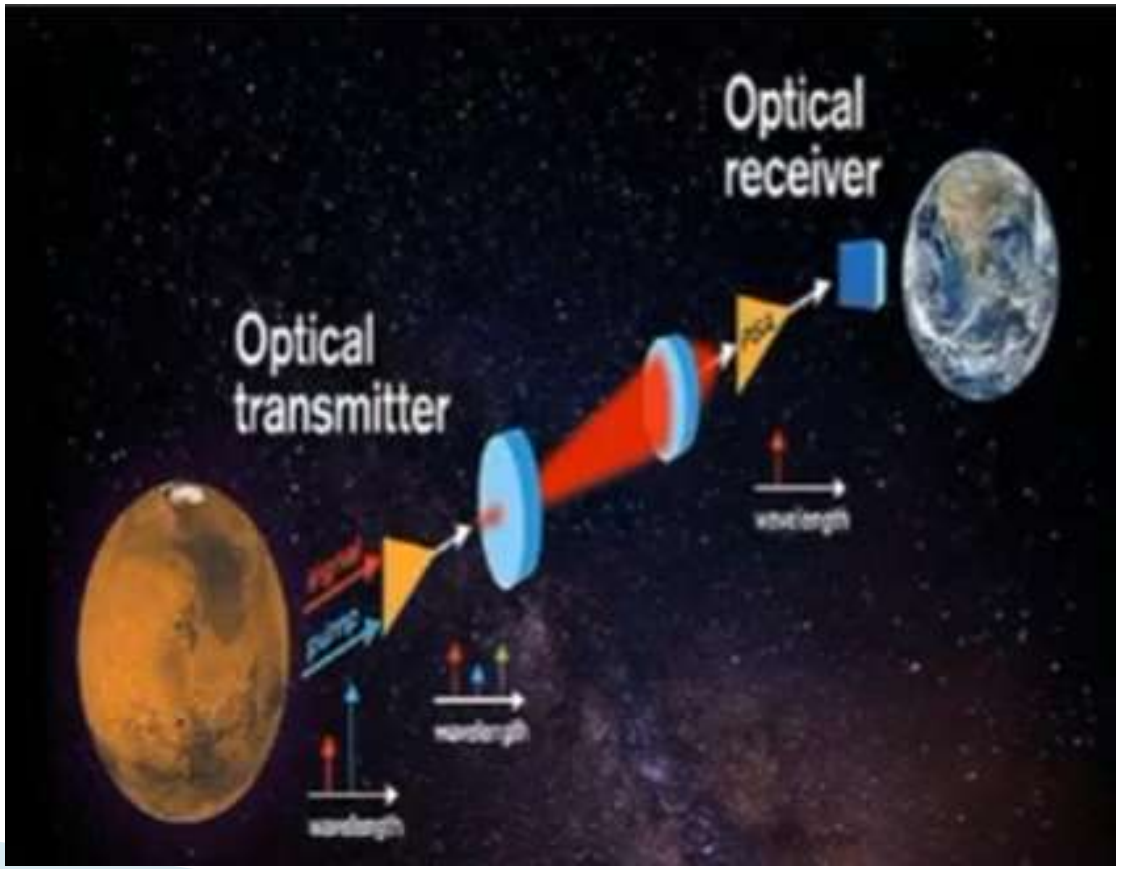
1950

- Development of analog techniques (analog circuits) for processing signals. Also the first digital computers is created during this year



1960

- Signal processing is driven by communications, oil exploration and space programs.

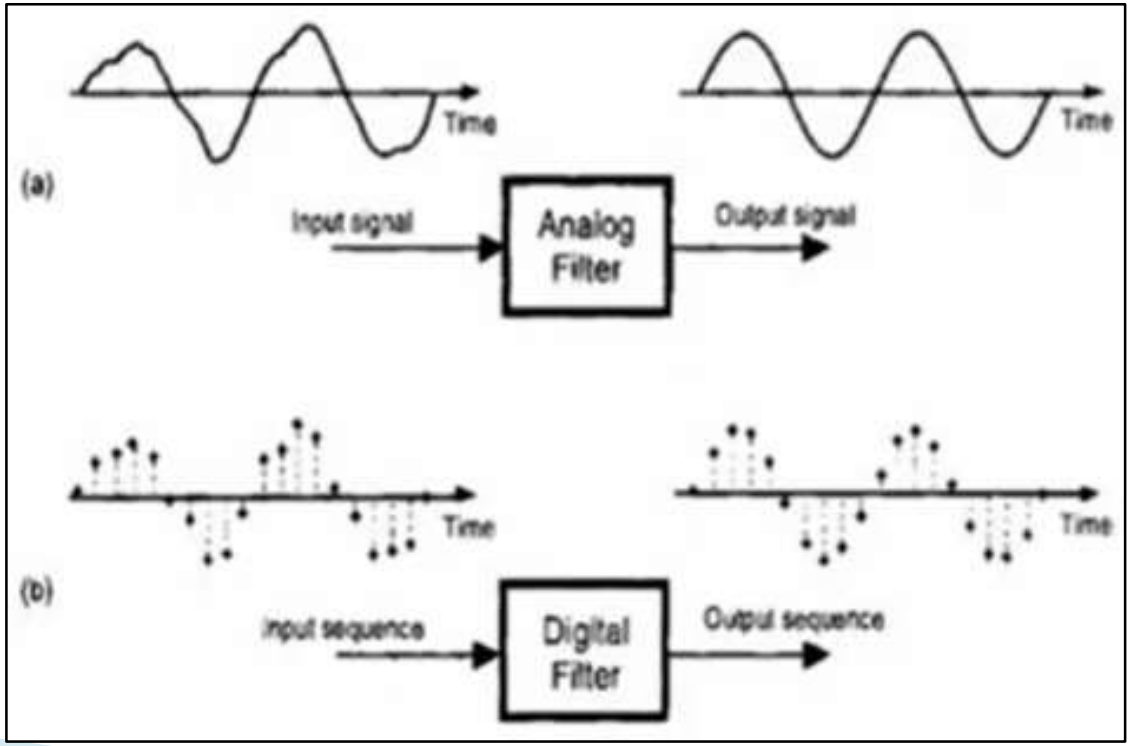


1965

- Coley and Tukey came up with the FFT (Fast Fourier Transform), a numerical technique to solve Fourier Representations.

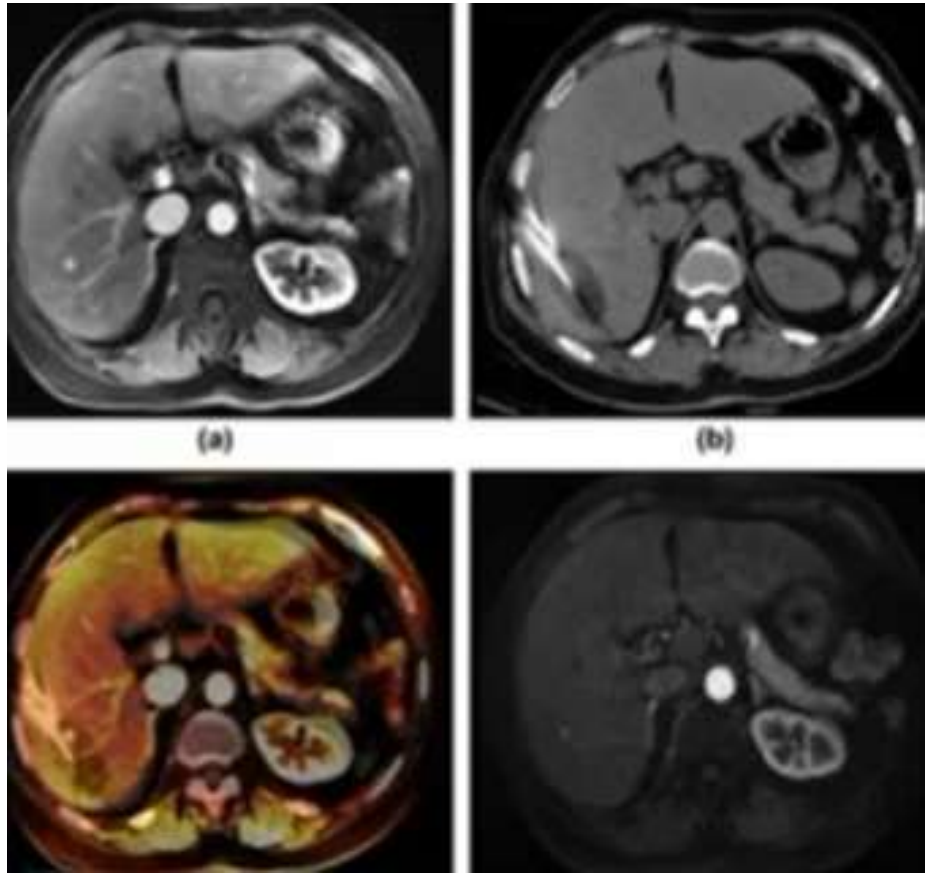
1970

- Development of FFT algorithms and digital filters. FFT is widely used for application in Engineering, Science and mathematics.



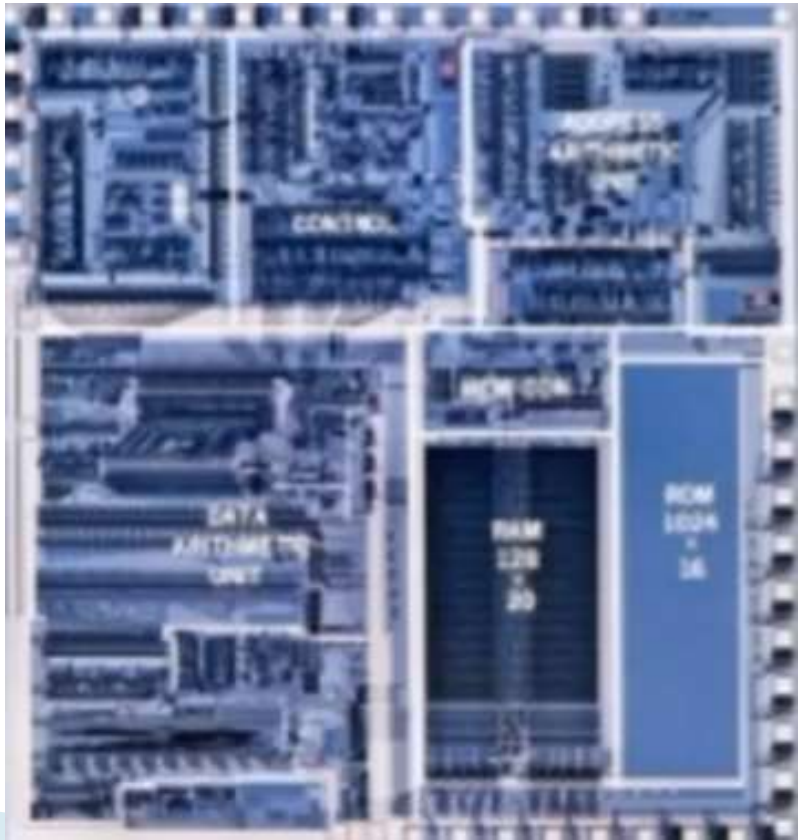
1980

- Development of remote sensing, medical imaging and DSP chipset.



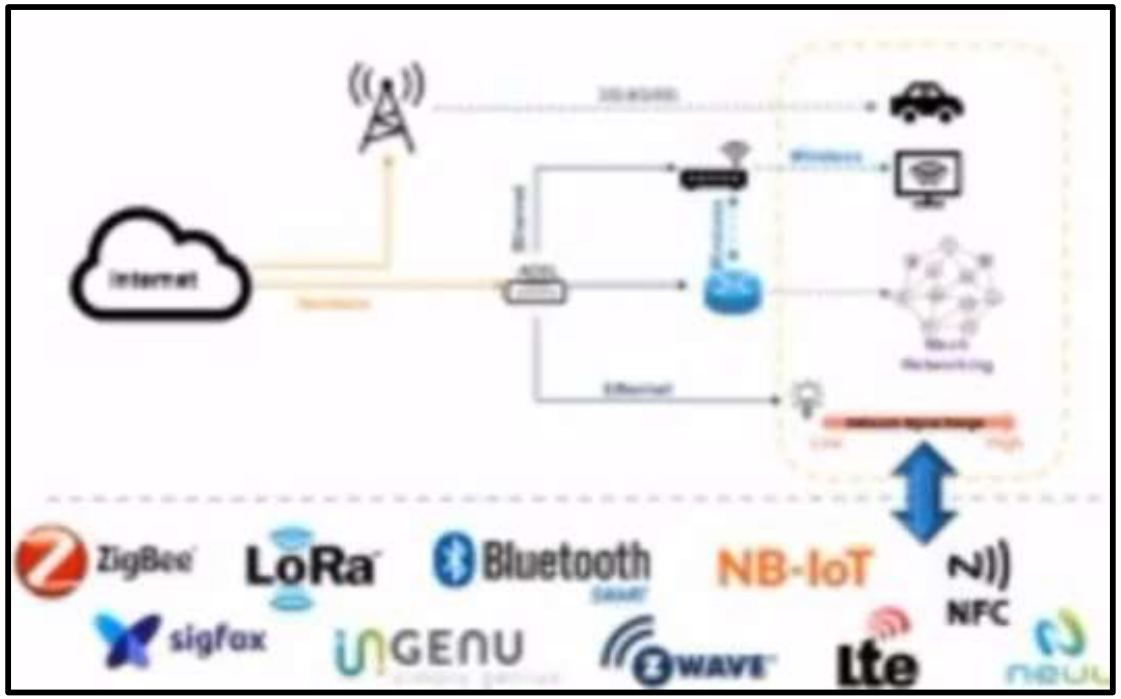
1990

- DSP become the standard technique, more use of DSP chipset and consumer electronics.



2000

- Wide range of applications including analyzations of networks, large sensing systems, smartphones and many more.



Representation of Discrete Time Signal

Discrete Time Signal

- The signals which are defined only at discrete instants of time are known as **Discrete time signals**. The discrete time signals are represented by x(n) where n is the independent variable in time domain.

Discrete time signal may be represented by the following:

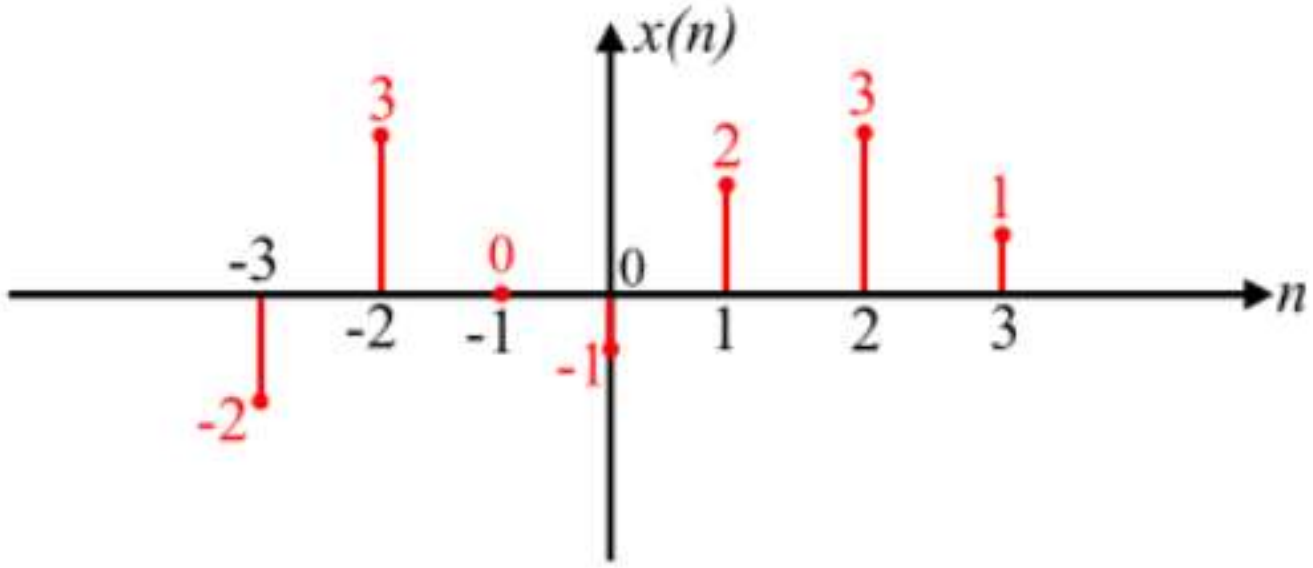
- Graphical Representation
- Functional Representation
- Tabular Representation
- Sequence Representation

Consider a discrete time signal x(n) with the values,

- $x(-3) = -2$
- $x(-2) = 3$
- $x(-1) = 0$
- $x(0) = -1$
- $x(1) = 2$
- $x(2) = 3$
- $x(3) = 1$

GRAPHICAL REPRESENTATION

- We can represent the discrete-time signal graphically



FUNCTIONAL REPRESENTATION

In the functional representation of discrete time signals, the magnitude of the signal is written against the values of n.

$$x(n)=\{-2 \text{ for } n=-3,\ 3 \text{ for } n=-2,\ 0 \text{ for } n=-1,\ -1 \text{ for } n=0,\ 2 \text{ for } n=1,\ 3 \text{ for } n=2,\ 1 \text{ for } n=3\ \}$$

TABULAR REPRESENTATION

In the tabular representation of discrete time signals, the sampling instant n and the magnitude of the discrete time signal at the corresponding sampling instant are represented in the form of a table.

n	−3	−2	−1	0	1	2	3
x(n)	−2	3	0	−1	2	3	1

SEQUENCE REPRESENTATION

The discrete time signal x(n) can be represented in the sequence representation as follows:

$$x(n)=\{-2,3,0,\underset{\uparrow}{-1},2,3,1\ \}$$

Here, the arrow mark (↑) denotes the term corresponding to n = 0. When no arrow is indicated in the sequence representation of a discrete time signal, then the first term of the sequence corresponds to n = 0.

Sum of Discrete time sequence

With the help of adding the corresponding element of the sequence, we can get the sum of two discrete-time sequences, which is described as follows:

$$\{Cn\} = \{an\} + \{bn\} \rightarrow Cn = an + bn$$

Sum of Discrete time sequence

With the help of adding the corresponding element of the sequence, we can get the sum of two discrete-time sequences, which is described as follows:

$$\{Cn\} = \{an\} + \{bn\} \rightarrow Cn = an + bn$$

Product of Discrete time sequence

With the help of adding the corresponding element of the sequence, we can get the sum of two discrete-time sequences, which is described as follows:

$$\{Cn\} = \{an\}\ \{bn\} \rightarrow Cn = an\ bn$$

With the help of multiplying each element of the sequence and a constant, we can get the product of a sequence and a constant k, which is described as follows:

$$\{C_n\} = k\{a_n\} \rightarrow C_n = ka_n$$

Mathematical Operations on Discrete Time Signal

SIGNAL ADDITION

- The Addition of two DTS is performed on a sample by sample basis. The sum of two signals $X_1(n)$ and $X_2(n)$ is a signal Y(n) whose value at any instant is equal to the sum of the samples of these two signals at that instant.

Example: Perform the addition of two discrete time signal shown below.

$X_1(n) = \{5,3,\underset{\uparrow}{-1},2,4\}$

$X_2(n) = \{2,6,\ 4,\underset{\uparrow}{1},3\}$

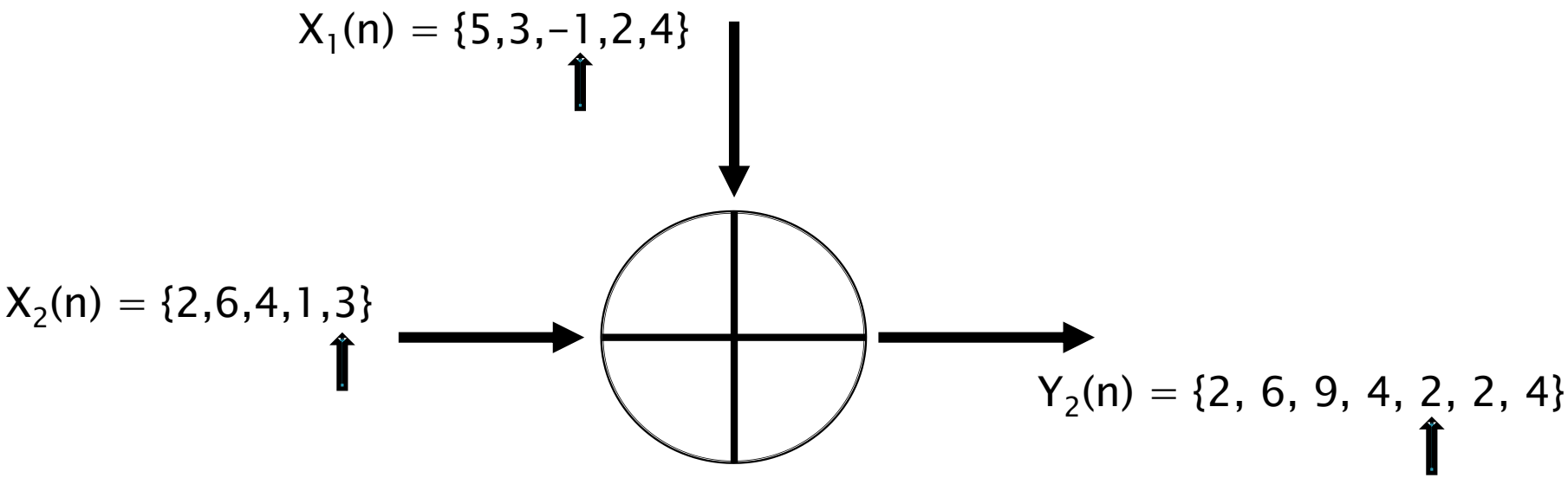
Solution:

$$X_1(n) = \{5,\ 3,\ \underset{\uparrow}{-1},\ 2,\ 4\}$$

$$X_2(n) = \{2,\ 6,\ 4,\ 1,\ \underset{\uparrow}{3}\}$$

$$Y_2(n) = \{2,\ 6,\ 9,\ 4,\ \underset{\uparrow}{2},\ 2,\ 4\}$$

Summation Block:



SIGNAL MULTIPLICATION

- The Multiplication of two DTS is performed also on a sample by sample basis. The product of two signals $X_1(n)$ and $X_2(n)$ is a signal Y(n) whose value at any instant is equal to the product of the samples these two signals at that instant.
- Note: The product is also called as modulation

Example: Perform the addition of two discrete time signal shown below.

$X_1(n) = \{5,3,-1,2,4\}$



$X_2(n) = \{2,6, 4,1,3\}$



Solution:

$$X_1(n) = \{5,\ 3,\ -1,\ 2,\ 4\}$$



$$X_2(n) = \{2,\ 6,\ 4,\ 1,\ 3\}$$



- $Y_2(n) = \{0,\ 0,\ 20,\ 3,\ -3,\ 0,\ 0\}$

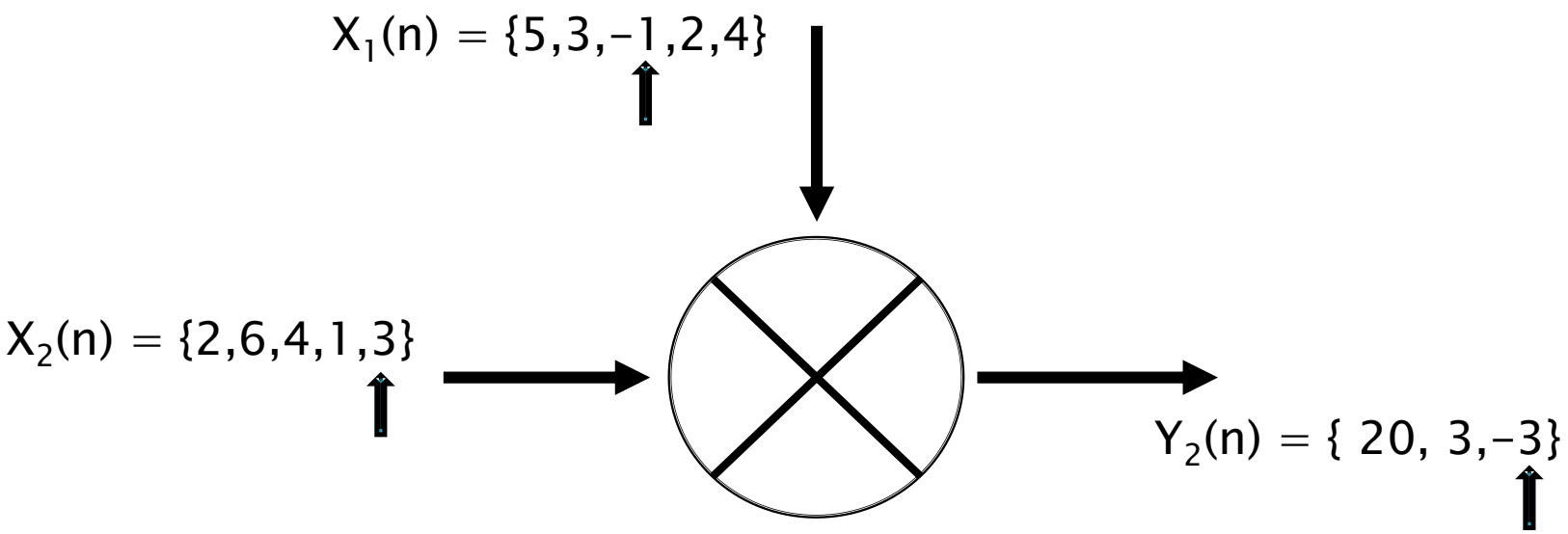


Or

$$Y_2(n) = \{20,\ 3,\ -3\}$$

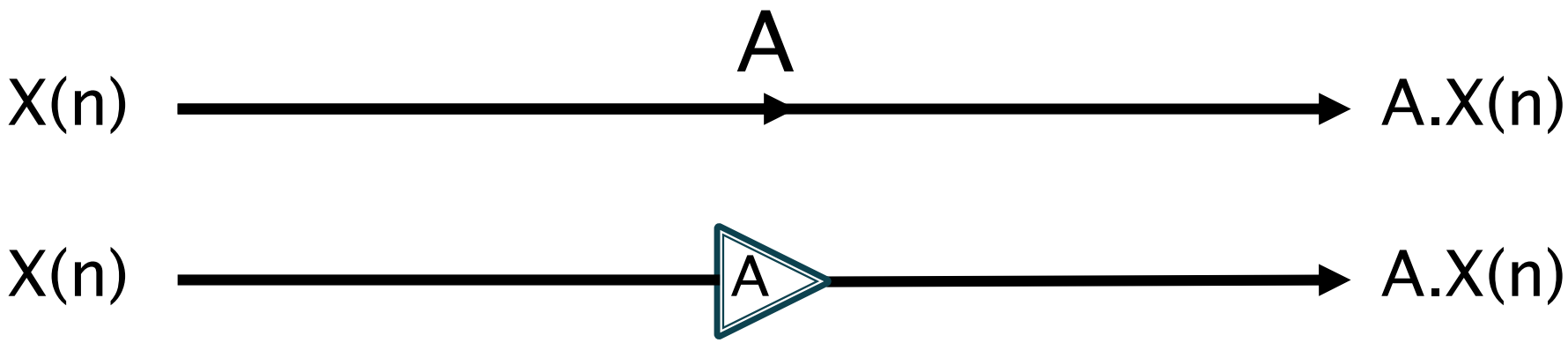


Multiplication Block:



Amplitude Scaling or Scalar Multiplication

- Amplitude Scaling of a discrete time signal (DTS) by a constant 'A' is accomplished by multiplying the value of every sample by constant 'A'.



$X(n) = \{5, 3, \underset{\uparrow}{-1}, 2, 4\}$

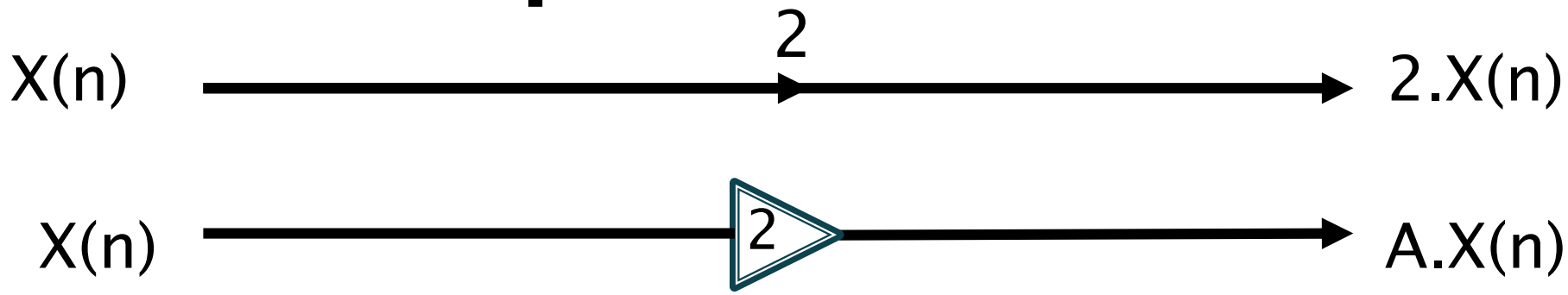
$A.X(n) = \{5A,\ 3A,\ \underset{\uparrow}{-1A},\ 2A,\ 4A\}$

After Amplitude Scaling

Example:

For the signal show below determine 2.X(n)

$X_1(n) = \{5,\ 3,\ \underset{\uparrow}{-1},\ 2,\ 4\}$



$2X(n) = \{(5X2),\ (3X2),\ \underset{\uparrow}{(-1X2)},\ (2X2),\ (4X2)\}$

$2X(n) = \{10,\ 6,\ \underset{\uparrow}{-2},\ 4,\ 8\}$

Time Shifting Operation

Time shifting or Shifting of a signal in time means that the signal may be either delayed in the time axis or advanced in the time axis. This is done by adding or subtracting a scalar "k" quantity to the time variable "n" in the function.

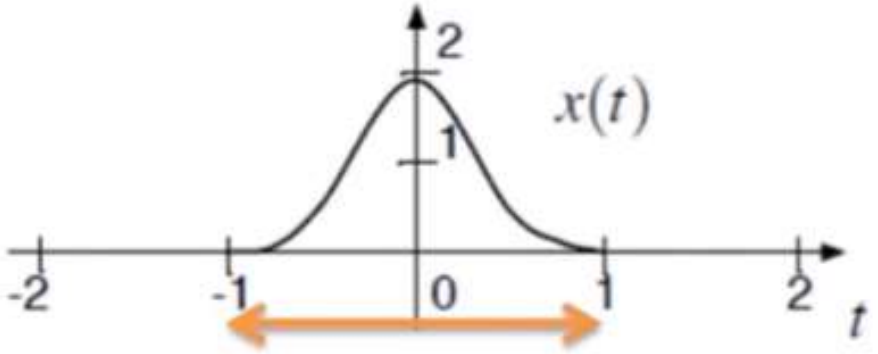
Time Delay Operation: x (n-k)

Subtracting a fixed positive number from the time variable will shift the signal to the right (Time delay) by the subtracted quantity.

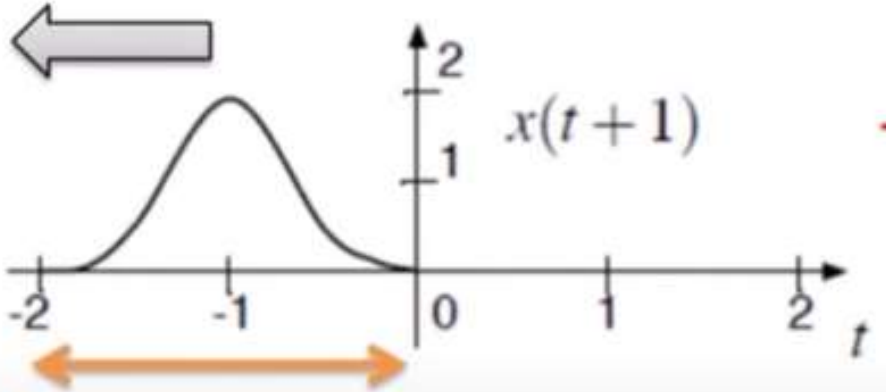
Time advance Operation: x (n+k)

Adding a fixed positive number from the time variable will shift the signal to the left (Time advance) by the added quantity.

Continuous Time Signal Shifting Mechanism:-

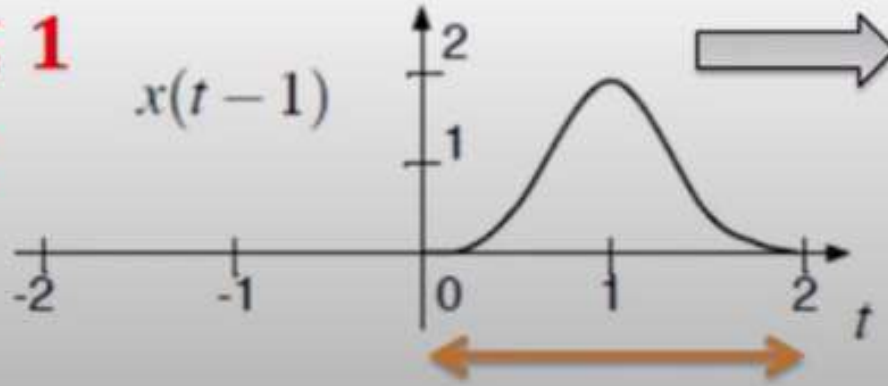


Left Shift by
1 unit.



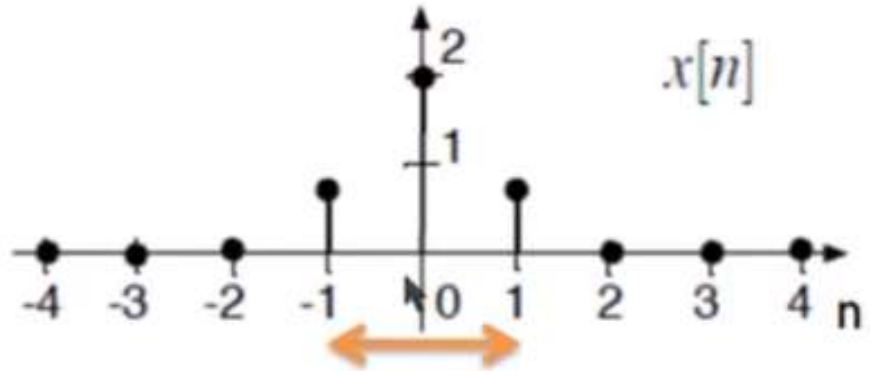
$$-1 \leq t+1 \leq 1$$
$$-2 \leq t \leq 0$$

$$-1 \leq t-1 \leq 1$$
$$0 \leq t \leq 2$$

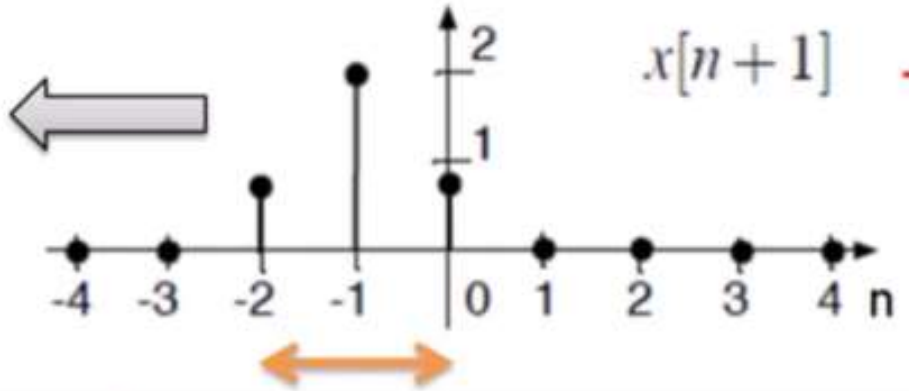


Right Shift by
1 unit.

Discrete Time Signal Shifting Mechanism:-



Left Shift of the signal by 1 unit.

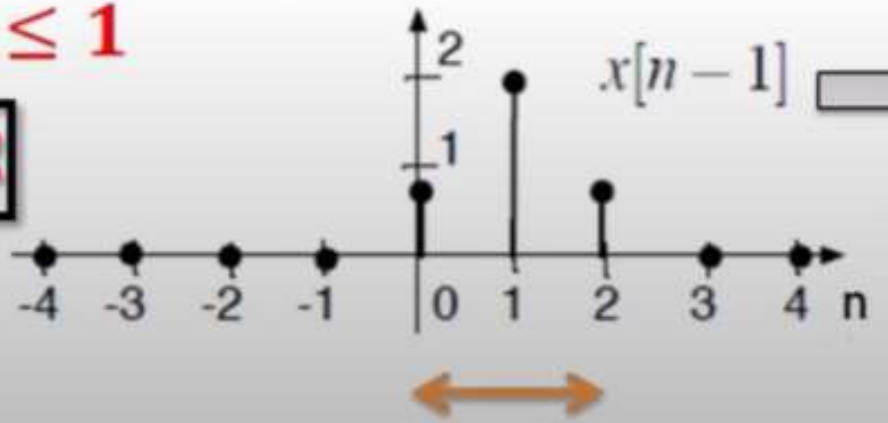


$$-1 \leq n+1 \leq 1$$

$$\boxed{-2 \leq n \leq 0}$$

$$-1 \leq n-1 \leq 1$$

$$\boxed{0 \leq n \leq 2}$$



Right Shift of the Signal by 1 unit.

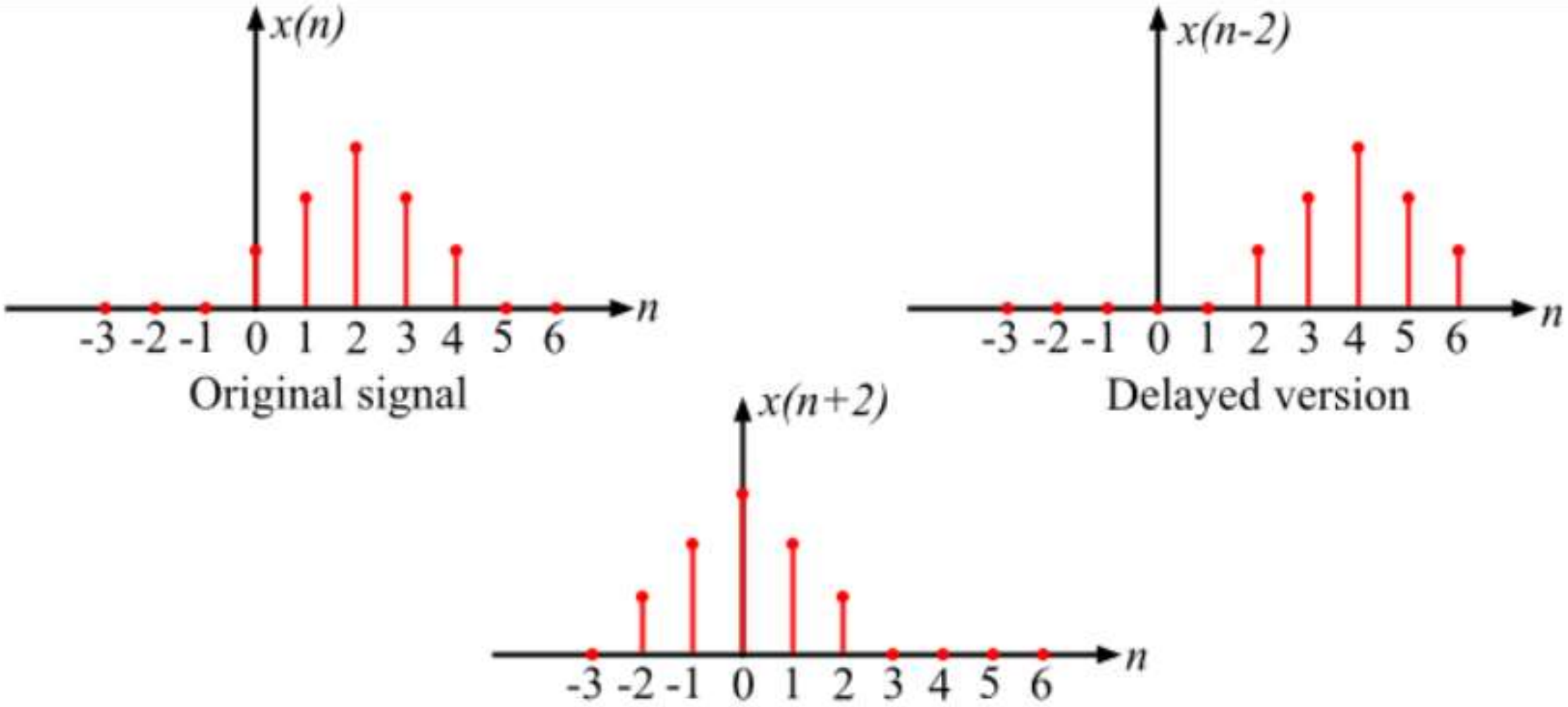

x(n)
n
-3 -2 -1 0 1 2 3 4 5 6
Original signal
x(n-2)
n
-3 -2 -1 0 1 2 3 4 5 6
Delayed version
x(n+2)
n
-3 -2 -1 0 1 2 3 4 5 6
Advanced version

Figure-2

Time Scaling Operation

Time scaling compresses or enlarges a signal by multiplying the time variable by some number. The time scaling of signal may be time compression or time expansion depending upon the value of the constant or scaling factor. The time scaling operation of signals is very useful when data is to be fed at some rate and is to be taken out at a different rate.

Decimation or Down Sampling

x(kn) where k>1

If that number is greater than one, the signal get compressed in time and the operation is called decimation or down sampling.

Interpolation or Up Sampling

x(kn) where k<1

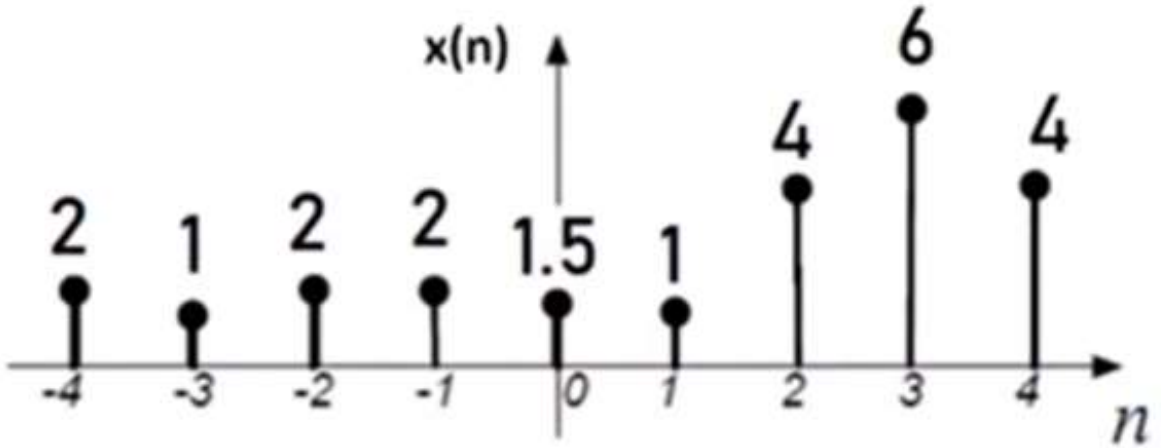
If that number is less than one, the signal becomes wider and the operation is called Interpolation or up sampling.

decimation or down sampling:-

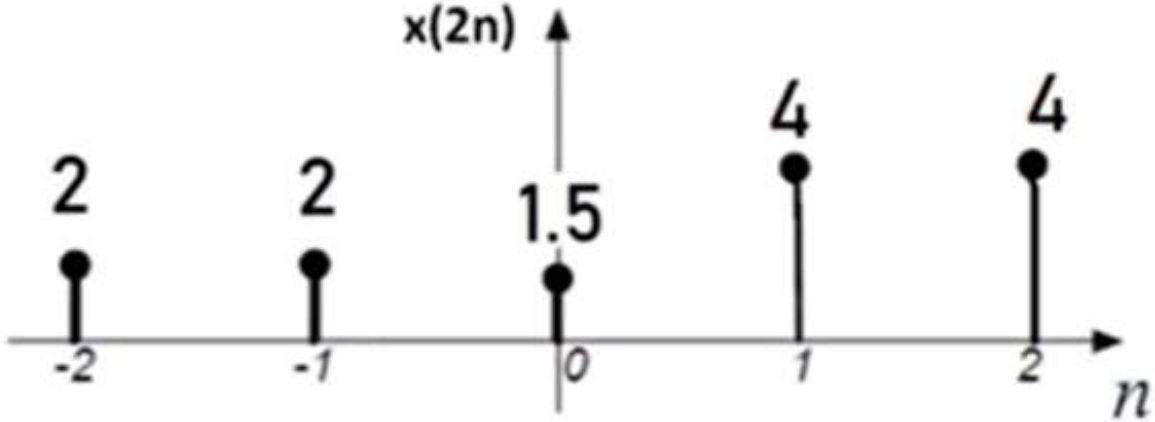
$$x(kn) where\ k > 1$$

	-4	-3	-2	-1	0	1	2	3	4
Divide by 2	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2

$$-4 \leq 2n \leq 4$$



$$-2 \leq n \leq 2$$

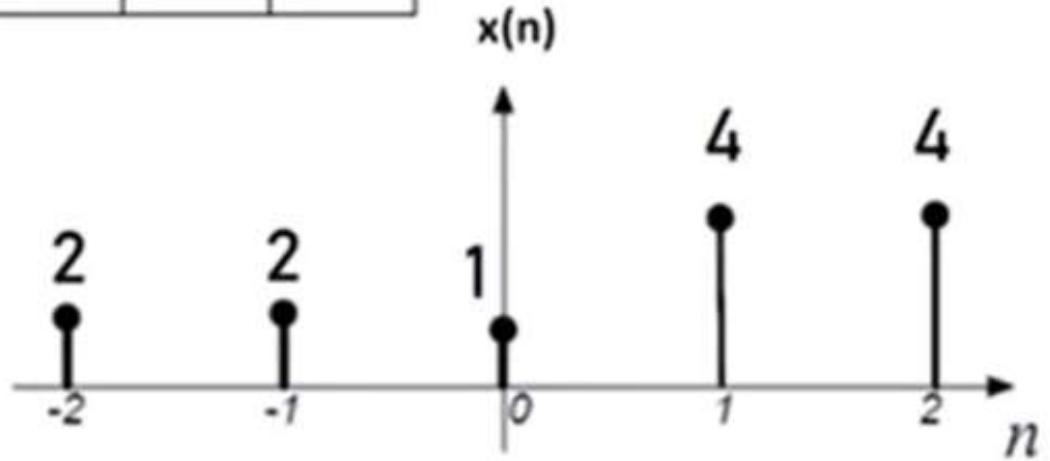


interpolation or up sampling:-

$x(kn) where\ k < 1$

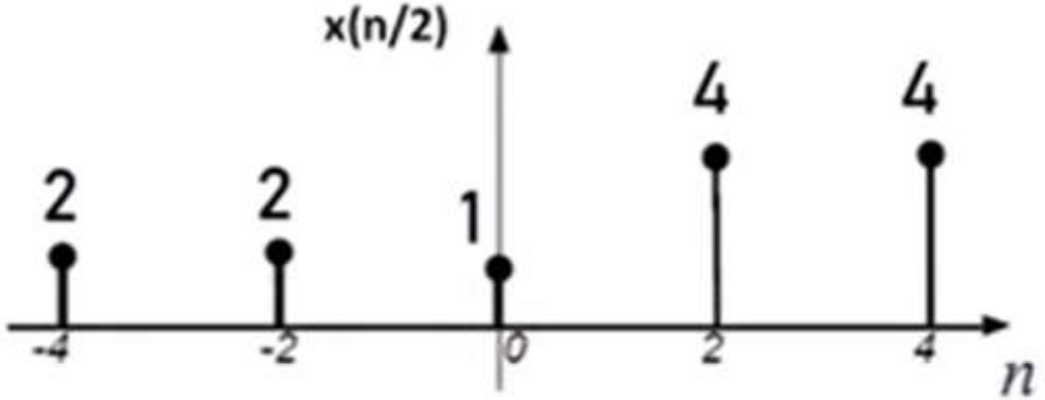
Multiply by 2

-2	-1	0	1	2
-4	-2	0	2	4



$$-2 \leq \frac{n}{2} \leq 2$$

$$-4 \leq n \leq 4$$



Folding Operation (Time Reversal)

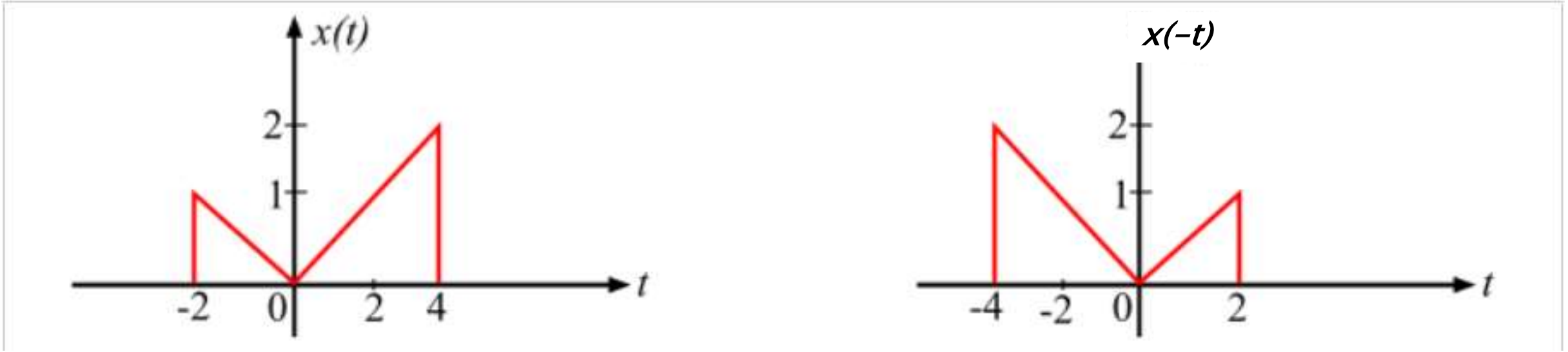
The time reversal of a signal is folding of the signal about the time origin (n=0). Time reversal of a signal can be achieved by taking mirror image of the signal about y axis. Time reversal of a signal is a useful operation on signals in convolution.

Time reversal is known as **folding or reflection**

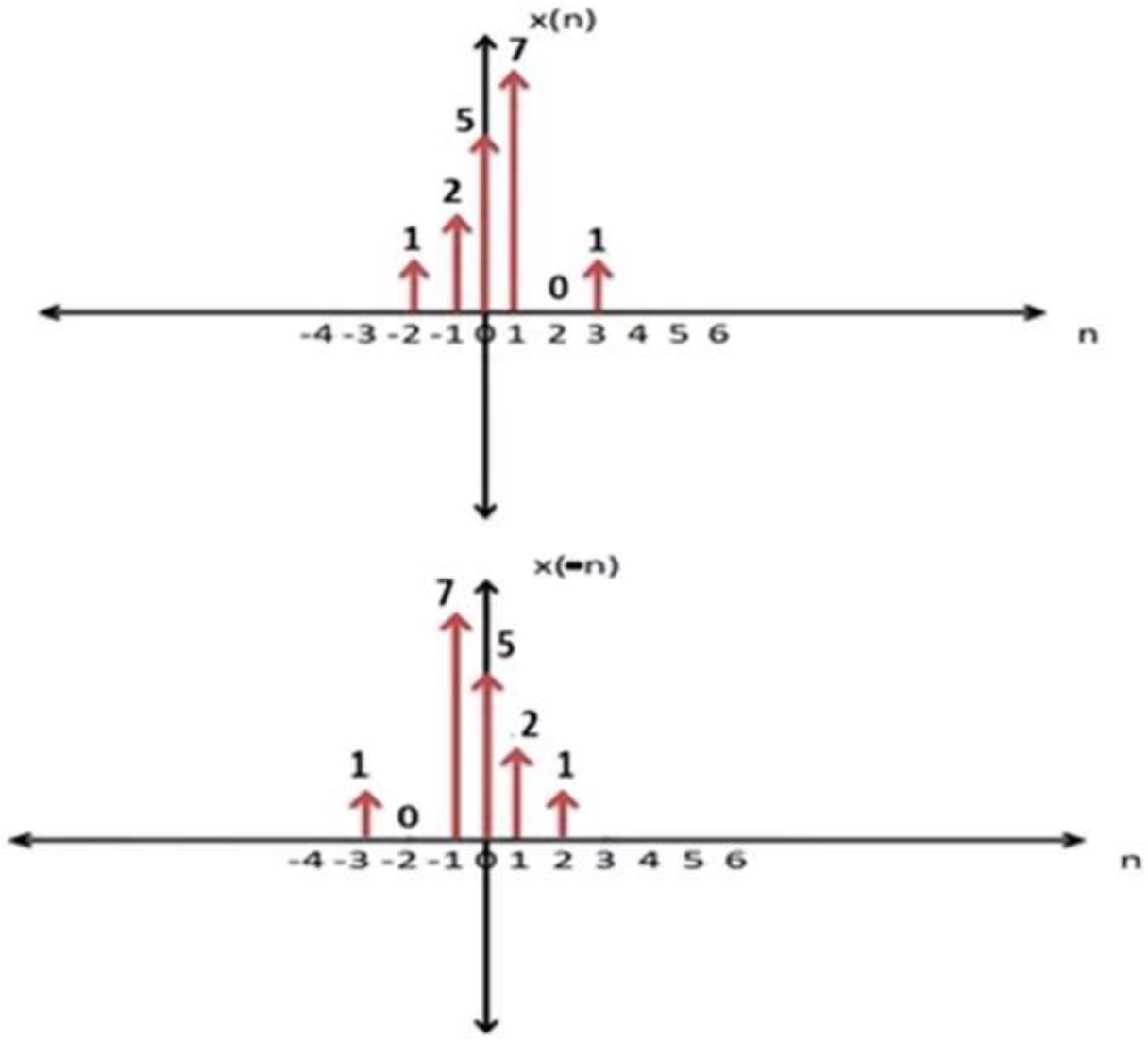
Time Reversal of a Continuous Time Signal

The time reversal of a continuous time signal x(t) is the rotation of the signal by 180° about the vertical axis. Mathematically, for the continuous time signal x(t), the time reversal is given as,

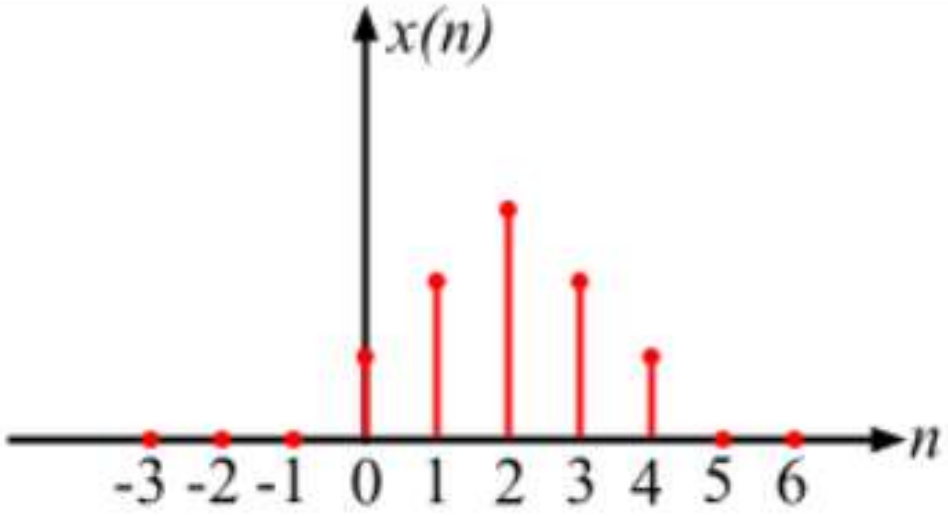
$$x(t) = x(-t)$$

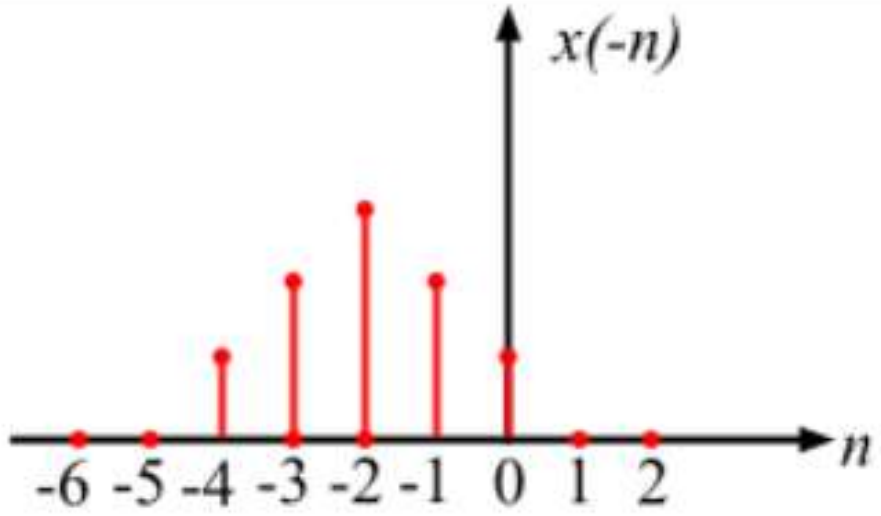


Time Reversal of a Discrete Time Signal



Time Reversal of a Discrete Time Signal





Folding + Shifting Operation

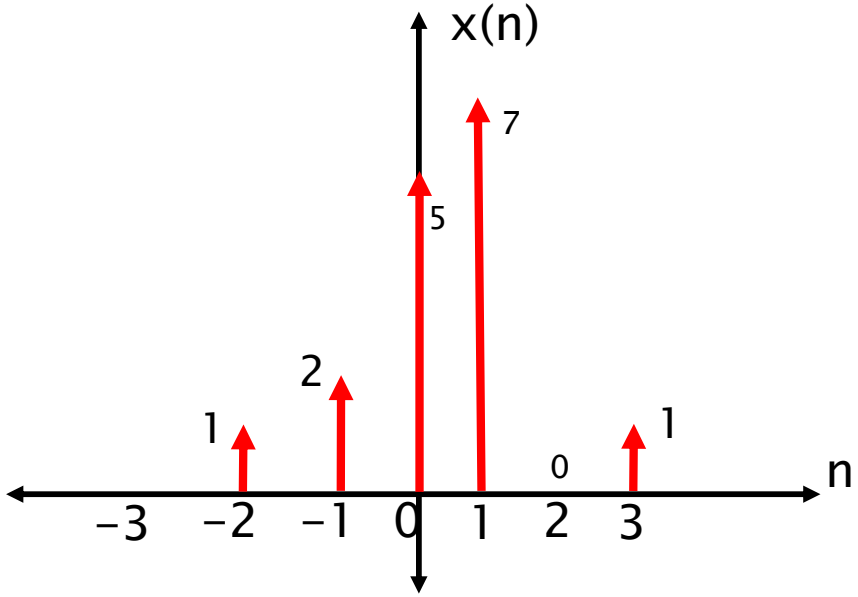
Plot **x(n)** the original signal

Plot **x(-n)** by performing time reversal or folding operation

Plot **x(-n-k)** by shifting **x(-n)** to the LEFT by **k** unit

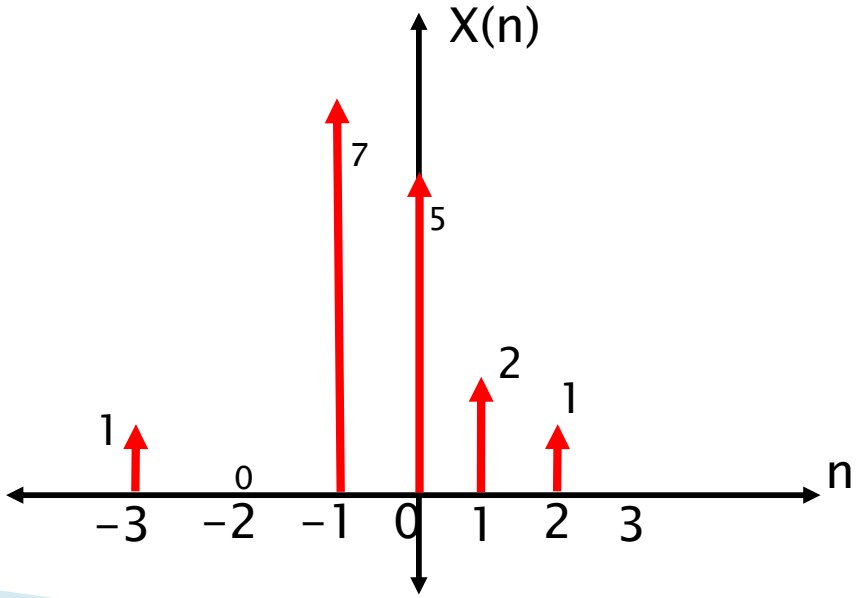
Plot **x(-n+k)** by shifting **x(-n)** to the RIGHT by **k** unit

ORIGINAL SIGNAL



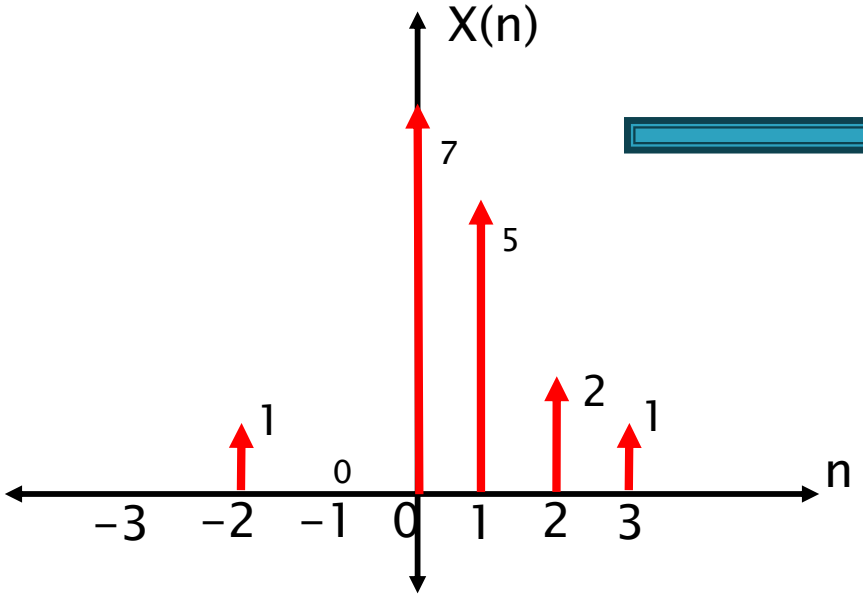
x(n)

Time Reversal (Folding)



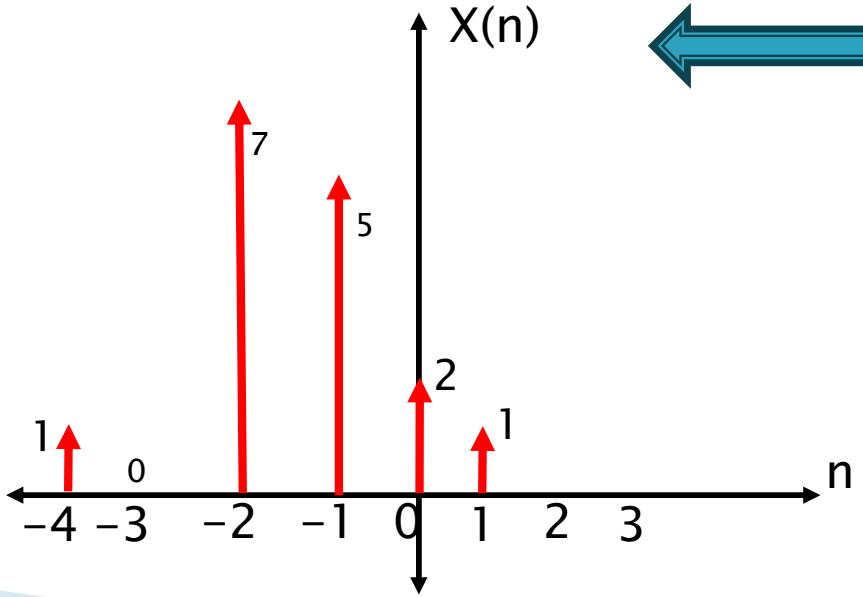
x(−n)

Folding + Shift right of Folded Signal



x(−n+1)

Folding + Shift left of Folded Signal



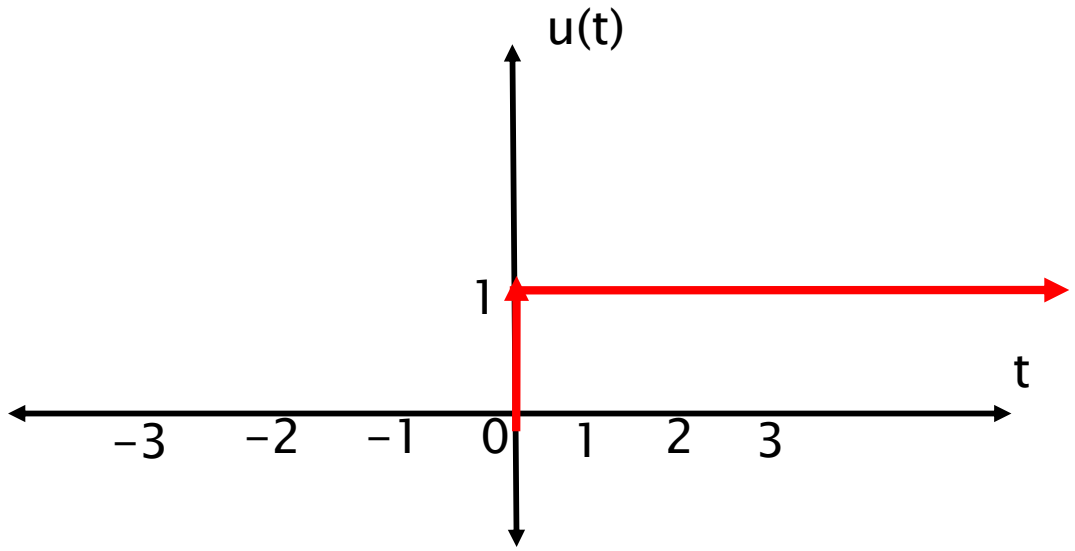
x(−n−1)

Standard/Basic Signals

Step Function

- The continuous time step function exists only for positive time and remains zero for negative time.
- Its amplitude remains constant.
- The step function whose magnitude is 1, is known as the Unit Step function u(t).

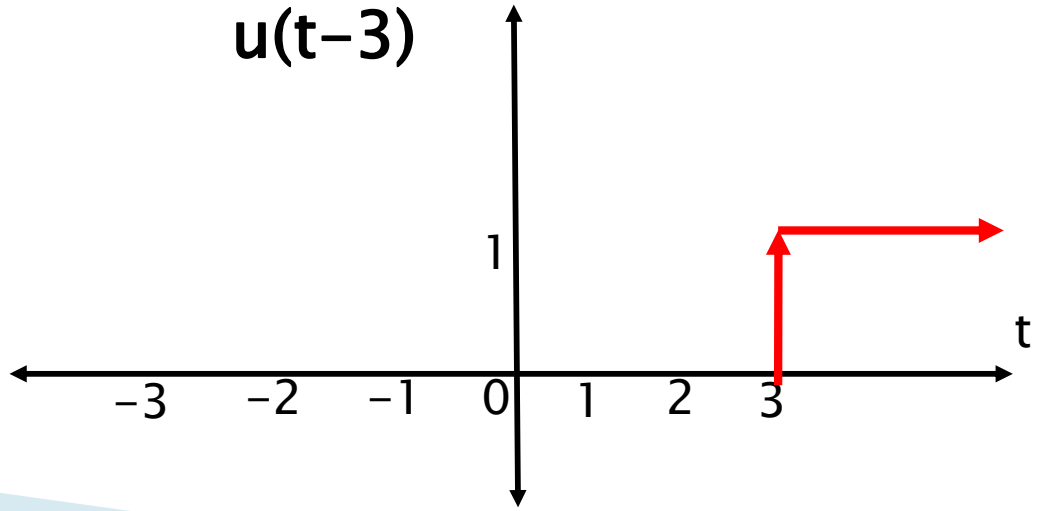
$$u(t) = \begin{cases} 1 \; ; \; t \geq 0 \\ 0 \; ; \; t < 0 \end{cases}$$


u(t)
1
t
−3
−2
−1
0
1
2
3

Unit Step Function

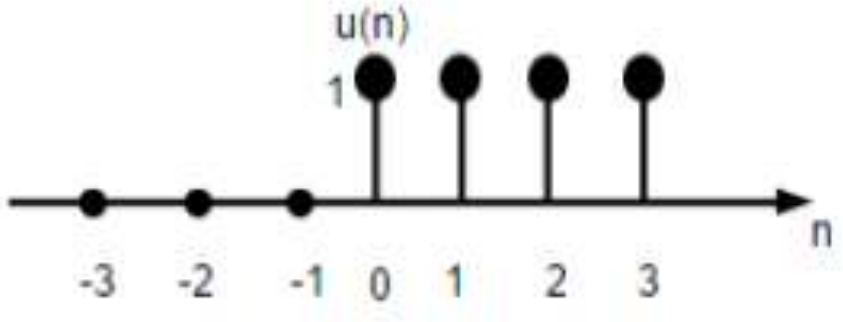
A Continuous time shifted unit step function is represented as

$$u(t-k) = \begin{cases} 1 \; ; \; t \geq k \\ 0 \; ; \; t < k \end{cases}$$

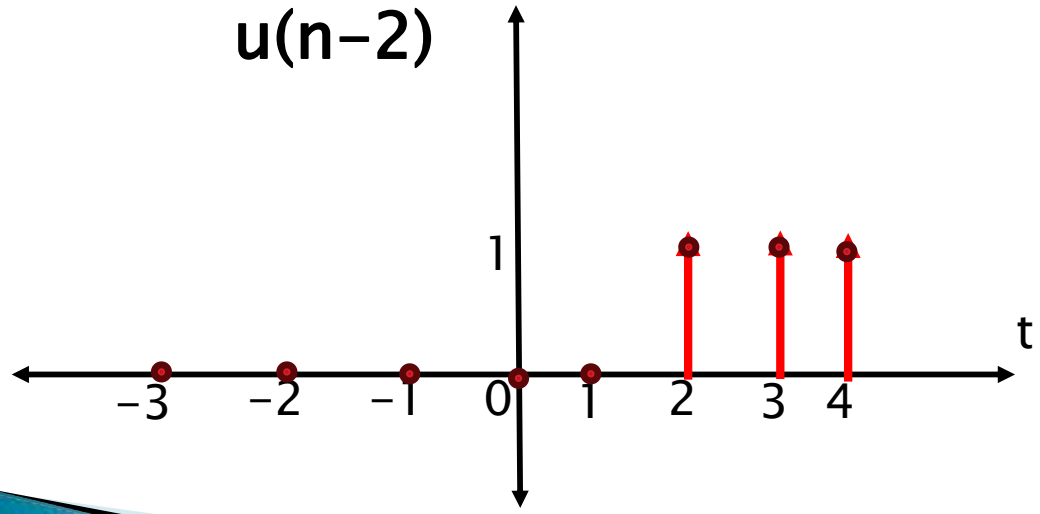


Unit Step Function

Discrete time unit step signal is defined as;



$$U(n) = \begin{cases} 1, & for \quad n \geq 0 \\ 0, & for \quad n < 0 \end{cases}$$



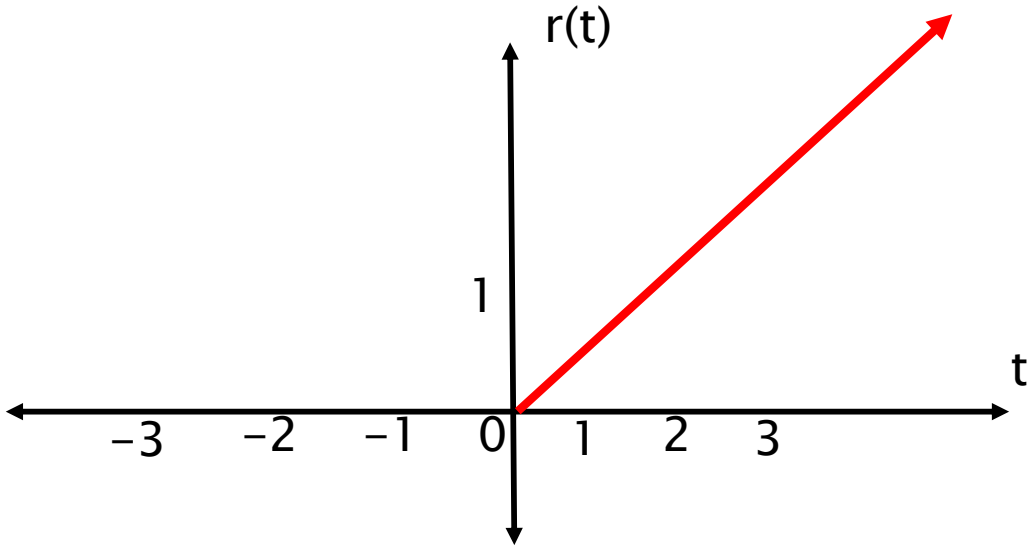
$$u(n-k) = \begin{cases} 1 \; ; \; n \geq k \\ 0 \; ; \; n < k \end{cases}$$

Ramp Function

- The continuous time ramp function starts at time=0 and increases linearly with respect to time.

- The ramp function whose slope is 1, is known as the unit ramp function, r(t)

$$r(t) = \begin{cases} t \; ; \; t \geq 0 \\ 0 \; ; \; t < 0 \end{cases}$$

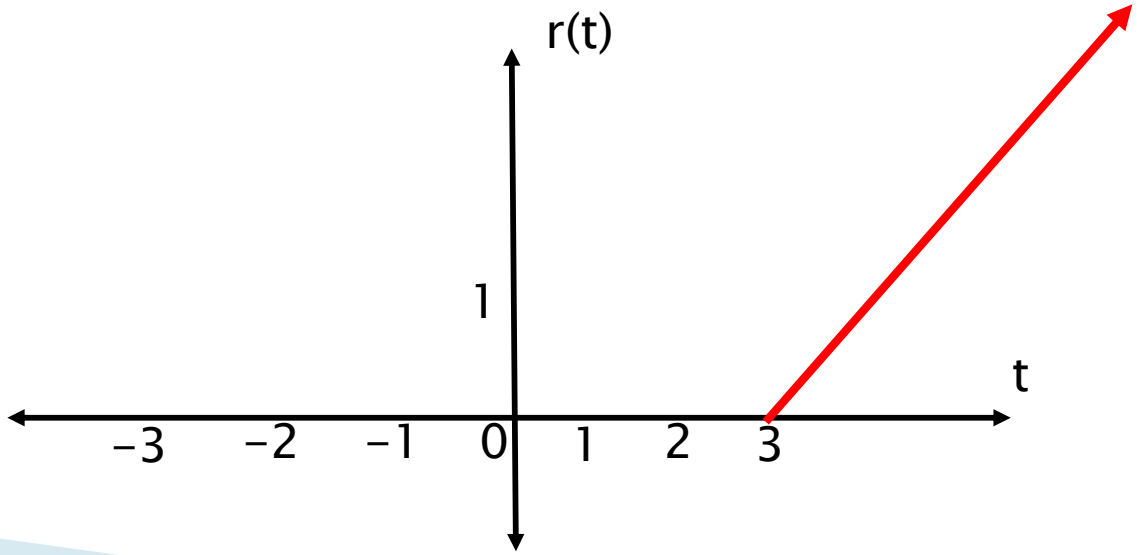

r(t)
1
t
−3
−2
−1
0
1
2
3

Unit Ramp Function

A Continuous time shifted unit ramp function is represented as

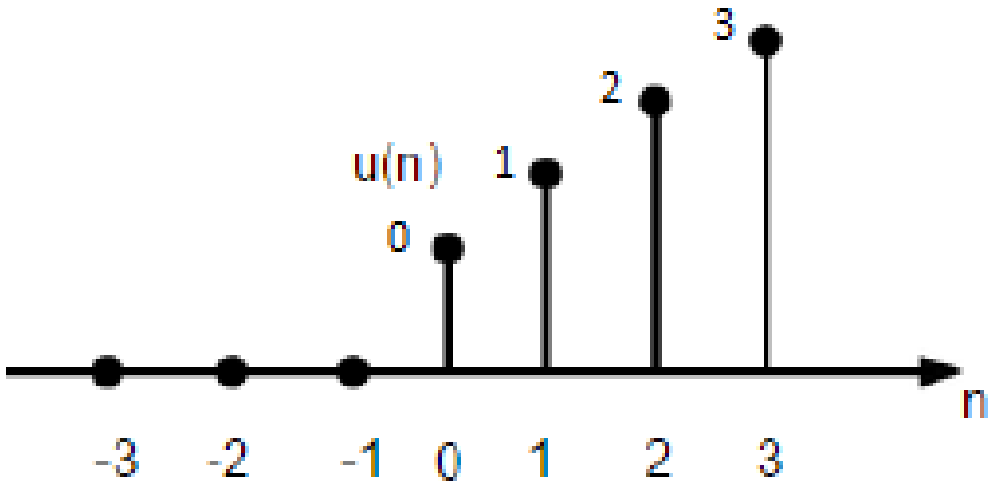
$$r(t-k) = \begin{cases} t-k \; ; \; t \geq k \\ 0 \quad ; \; t < k \end{cases}$$

r(t−3)



Unit Ramp Function

A discrete unit ramp function can be defined as –

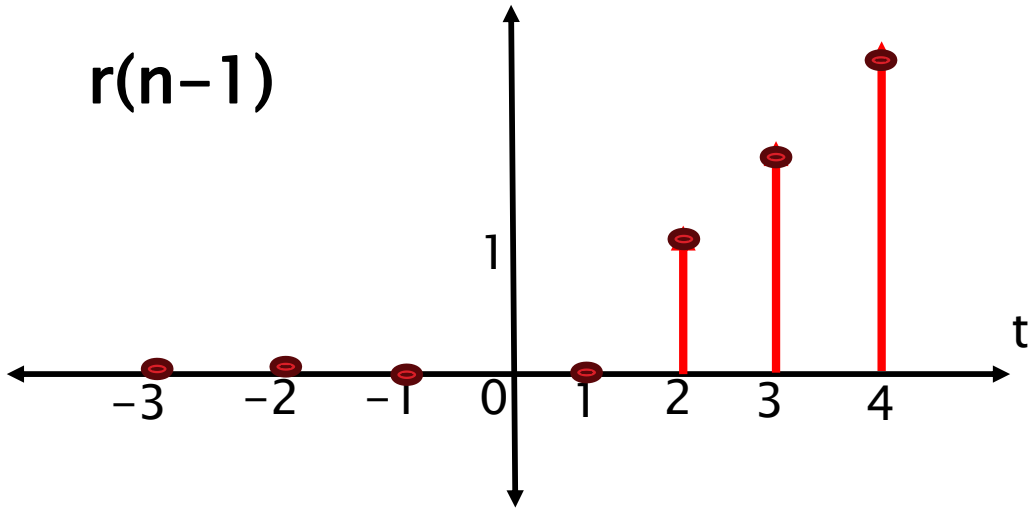


$$r(n) = \begin{cases} n, & for \quad n \geq 0 \\ 0, & for \quad n < 0 \end{cases}$$

$$r(n-k) = \begin{cases} n-k \; ; \; n \geq k \\ 0 \; ; \quad n < k \end{cases}$$

Unit Ramp Function

A discrete unit ramp function can be defined as –



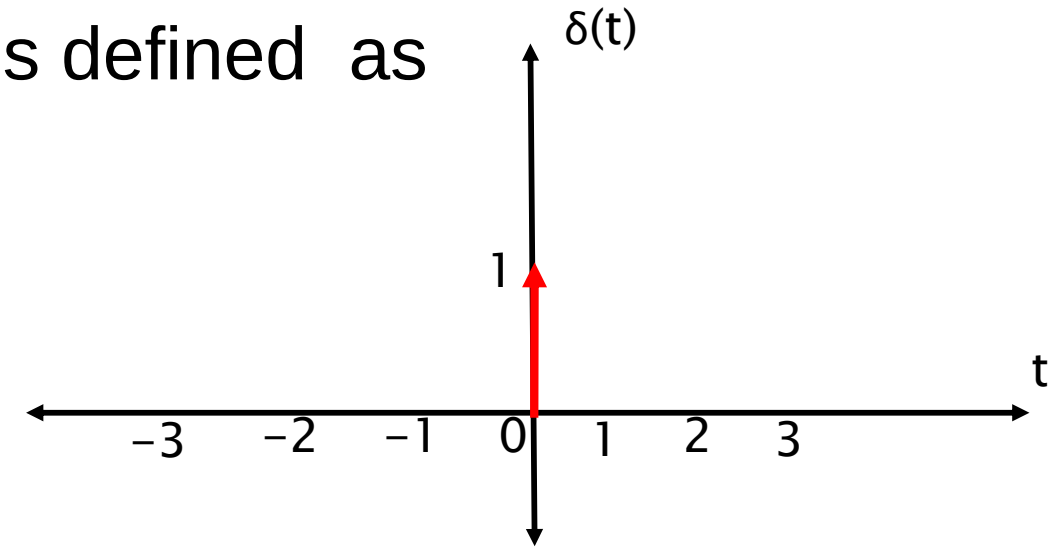
$$r(n) = \begin{cases} n, & for \quad n \geq 0 \\ 0, & for \quad n < 0 \end{cases}$$

$$r(n-k) = \begin{cases} n-k \; ; \; n \geq k \\ 0 \; ; \quad n < k \end{cases}$$

Unit Impulse Function

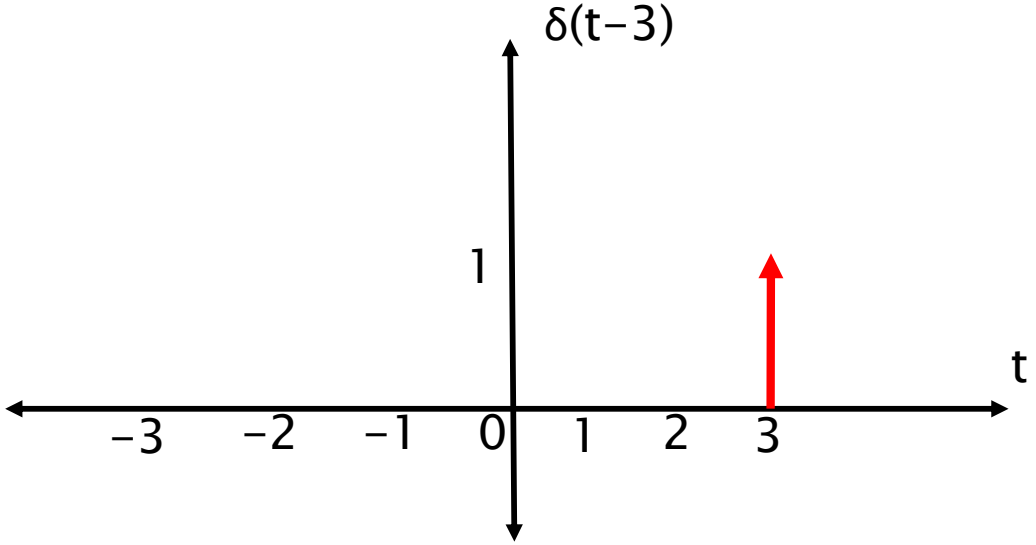
A Continuous time unit impulse function is defined as

$$\delta(t) = \begin{cases} 1 \; ; \; t = 0 \\ 0 \; ; \; t \neq 0 \end{cases}$$



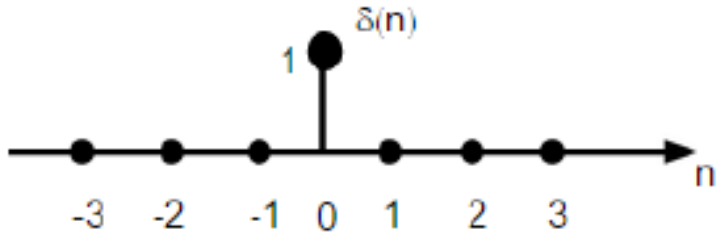
Shifted unit impulse function is defined as

$$\delta(t-k) = \begin{cases} 1 \; ; \; t = k \\ 0 \; ; \; t \neq k \end{cases}$$



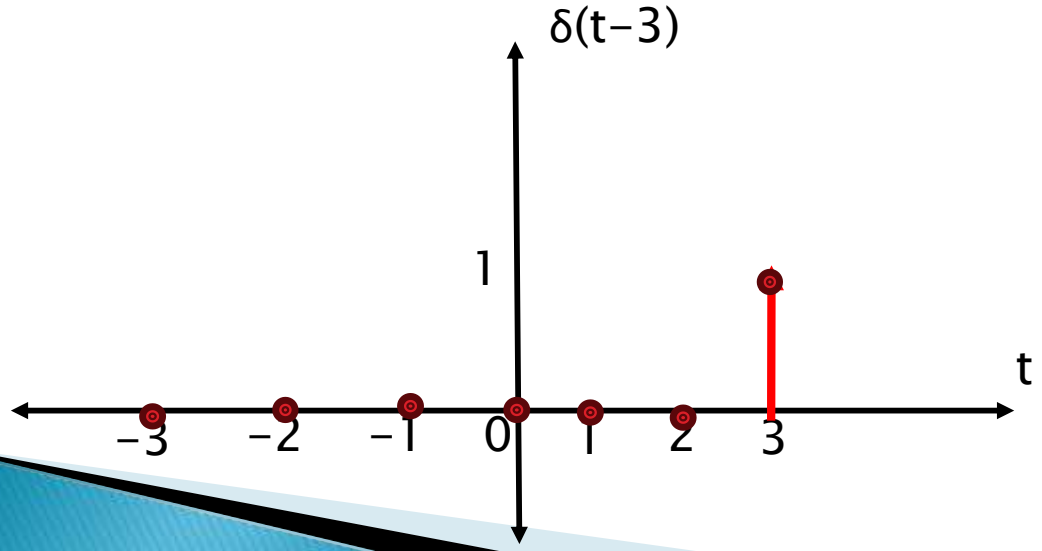
Unit Impulse Function

It is denoted as δn in discrete time domain and can be defined as;



$$\delta(n) = \begin{cases} 1, & for \quad n = 0 \\ 0, & Otherwise \end{cases}$$

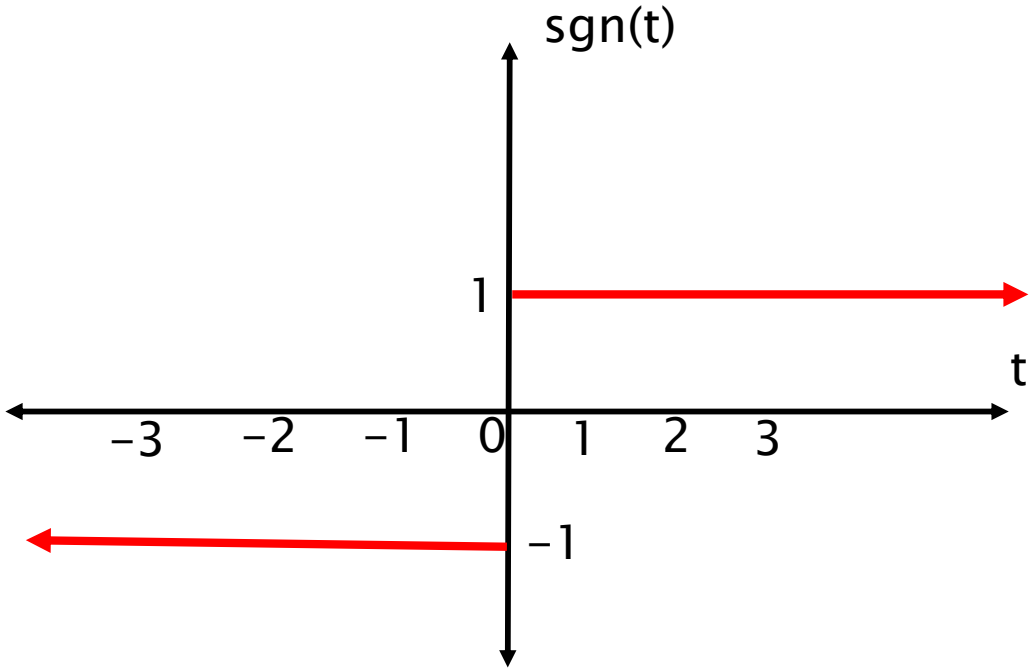
Shifted unit impulse function is defined as



$$\delta(t-k) = \begin{cases} 1 \; ; n = k \\ 0 \; ; n \neq k \end{cases}$$

Signum Function

The signum function is denoted as sgn (t). It is defined as



$$\text{sgn}(t) = \begin{cases} 1 & ; t > 0 \\ 0 & ; t = 0 \\ -1 & ; t < 0 \end{cases}$$

Rectangular Function

The rectangular function is denoted as rec(t). It is defined as

$$x(t) = A \, \text{rect}\left(\frac{t}{T}\right)$$

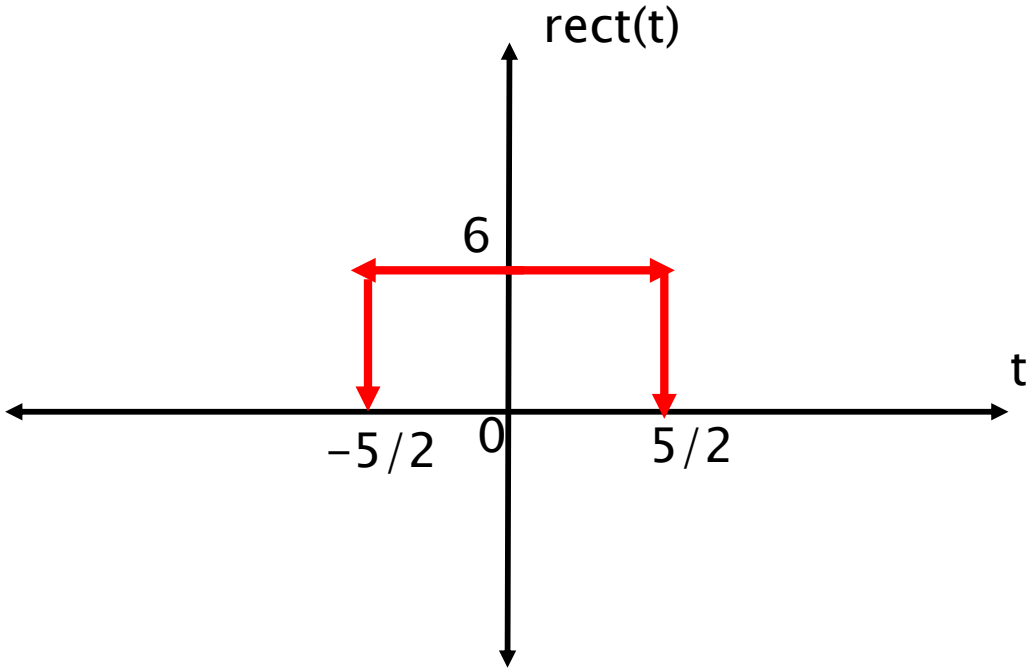
where A= Amplitude

T= period of rectangle

The rectangular signal is also known as the unit pulse, gate function or normalized boxcar function. Also, the rectangular function is an even function of time.

Rectangular Function

$$x(t) = 6\,\mathrm{rect}\left(t/5\right)$$



Triangular Function

The triangular function is defined as

$$x(t) = A\left[1 - \frac{|t|}{T}\right]$$

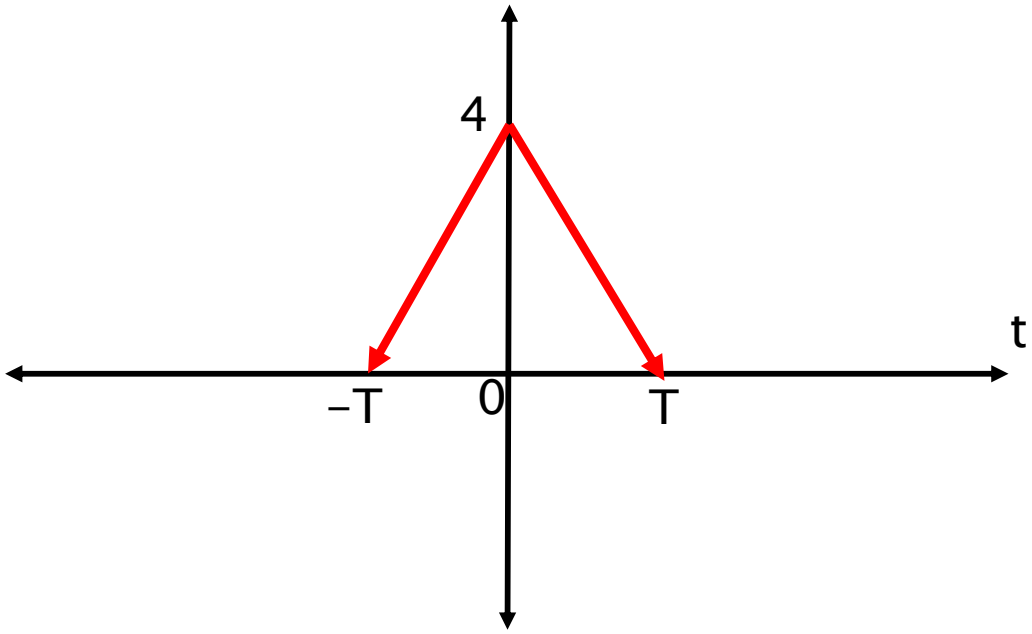
where A= Amplitude

T= period of triangle

The triangle is also an even function of time.

Triangular Function

$$x(t)=A[\ 1-(/t//6\}]$$



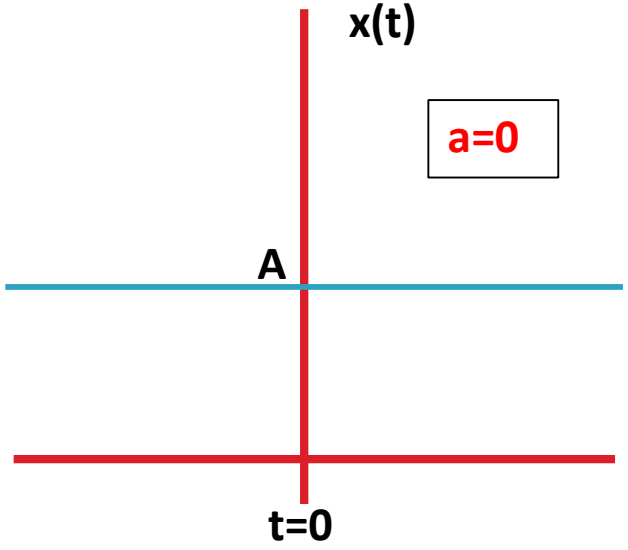
Exponential Signals

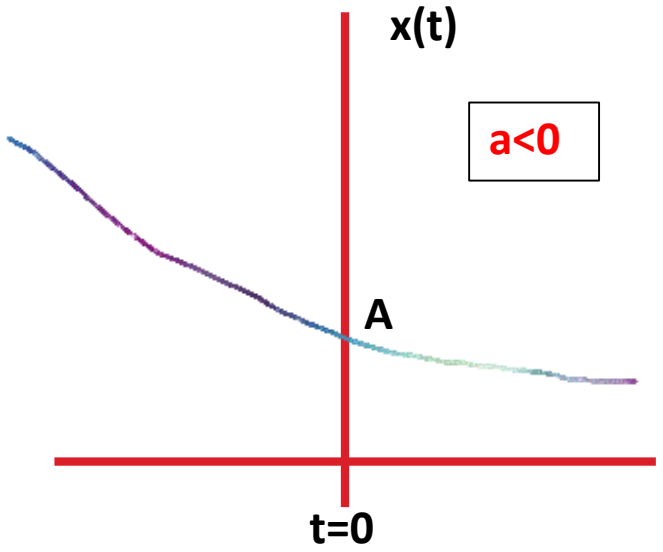
The continuous time real exponential signal is defined as

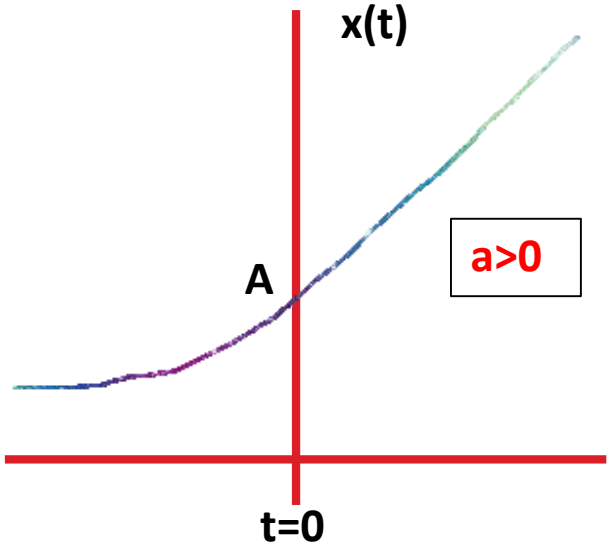
$$x(t) = Ae^{\alpha t}$$

Here both 'A' and 'α' are both real.

- The value of x(t) is 'A' at t=0.
- The value of 'α 'can be either positive or negative
- If 'α = 0' then, exponential signal x(t) has constant amplitude for all 't'
- If 'α < 0' then, exponential signal x(t) is a decaying exponential signal
- If 'α > 0' then, exponential signal x(t) is a growing exponential signal


x(t)
a=0
A
t=0


x(t)
a<0
A
t=0


x(t)
a>0
A
t=0

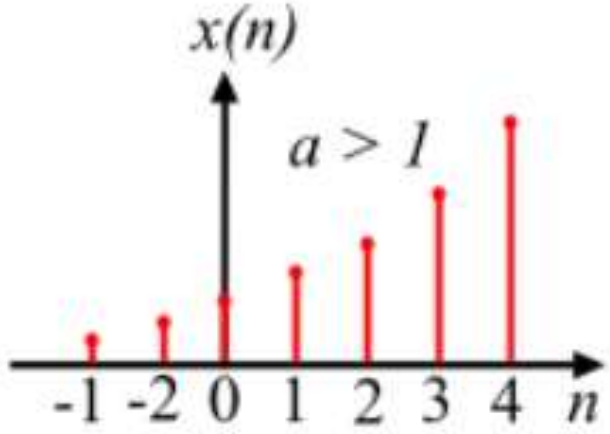
Exponential Signals

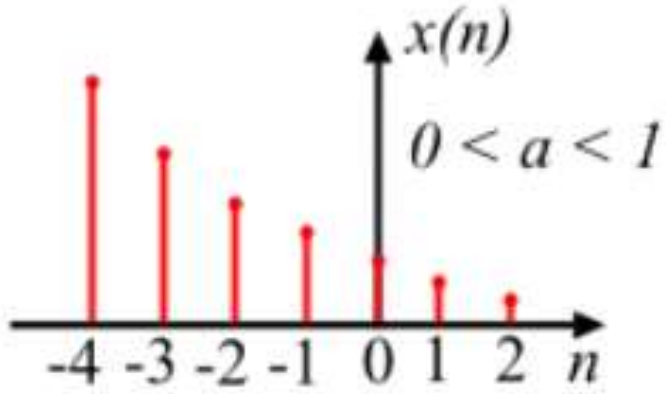
The discrete time real exponential signal is defined as

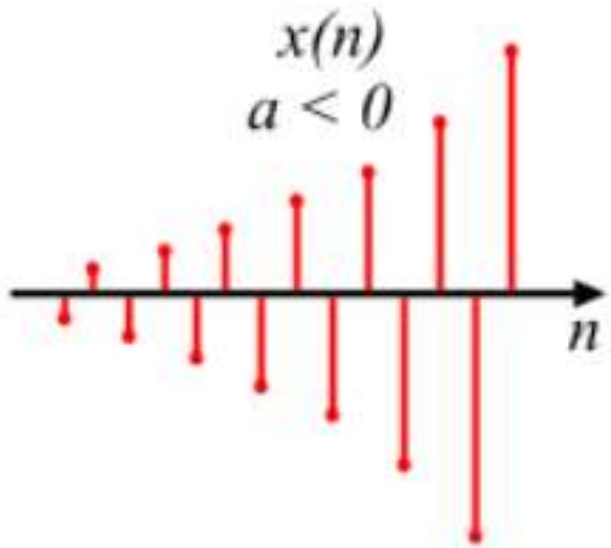
$$x_n(n)=A\alpha$$

Here both 'A' and 'α' are both real.

- The value of x(n) is 'A' at n=0.
- If 'α < 0' then, exponential signal x(n) has alternating sign
- If 'α > 1' then, exponential signal x(n) is growing exponential signal
- If '0<α < 1' then, exponential signal x(n) is a decaying exponential signal


x(n)
a > 1
-1 -2 0 1 2 3 4 n


x(n)
0 < a < 1
-4 -3 -2 -1 0 1 2 n


x(n)
a < 0
n

Sinusoidal Signal

The continuous time sinusoidal signal is represented

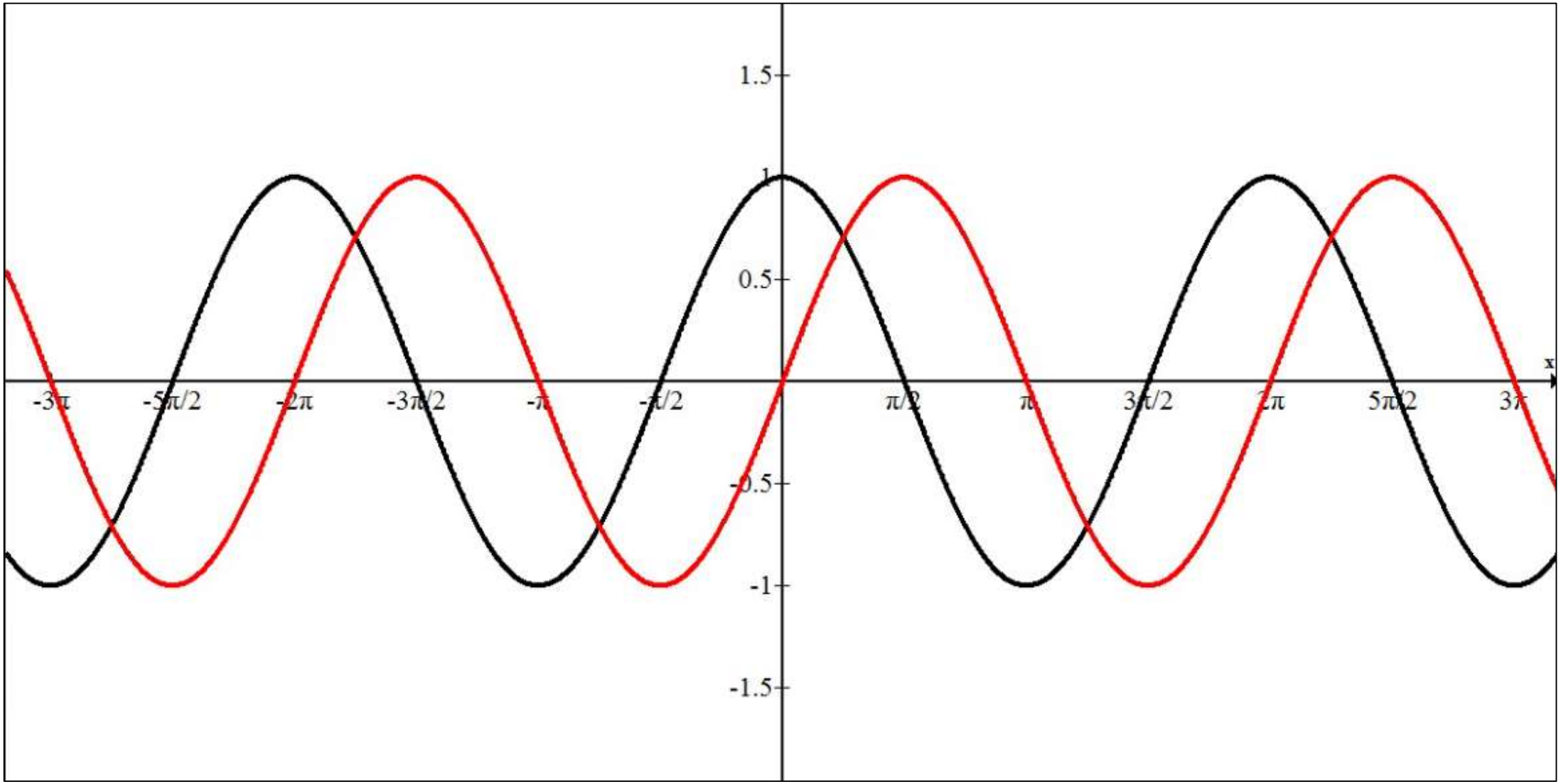
$x(t) = A \sin(\omega t \pm \Phi)$

$x(t) = A \cos(\omega t \pm \Phi)$

Where, A=> Amplitude of the sinusoidal signal

ϕ=> Phase angle in radians.

ω=> Angular frequency in radians/second



cos

sin

Sinusoidal Signal

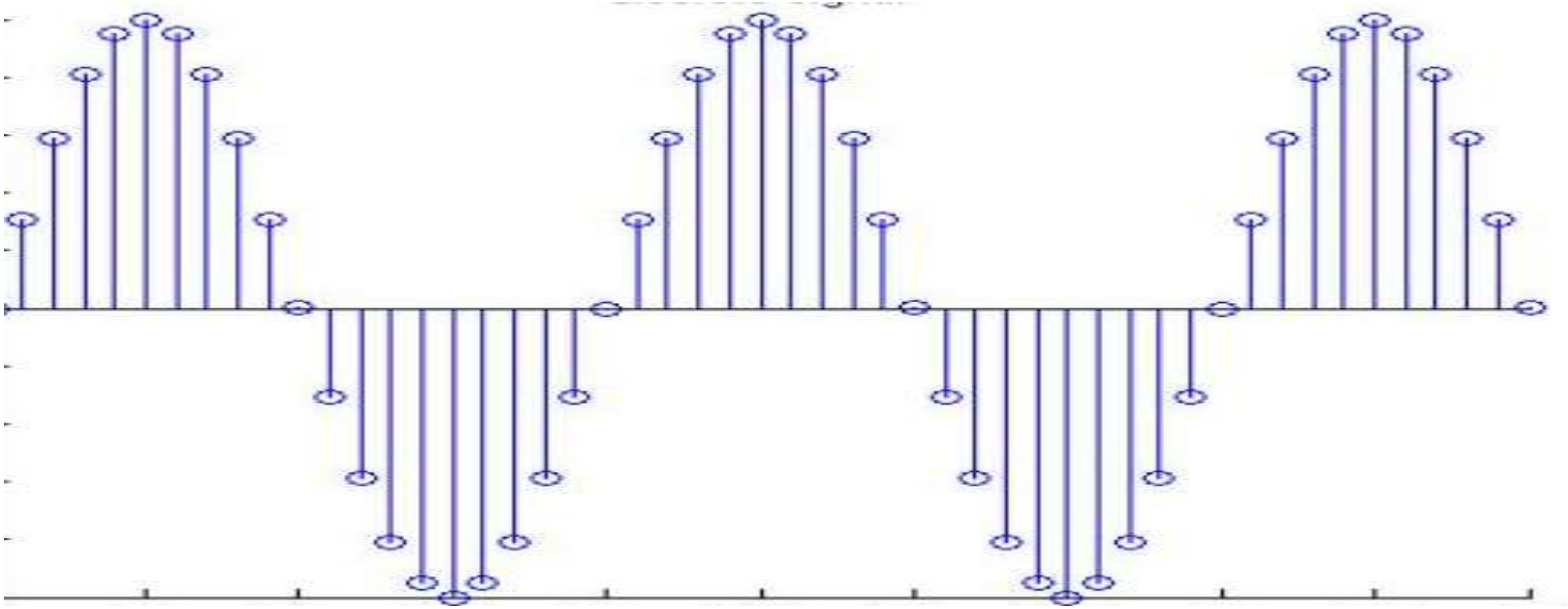
The discrete time sinusoidal signal is represented

$x(n) = A \sin(\omega n \pm \Phi)$ $x(n) = A \cos(\omega n \pm \Phi)$

Where, A=> Amplitude of the sinusoidal signal

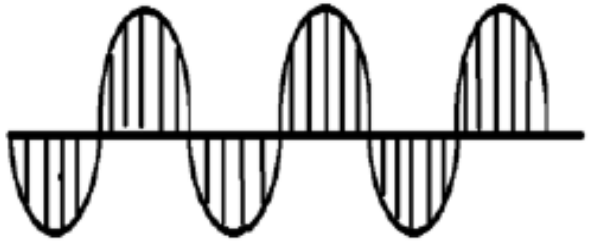
ϕ=> Phase angle in radians.

ω=> Angular frequency in radians/second



Sinusoidal Signal

All continuous-time signals are periodic. The discrete-time sinusoidal sequences may or may not be periodic. They depend on the value of ω. For a discrete time signal to be periodic, the angular frequency ω must be a rational multiple of 2π.



Discrete sinusoidal signal

A discrete sinusoidal signal is shown in the figure above.

Discrete form of a sinusoidal signal can be represented in the format −

$$x(n) = A\sin(\omega n + \phi)$$

Here A,ω and φ have their usual meaning and n is the integer. Time period of the discrete sinusoidal signal is given by −

$$N = \frac{2\pi m}{\omega}$$

Where, N and m are integers.

Linear Time-Invariant Systems (LTI systems)



Linear Time-invariant Systems

LTI systems are a class of systems used in signals and systems. They have an output that is linearly related to their input and whose characteristics do not change over time.

LTI Systems Examples

- Electrical circuits
- Audio processing systems
- Communication systems

Properties of LTI Systems

❑ Linearity Properties

When a system's outputs for a linear combination of inputs match the outputs of a linear combination of each input response separately, the system is said to be linear.

Linear systems are systems whose outputs for a linear combination of inputs are the same as a linear combination of individual responses to those inputs.

Properties of LTI Systems

- **Time invariance**

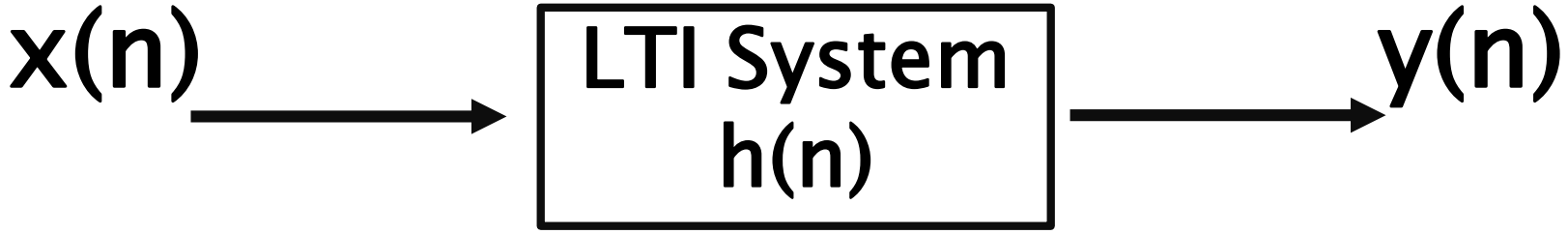
Time-invariant systems are ones whose output is independent of the timing of the input application.

Time-invariant systems are systems where the output does not depend on *when* an input was applied.

Convolution

Convolution is a mathematical operation used to determine the output of an LTI system given its input and impulse response.

Convolution combines two sequences to produce a third sequence, representing the system's output.



$$y(n) = x(n) * h(n)$$

Convolution formula:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Classification of Discrete Time Signal

Classification of Discrete Time Signals

1. Deterministic and Nondeterministic Signals
2. Causal, Non-causal and Anti-causal Signals
3. Symmetric (Even) and Anti-Symmetric (Odd) Signals
4. Periodic and Aperiodic (Non-periodic) Signals
5. Energy and Power Signals

Deterministic and Nondeterministic Signals

Deterministic Signals

- A discrete time signal is said to be deterministic if it can be completely specified by *mathematical equations*.

- The signal can be determined or analyzed at every instant in time.

- It is a signal in which each value of the signal is fixed, being determined by a mathematical expression, rule, or table.

Examples of Deterministic Signals :

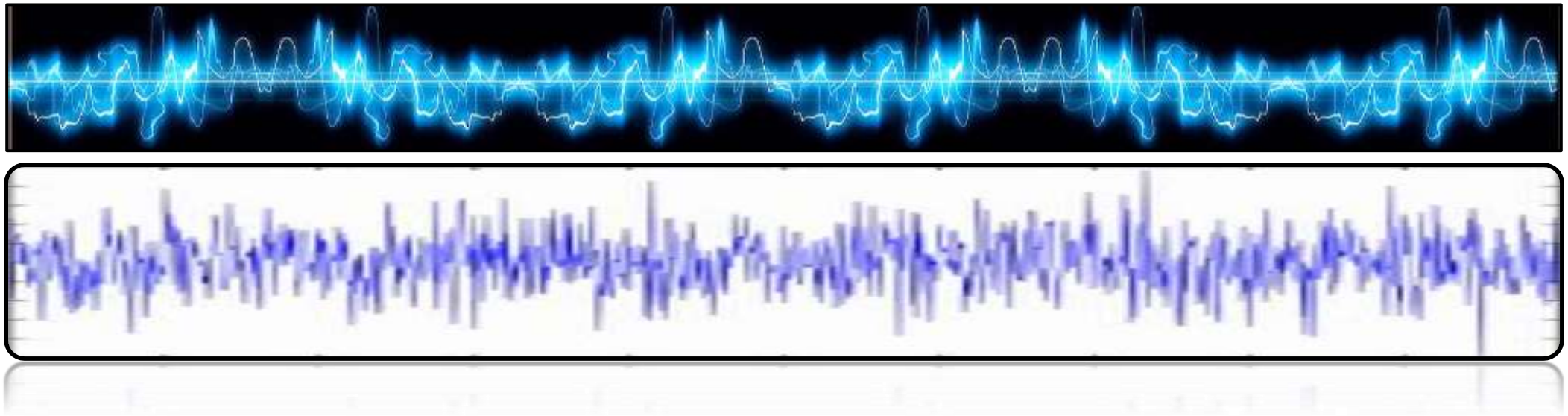
- Step Signal
- Exponential Signal
- Ramp Signal
- Sinusoidal Signal

Nondeterministic Signal / Random Signal

- A DTS is said to be nondeterministic if it can't be specified by any *mathematical equations*.
- The signal can't be determined or analyzed at every instant in time.
- It is a signal where there is uncertainty at any particular instant in time.

Examples of Nondeterministic Signals :

- The noise signals from various source

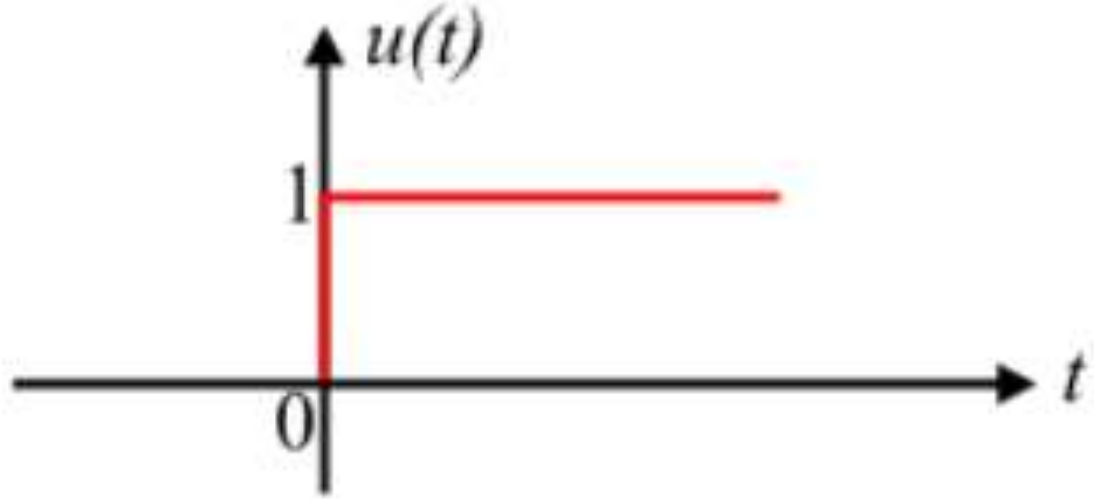


Causal, Anti-causal and Non-causal Signals

Causal Signals

- A discrete time signal is said to be *causal* if it is finite or defined for $n \geq 0$, and zero for $n < 0$
- The causal signals are signals that are zero for all negative times or signals does not exist for negative time.
- A causal signal can also be called as Right side sequence signals.

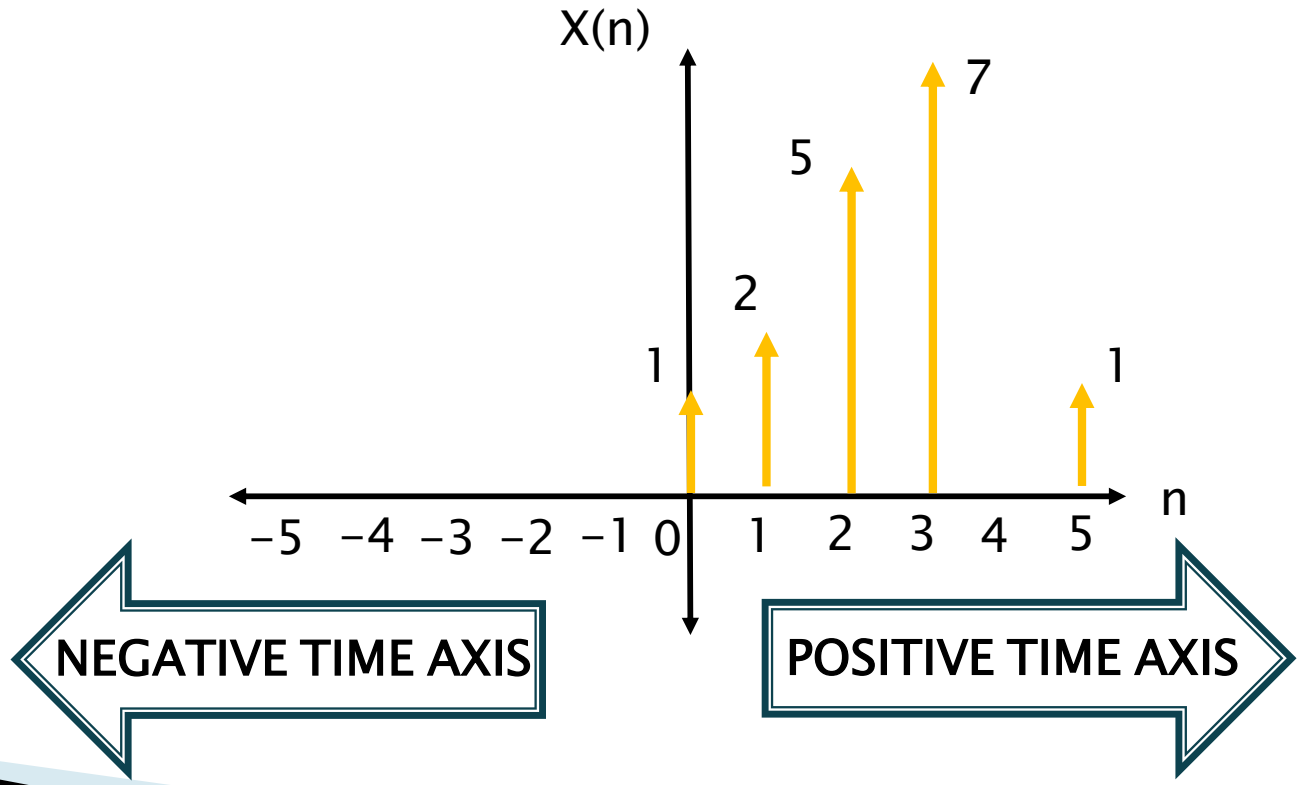
Examples:



The unit step signal u(t)

Causal Signals are signals that are zero for all negative axis.

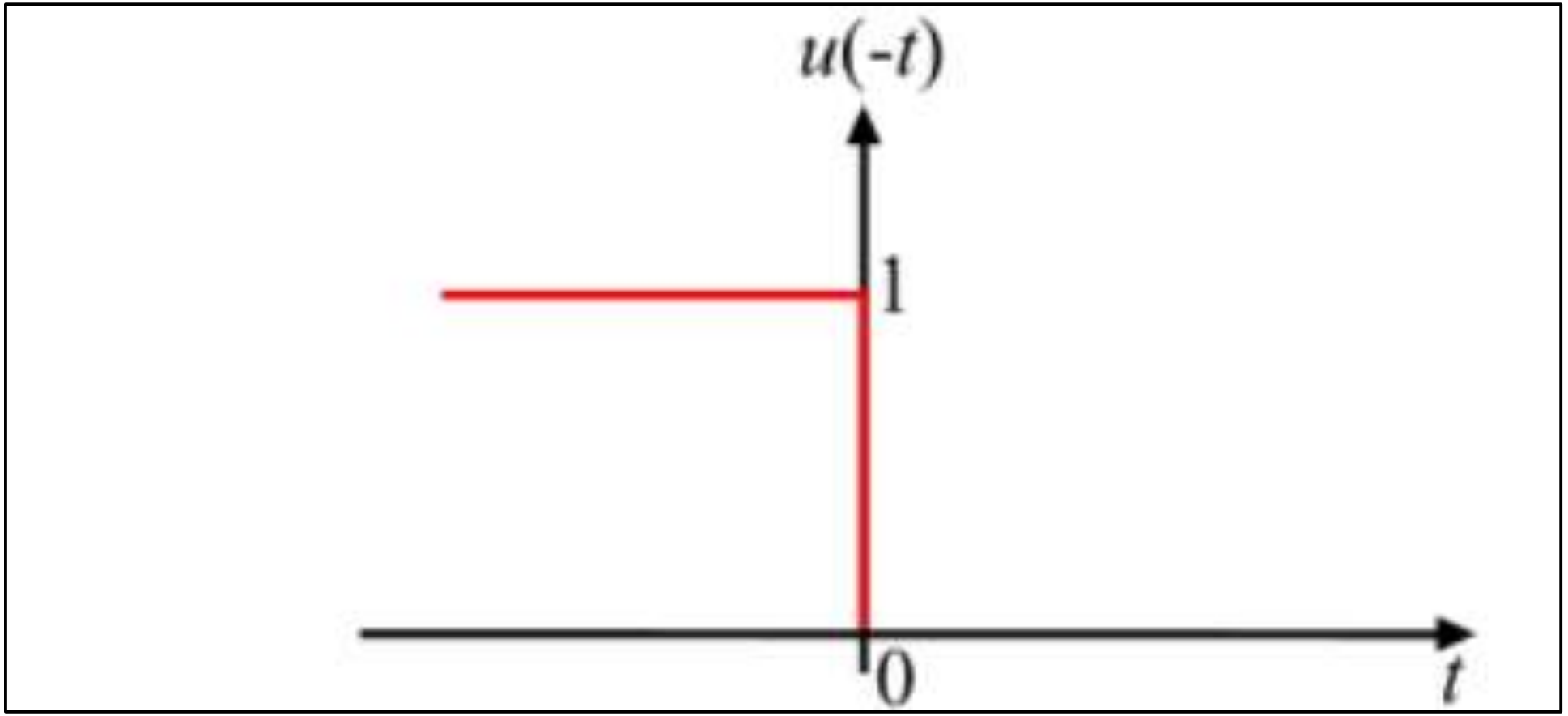
X(n)= { 1, 2, 5, 7, 0, 1 } ⬆



Anti-Causal Signals

- A discrete time signal is said to be *anti-causal* if it is finite or defined for $n \leq 0$, and zero for $n > 0$
- The anti-causal signals are signals that are zero for all positive times or signals does not exist for positive time.
- A causal signal can also be called as Left side sequence signals.

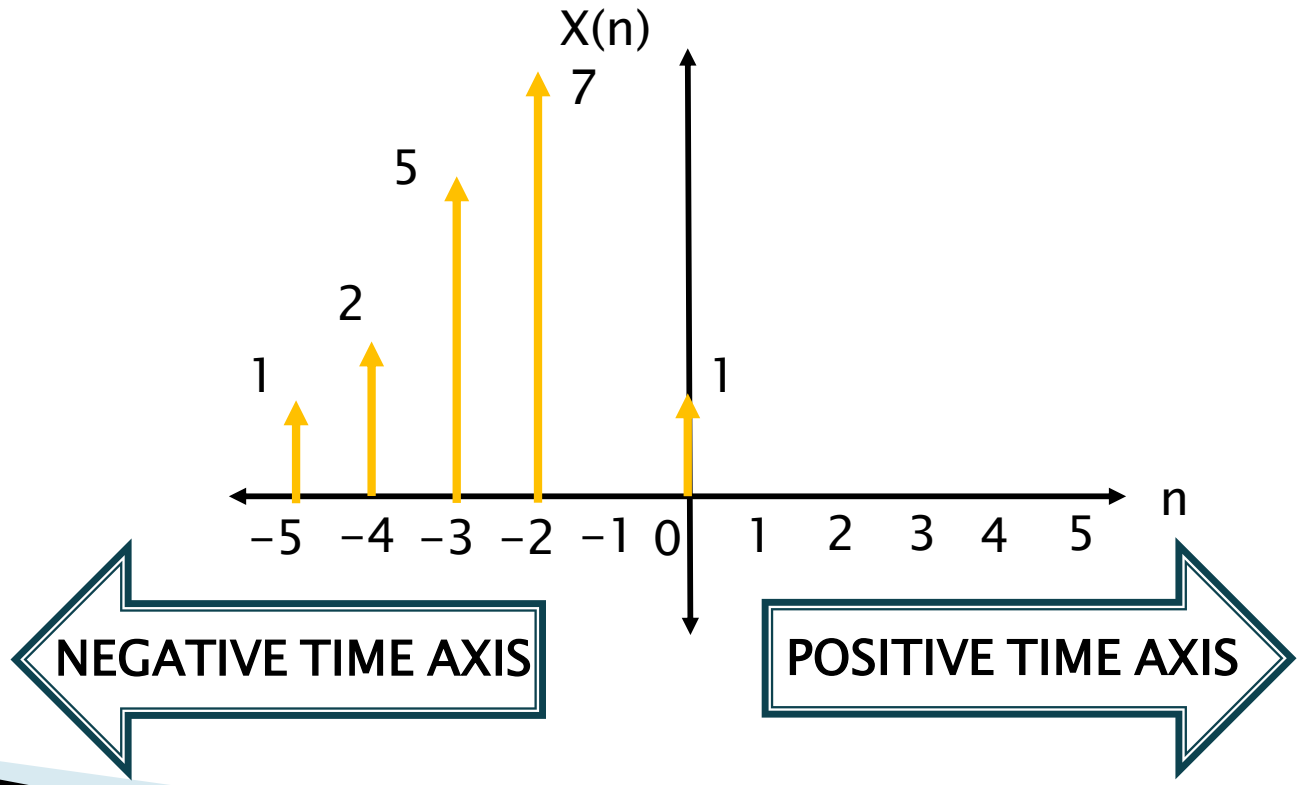
Examples:



The reversed unit step signal u(-t)

Anti-Causal Signals are signals that are zero for all positive times

$$X(n) = \{ 1, 2, 5, 7, \underset{\uparrow}{0}, 1 \}$$



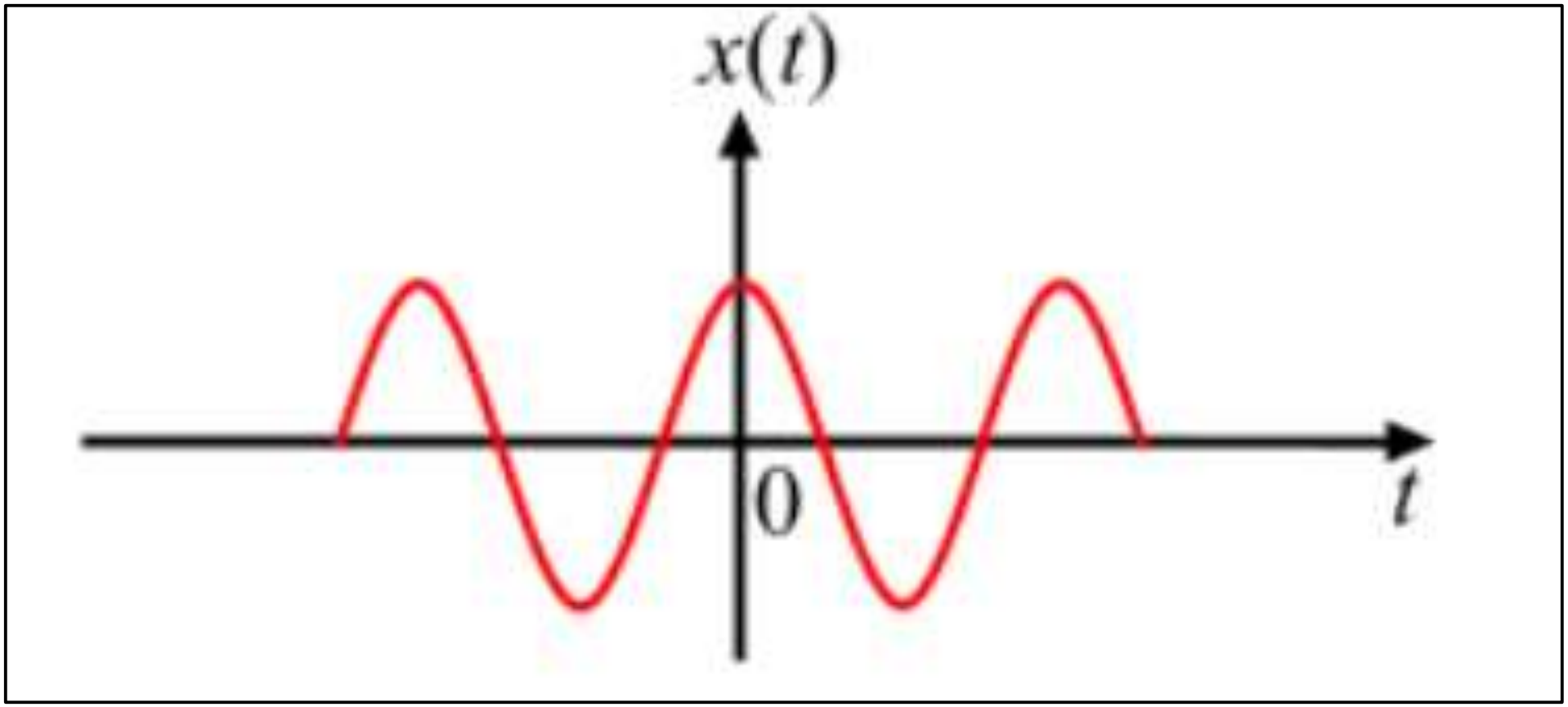
Non-Causal Signals

- A discrete time signal is said to be *non-causal* if it is finite or defined for either $n \leq 0$, or for both $n \leq 0$ and $n > 0$

- The Non-causal signals are signals that have non zero (finite) values in negative times but may or may not have finite values in positive times.

- A Non-causal signal can also be called as Both side sequence signals.

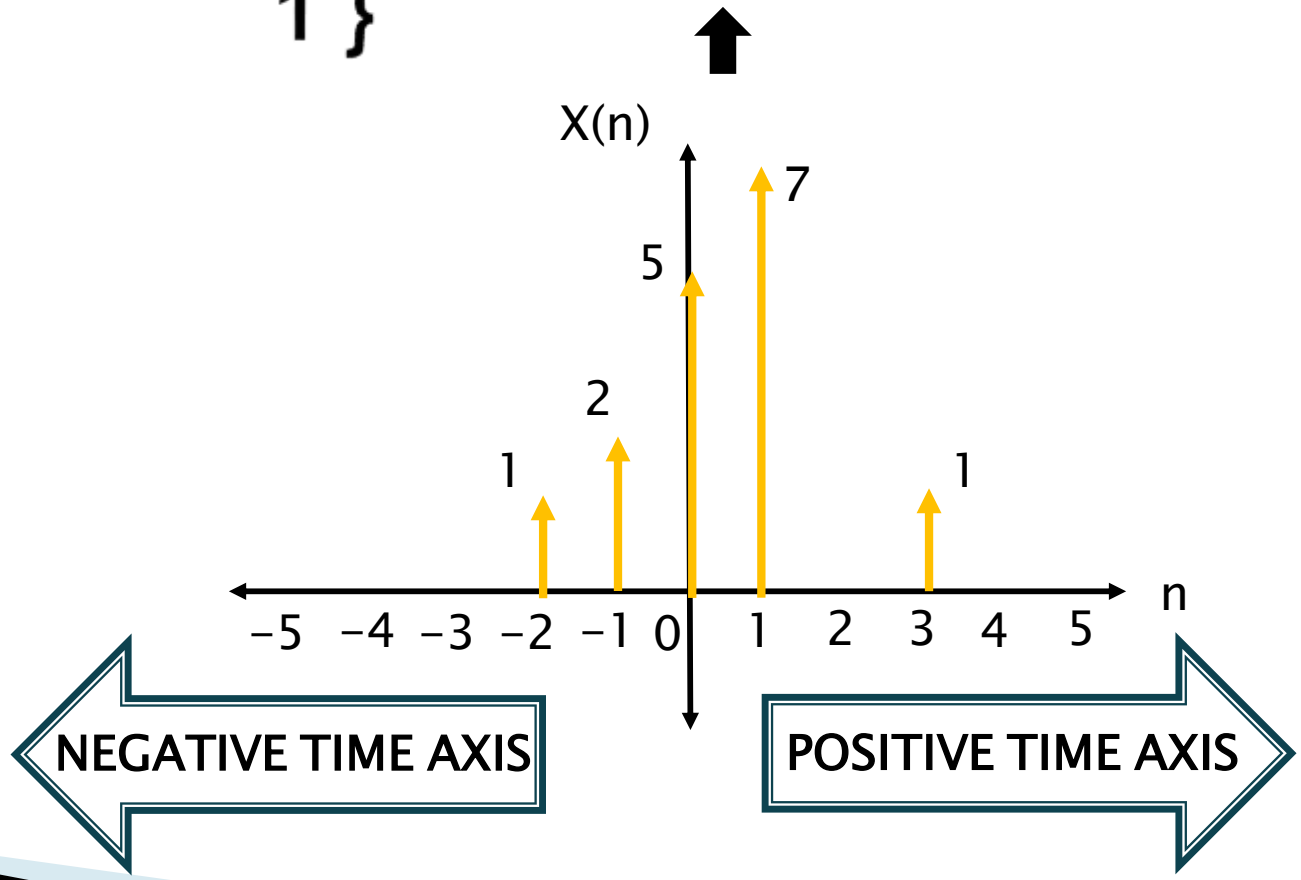
Examples:



The sine and cosine signals

Non-causal signals are signals that have non zero (finite) values in negative times but may or may not have finite values in positive times.

$$X(n) = \{ 1, 2, 5, 7, 0, 1 \}$$



Note: All Anti-causal signals are non causal but not all non causal signals are anti-causal

Symmetric and Anti-Symmetric signals

Symmetric Signals

A signal which is symmetrical about the vertical axis or amplitude axis is known as symmetric signal . Therefore, the symmetric signals are also called the **even signal** or **even function**.

Cosine function is an example of even signal.

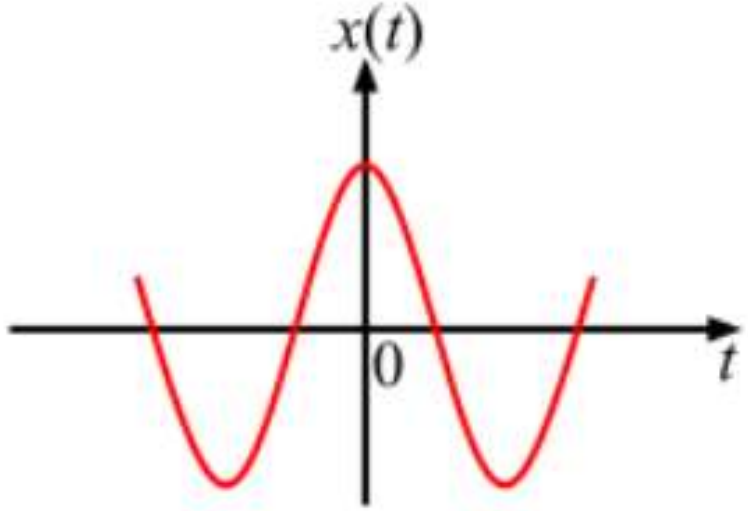
A signal is referred to as an even if it is identical to its time-reversed counterparts :

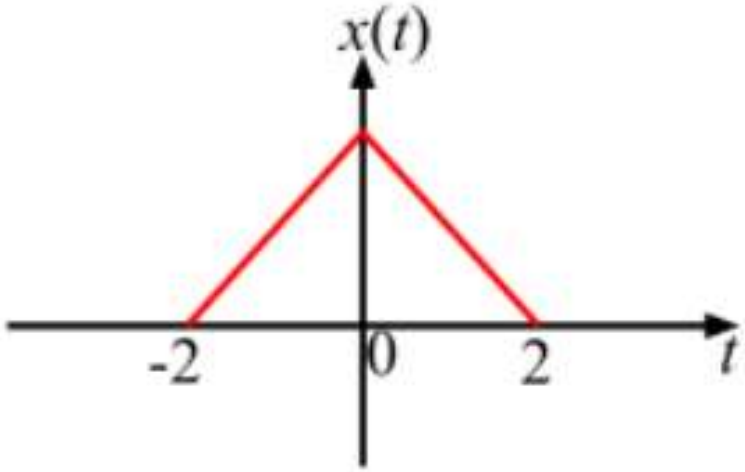
$$x(t) = x(-t)$$

Continuous-time Even Signal

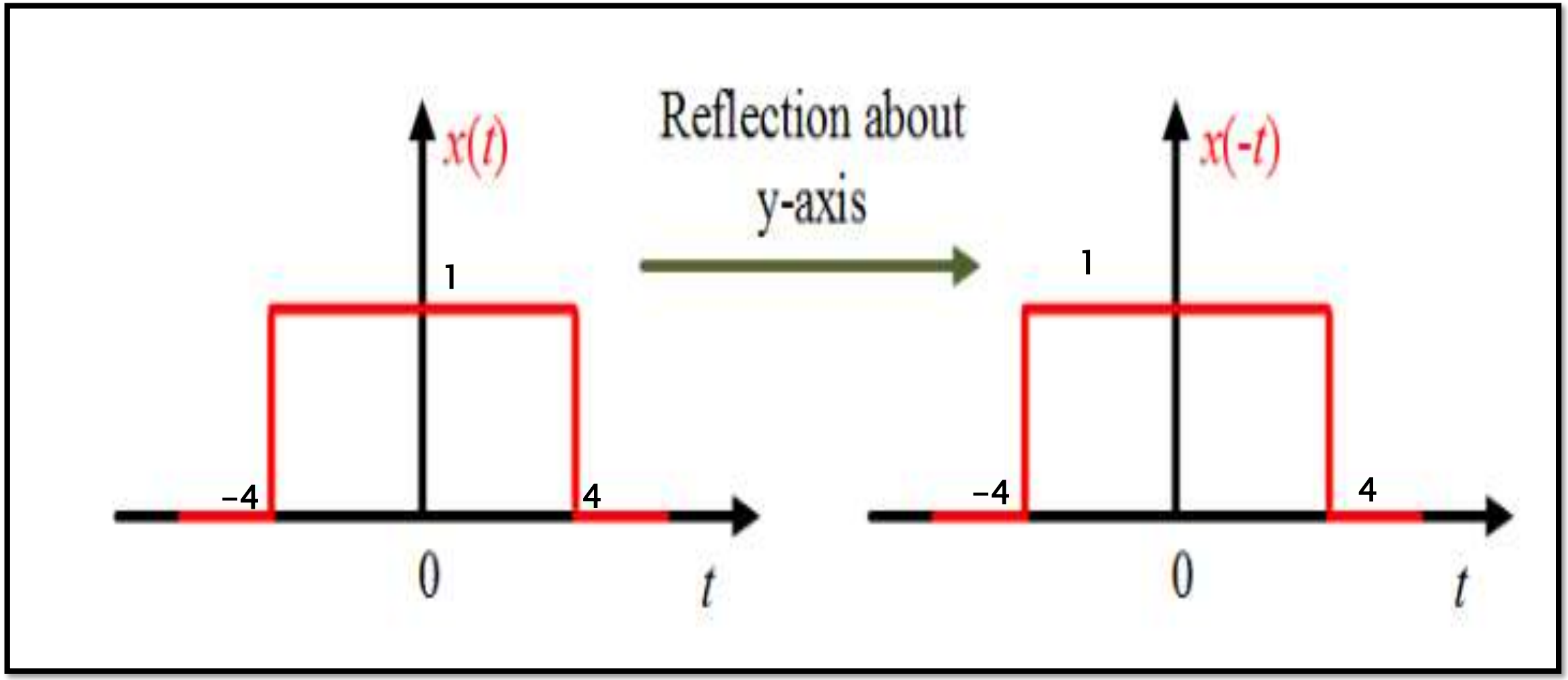
A continuous-time signal x(t) is called the even signal or symmetrical signal if it satisfies the following condition,

$$x(t) = x(-t); \text{ for } -\infty < t < \infty$$





-1


x(t)
1
−4
4
0
t
Reflection about
y-axis
x(-t)
1
−4
4
0
t

Properties of Symmetric (even) Signals

The properties of the even signals are given as follows −

- The even signals are symmetrical about the y axis or amplitude axis.
- The value of an even signal at time (t) is same as at time (−t).
- The even signal is identical with its reflection about the origin.
- The area under the even signal is two time of its one side area.

Discrete-time Even Signal

A discrete-time signal x(n) is said to be even signal or symmetrical signal if it satisfies the condition,

$$x(n) = x(-n); \text{ for } -\infty < n < \infty$$

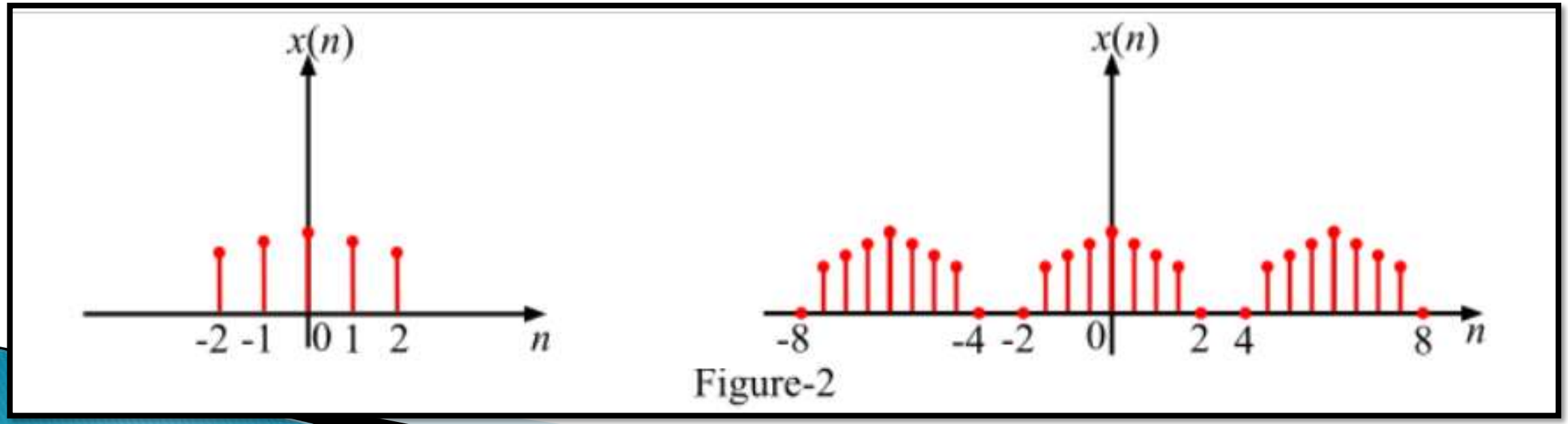
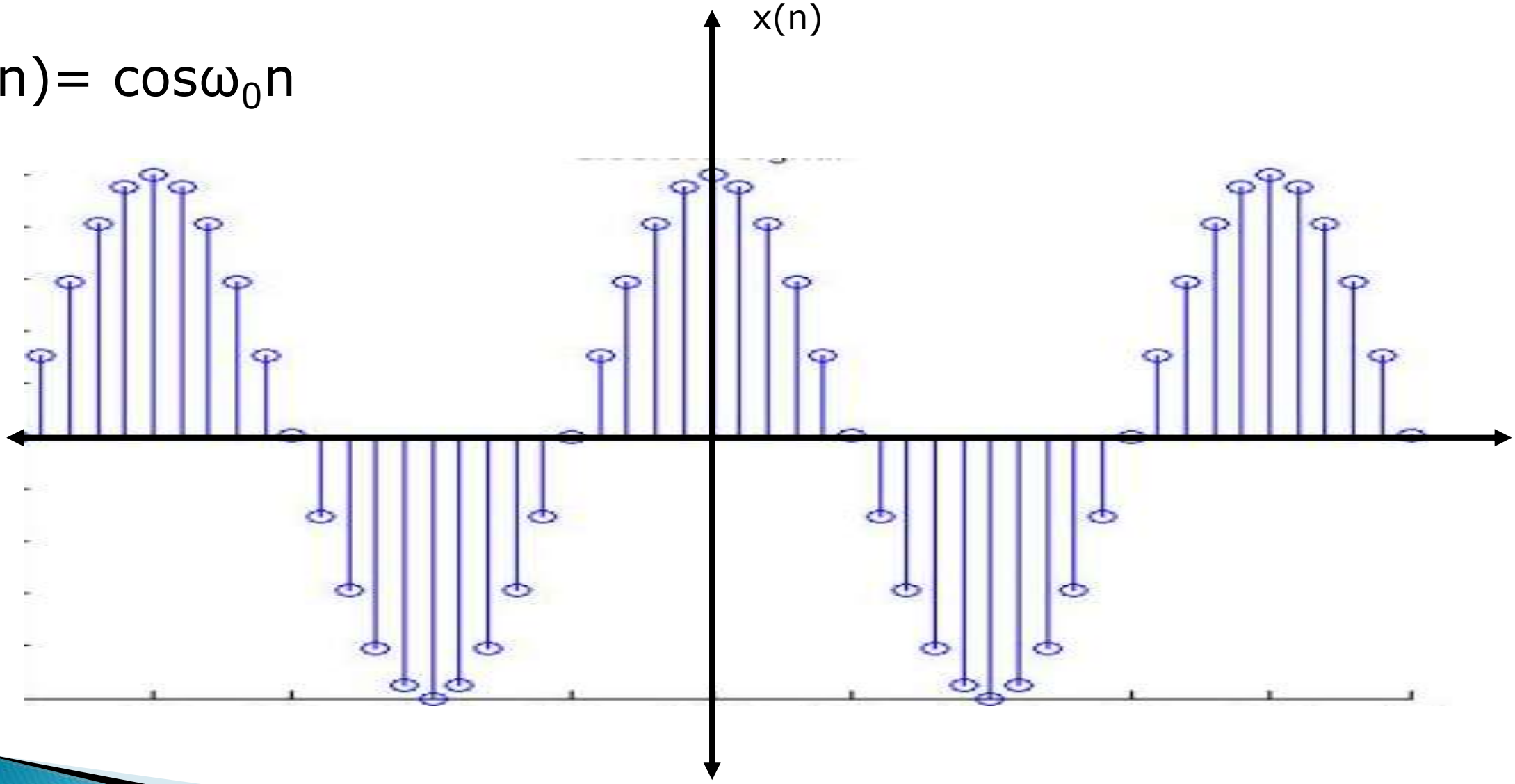


Figure-2

Discrete-time Even Signal

$X(n) = \cos\omega_0 n$



Discrete-time Even Signal

$X(n) = n^2$

$X(n) = n^4$

$X(n) = n^6$

$X(n) = n^8$

Anti-symmetric Signals

A signal which is symmetric about the origin is known as anti-symmetric signal . Therefore, the anti-symmetric signals are also called the **odd signal** or **odd function**.

Sine function is an example of odd signal.

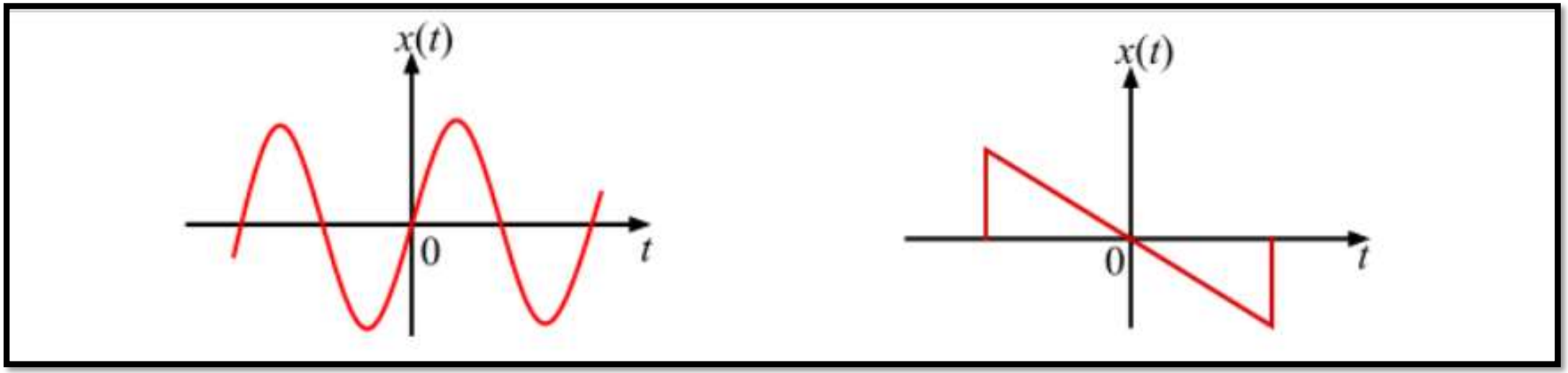
A signal is an odd signal if $x(t) = -x(-t)$ or $x(-t) = -x(t)$

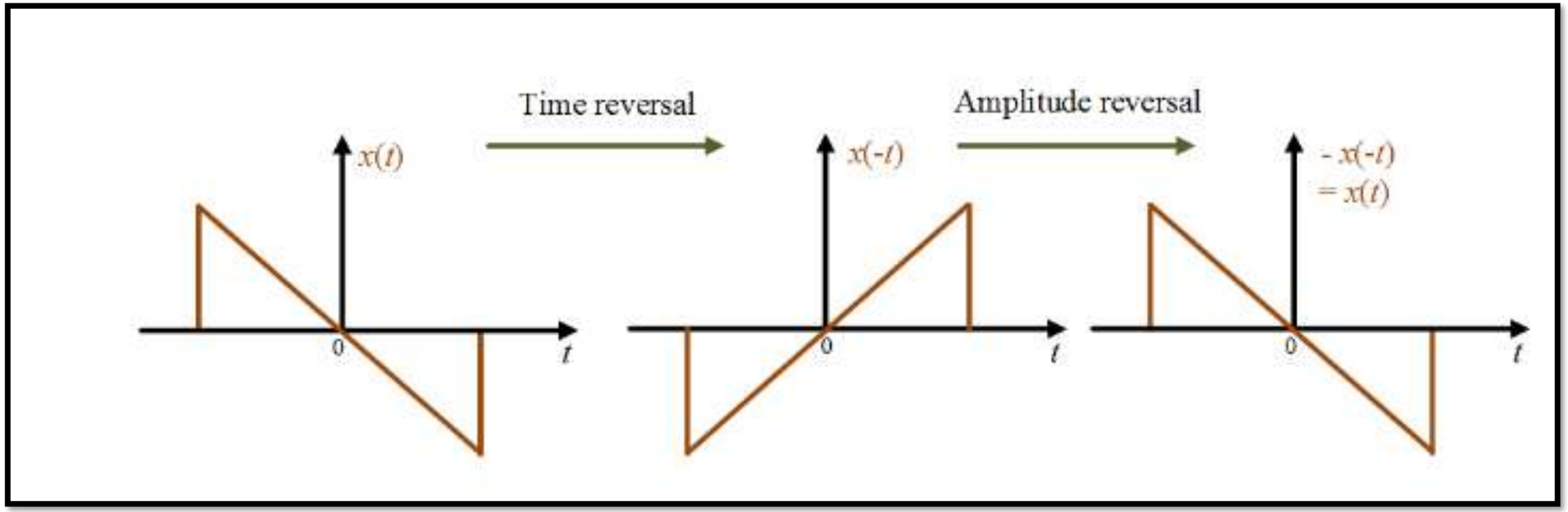
An odd signal must be 0 at t=0, in other words, odd signal passes the origin.

Continuous-time Odd Signal

A continuous time signal x(t) is called an odd signal or antisymmetric signal if it satisfies the following condition,

$$x(-t) = -x(t); \text{ for } -\infty < t < \infty$$

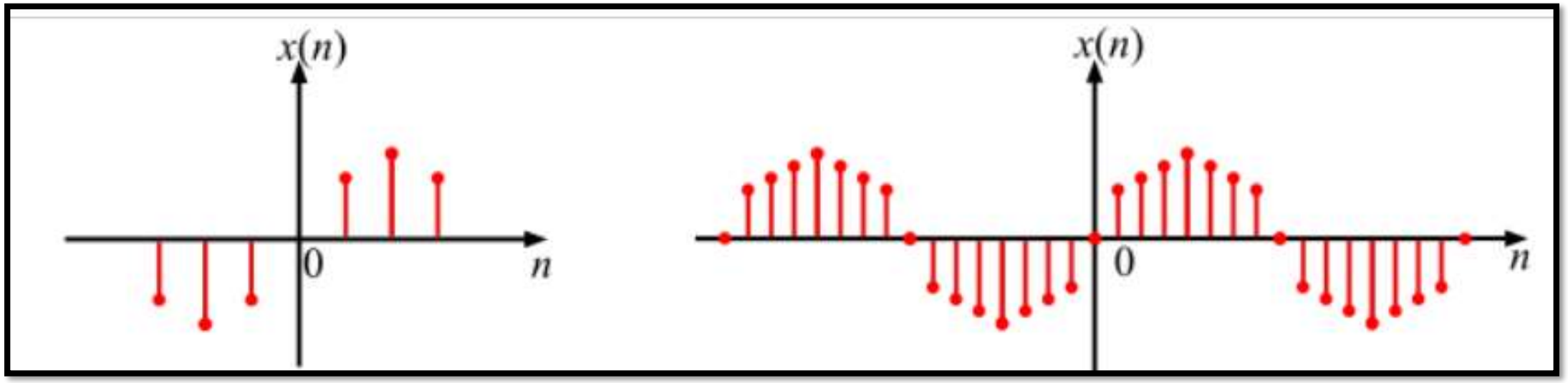



Time reversal
Amplitude reversal
x(t)
0
t
x(-t)
0
t
- x(-t)
= x(t)
0
t

Discrete-time Odd Signal

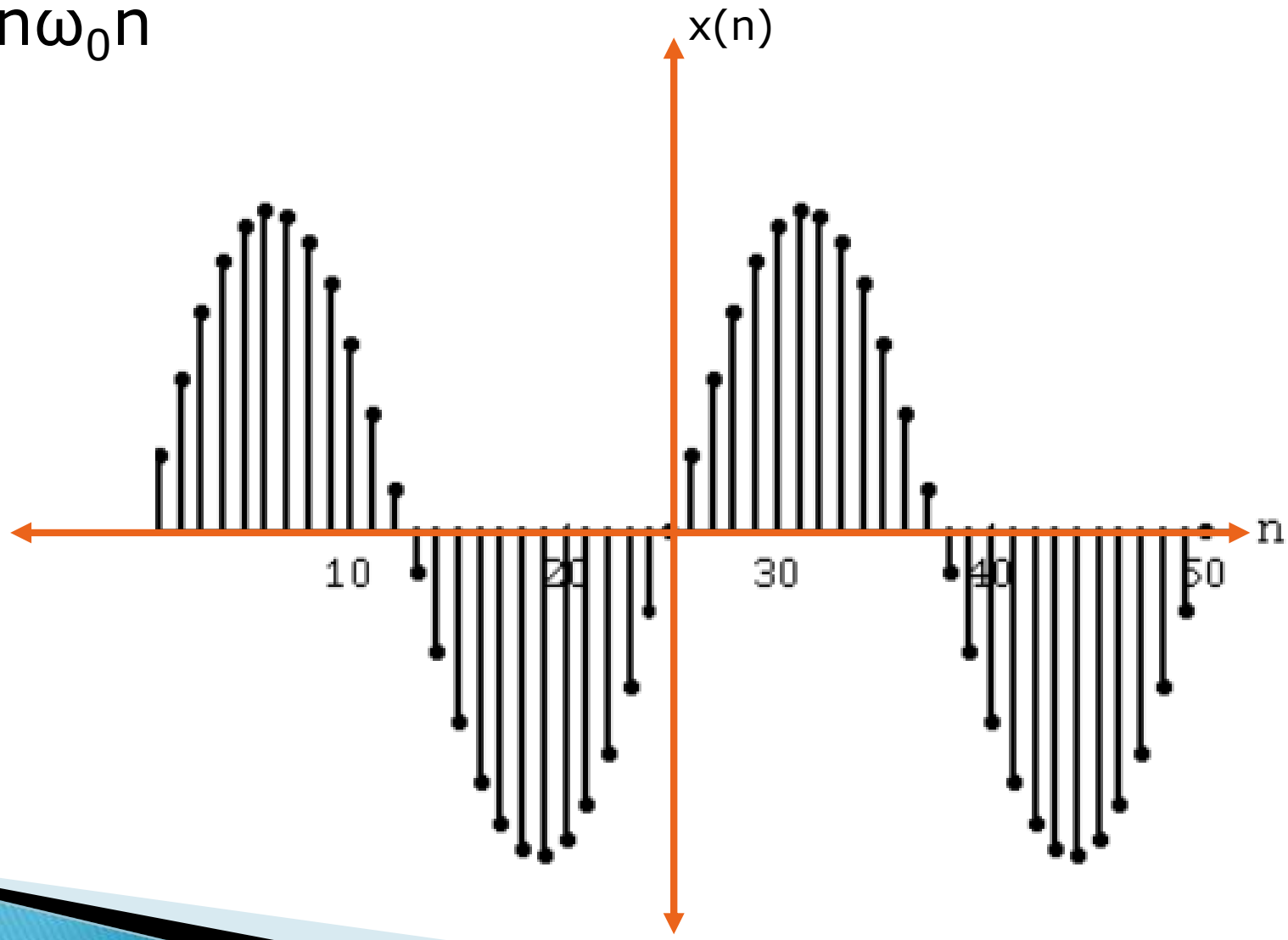
A discrete time signal x(n) is said to be an odd signal or antisymmetric signal, if it satisfies the following condition,

$$x(n) = -x(-n) \text{ or } x(-n)=-x(n); \text{ for } -\infty < n < \infty$$



Discrete-time Odd Signal

$X(n) = \sin\omega_0 n$



Discrete-time Odd Signal

$X(n) = n^3$

$X(n) = n^5$

$X(n) = n^7$

$X(n) = n^9$

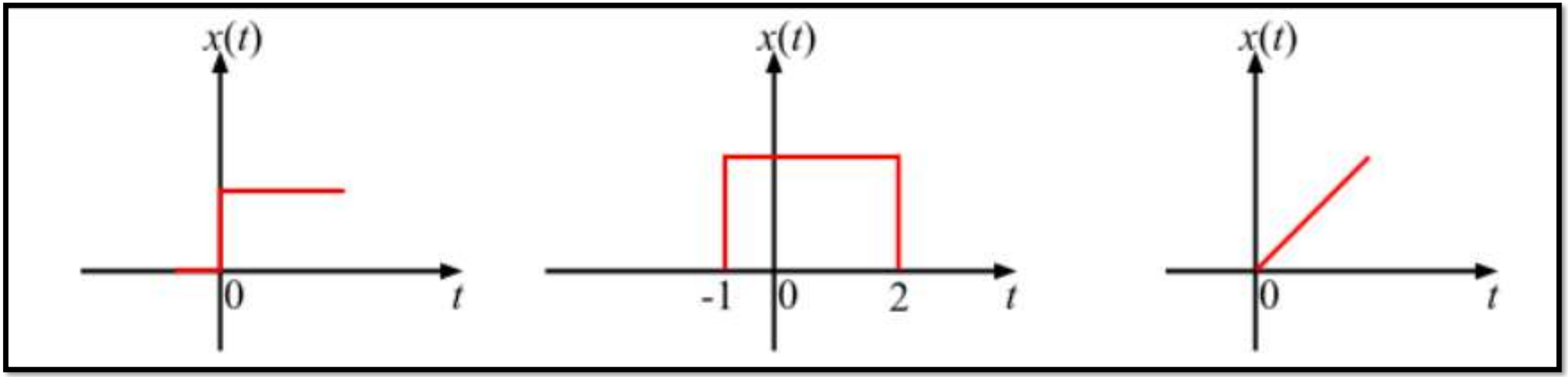
$X(n) = n^{15}$

Properties of Anti-symmetric (odd) Signals

The properties of the odd signals are given as follows −

- The odd signal is antisymmetric about the origin.
- The value of odd signal at time (t) is negative of its value at time (-t) for all t, i.e., $-\infty < t < \infty$.
- The odd signal must necessarily be zero at time t = 0 to hold $x(0) = -x(0)$.
- Area under the odd signal is always zero.

Note – A continuous-time signal or DTS is said to be **neither even nor odd** if it does not satisfy the condition of the even signal and the odd signal.



EVEN AND ODD COMPONENTS OF A SIGNAL

Any discrete time signal x(n) can be represented by the summation of its even part $x_e(n)$ and odd part $x_o(n)$

$$x(n)= x_e(n) + x_o(n)$$

Equation 1

$$x(-n)= x_e(-n) + x_o(-n)$$

$$x(-n)= x_e(n) - x_o(n)$$

Equation 2

By adding the equation 1 and eq. 2

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

So, this is the even component or even part of a signal

By subtracting the equation 1 and eq. 2

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

So, this is the odd component or odd part of a signal

Determine the even part and odd part of the signal shown below

1. $x(n) = a^n$

2. $x(n) = 2e^{j\frac{\pi}{3}n}$

3. $x(n) = \{ 3, -3, 2, -2 \}$

1. $x(n) = a^n$

$$x_e(n) = \frac{x(n) + x(-n)}{2} = \frac{a^n + a^{-n}}{2}$$ **Even Part**

$$x_o(n) = \frac{x(n) - x(-n)}{2} = \frac{a^n - a^{-n}}{2}$$ **Odd Part**

2. $x(n) = 2e^{j\frac{\pi}{3}n}$

$$x(n) = 2\cos\frac{\pi}{3}n + j\,2\sin\frac{\pi}{3}n$$

$$x(-n) = 2\cos\frac{3\pi}{3}n - j\,2\sin\frac{\pi}{3}n$$

Even Part

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

Odd Part

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

$$x_e(n) = \frac{1}{2}[2\cos\frac{\pi}{3}n + j\,2\sin\frac{\pi}{3}n + 2\cos\frac{\pi}{3}n - j\,2\sin\frac{\pi}{3}n]$$

$$x_e(n) = 2\cos\frac{\pi}{3}n$$

$$x_o(n) = \frac{1}{2}[2\cos\frac{\pi}{3}n + j\,2\sin\frac{\pi}{3}n - 2\cos\frac{\pi}{3}n + j\,2\sin\frac{\pi}{3}n]$$

$$x_o(n) = j2\sin\frac{\pi}{3}n$$

3. x(n) = { 3,-3, 2,-2}

Even Part

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

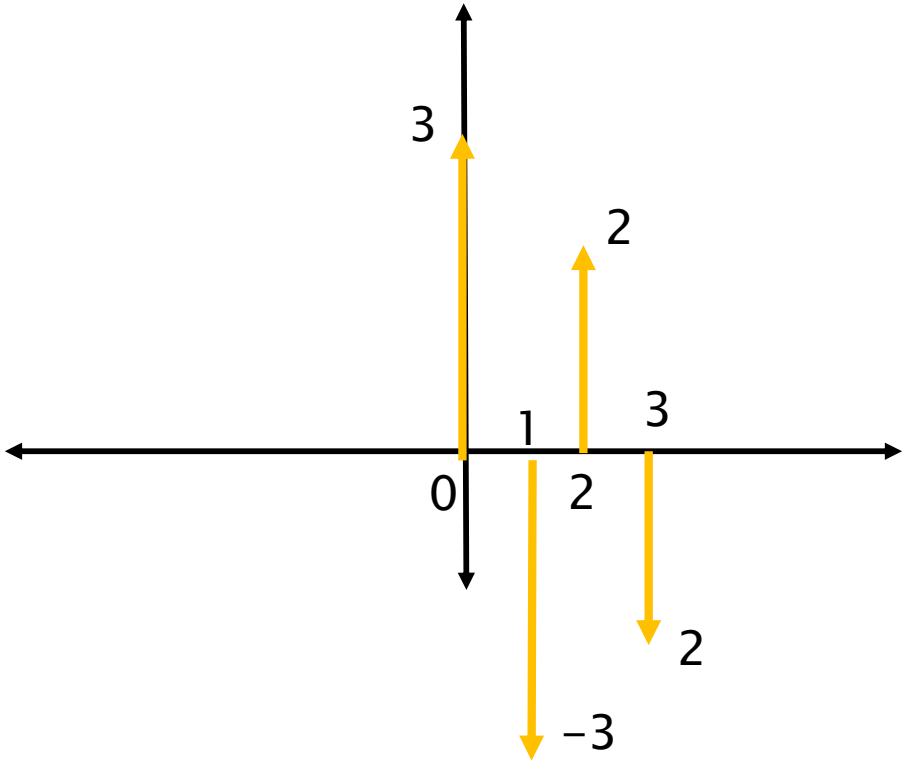
Odd Part

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

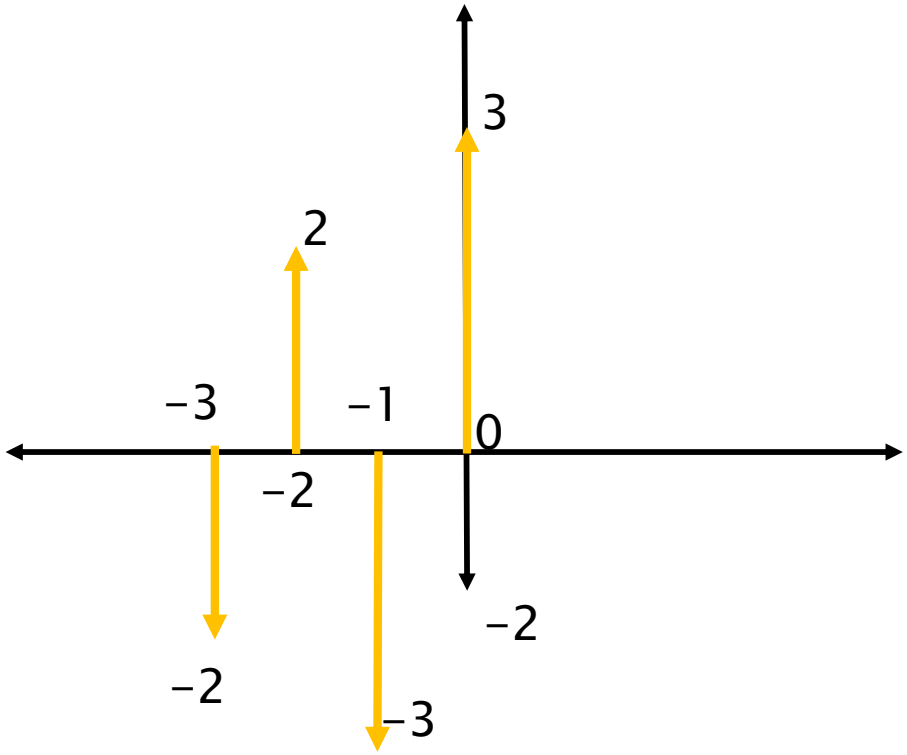
x(n) = { 3,-3, 2,-2}

x(-n) = { 3,-3, 2,-2}

3. x(n) = { 3,-3, 2,-2}



x(n) = { 3,-3, 2,-2}



x(-n) = { -2, 2, -3, 3}

Energy and Power Signal

ENERGY OF A SIGNAL

In signal processing, a signal is viewed as a function of time. The term “size of a signal” represents “strength of the signal. ” Knowing the “size” of a signal used in a certain application is crucial. For example, we may be interested in knowing the amount of electricity needed to power an LCD monitor instead of a CRT monitor. Both applications are different and have different tolerances. Thus, the amount of electricity driving these devices will also be different.

In the discrete domain, the energy of the signal is given by:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|2$$

The energy is finite only if the above sum converges to a finite value. This implies that the signal is "**squarely-summable**". Such a signal is called a finite energy signal.

What if the given signal does not decay concerning time (as in a continuous sine wave repeating its cycle infinitely)? The energy will be infinite and such a signal is "**not squarely-summable**". We need another measurable quantity to circumvent this problem. This leads us to the notion of "Power"

POWER OF A SIGNAL

Power is defined as the amount of energy consumed per unit time. This quantity is useful if the signal's energy goes to infinity, or the signal is "not-squarely-summable". For "non-squarely-summable" signals, the power is calculated by taking a snapshot of the signal over a specific interval of time as follows:

1) Take a snapshot of the signal over some finite time duration
2) Compute the energy of the signal E
3) Divide the energy by the number of samples taken for computation N
4) Extend the limit of several samples to infinity $N \rightarrow \infty$. This gives the total power of the signal.

In the discrete time domain, the total power of the signal is given by:

$$P = \lim_{n \to \infty} \frac{1}{2n+1} \sum_{n=-N}^{N} |x(n)|^2$$

Energy Signal

A signal can be classified based on its power or energy content. Signals having finite energy are energy signals. Power signals have finite and non-zero power.

- A DTS is said to be an Energy signal if its energy E is finite. (i.e., 0 <E< ∞); E≠ 0; E ≠ ∞; P=0

- An energy signal will have always zero TOTAL power.

- The energy of a DTS x(n) can be calculated as:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|2$$

- The energy of a CTS x(t) can be calculated as:

$$E = \int_{-\infty}^{\infty} /x(t)/2 \, dt$$

- The energy of a DTS x(n) can be calculated as:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|2$$

- These signals possess finite energy but have zero average power due to their behavior over infinite time intervals. Examples of energy signals include non-periodic signals that decay or approach zero as time progresses. Some common examples are exponential signals, Gaussian pulses, and finite-duration pulses.

Power Signal

- A DTS is said to be a Power signal if its average power P is finite. (i.e., $0 < P < \infty$); $P \neq 0$; $P \neq \infty$; $E = \infty$

- A power signal will have always infinite energy.

- The average power of a DTS x(n) can be calculated as:

$$P = \lim_{n \to \infty} \frac{1}{2n+1} \sum_{n=-N}^{N} |x(n)|^2$$

- The average power of a CTS x(t) can be calculated as:

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-t/2}^{t/2} /x(t)/2 \, dt$$

- The average power of a DTS x(n) can be calculated as:

$$P = \lim_{n\to\infty} \frac{1}{2n+1} \sum_{n=-N}^{N} |x(n)|^2$$

- Several examples of power signals include periodic signals such as sinusoids, square waves, and triangle waves. These signals have a repetitive pattern over time and exhibit finite power but infinite energy.

- Mutually Exclusive

Difference between Energy and Power Signal:

PARAMETER	ENERGY SIGNAL	POWER SIGNAL
Explanation	Energy signal is a measure of total energy contained in a signal.	Power signal is a measure of the rate at which energy is transmitted or used.
Representation	Energy represents the capacity to do work.	Power represents the rate at which the work is done.
Quantity Type	It is a scalar quantity.	It is a vector a quantity.
Type	It can be positive or negative.	It is always positive.
Energy/Power	Energy Signals have no power. (Power=0)	Power Signals have infinite energy. (Energy=∞)
Signal Type	Energy signals are non-periodic.	Power signals are periodic.
Example	Exponential Signal.	Sinusoidal or Square Wave signal.

Neither an Energy signal nor a Power signal:

Signals can also be a cat on the wall – neither an energy signal nor a power signal. Consider a signal of increasing amplitude defined by,

$$x(n)=n;\ P=E=\infty$$

For such a signal, both the energy and power will be infinite. Thus, it cannot be classified either as an energy signal or as a power signal.

Examples of Energy Signal

- $x(n) = \left(\frac{1}{2}\right)^n$ u(n)
- $x(n) = \delta(n)$

Examples of Power Signal

- $x(n) = e^{j\pi n/2}$
- $\mathrm{x(n)} = \sin\left(\frac{\pi n}{4}\right)$
- $\mathrm{x(n)} = \cos\left(\frac{\pi n}{4}\right)$
- $\mathrm{x(n)} = \mathrm{u(n)}$

Examples of Neither Energy nor Power Signal

- $x(n) = e^{2n}$u(n)
- $x(n) = 2^{n}$u(n)
- $x(n) = r$ (n)

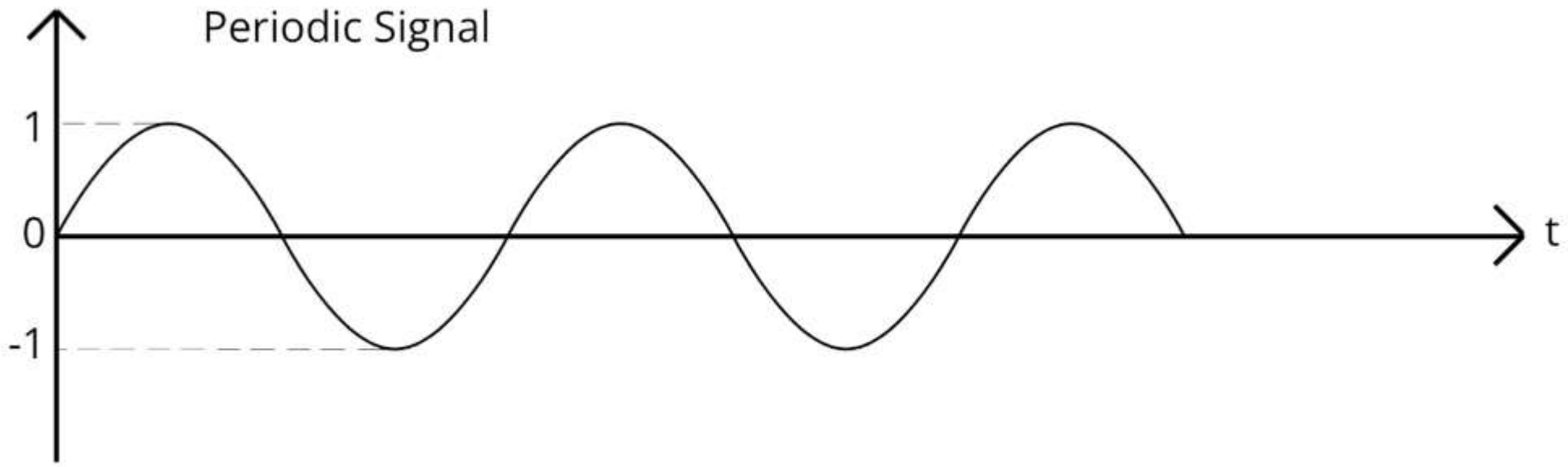
Periodic Signal
and
Aperiodic Signal

Periodic Signal

A signal can be classified as a Periodic signal if it has a definite pattern and repeats itself at a regular interval of time.

- A signal is a periodic signal if it completes a pattern within a measurable time frame, called a period, and repeats that pattern over identical subsequent periods.
- The completion of a full pattern is called a cycle.
- A period is defined as the amount of time (expressed in seconds) required to complete one full cycle.
- The duration of a period represented by T may be different for each signal but it is constant for any given periodic signal.

Periodic Signal



Periodic Signal

Periodic signals are characterized by their period, which is the time it takes for the signal to repeat itself, and their frequency, which is the number of cycles the signal goes through in a given period.

The period and the frequency of a periodic signal are inversely related: A signal with a short period has a high frequency, and a signal with a long period has a low frequency.

Periodic signals are important in many applications because they have a well-defined frequency spectrum and they can be easily analyzed using tools from Fourier analysis.

Continuous time Periodic Signal

A continuous time signal x(t) is said to be periodic if and only if

$$x(t + T) = x(t) \text{ for } -\infty < t < \infty$$

Where, T is a positive constant that represents the time period of the periodic signal. The smallest value of the time period (T) which justifies the definition of the periodic signal is known as **fundamental time period** of the signal and is denoted by (T_0).

Also,

$$x(t + mT) = x(t)$$

Where, m is an integer. This means if the definition is satisfied for $T = T_0$, then it is also satisfied for $T = 2T_0$, $T = 3T_0$ … and so on with T_0 as the fundamental period. Therefore, the fundamental period defines the duration of one complete cycle of the periodic signal x(t).

The reciprocal of the period (T) of the periodic signal is called the frequency (f) of the signal, i.e.,

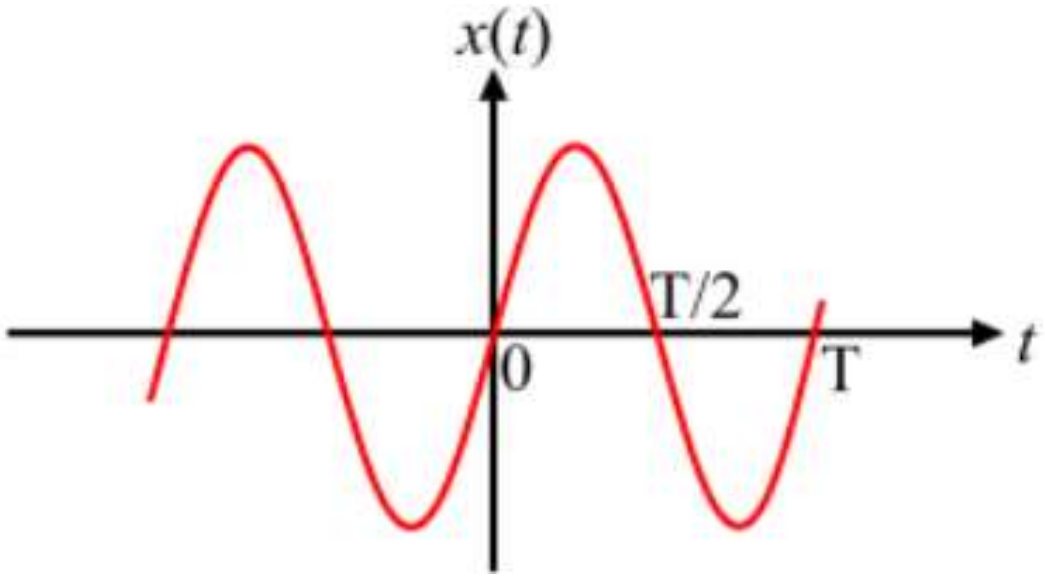
$$f = \frac{1}{T}$$

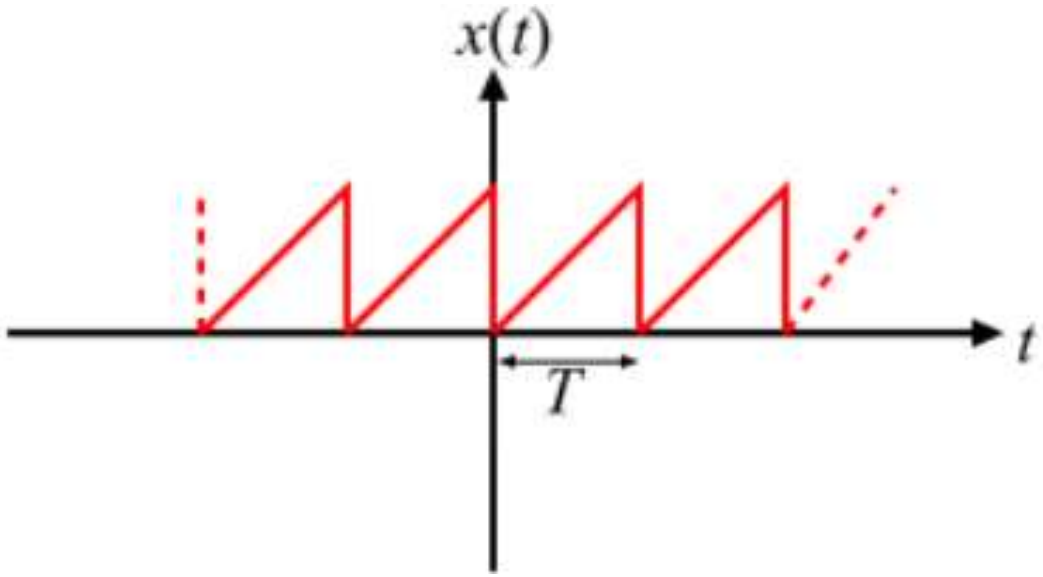
Since the angular frequency is given by

$$\omega = 2\pi f$$

$$\therefore \textbf{Period, } T = \frac{2\Pi}{\omega}$$

Some examples of continuous-time periodic signals are shown





Discrete time Periodic Signal

A discrete-time signal x(n) is said to be periodic if it satisfies the following condition

$$x(n) = x(n + N); \quad \text{for all integers } n$$

Here, N is the time period of the periodic signal and a positive integer. The smallest value of the time period (N) which satisfies the above condition is known as fundament period of the signal.

The fundamental period (N) may be defined as the minimum number of samples taken by the signal to repeat itself.

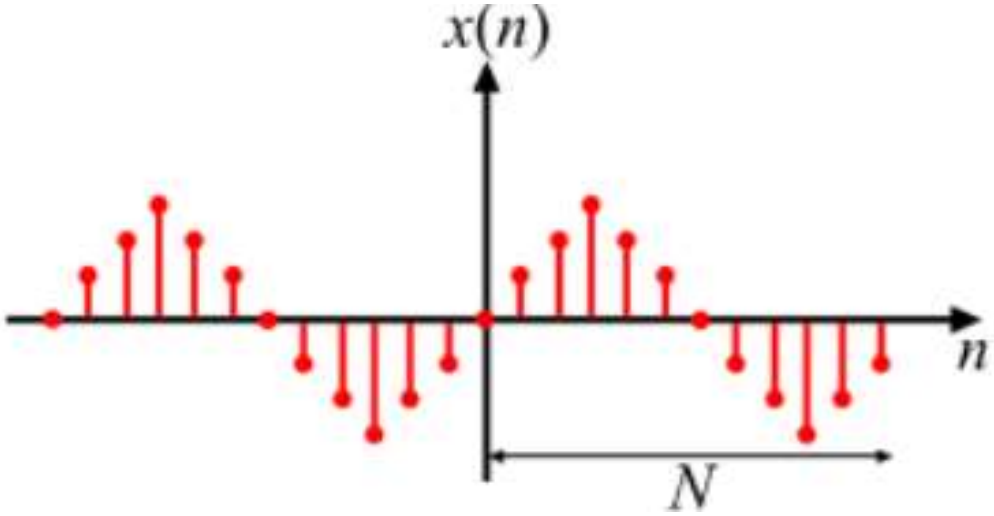
The angular frequency of the discrete-time periodic sequences is given by,

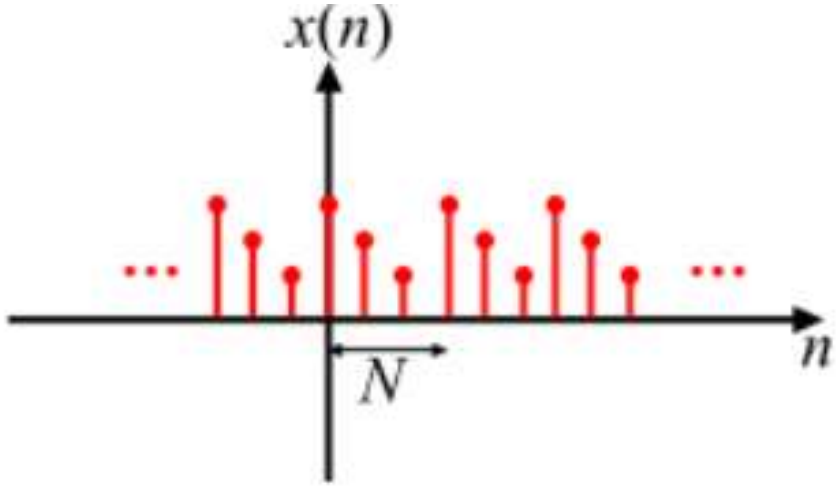
$$\omega = \frac{2\Pi}{N}$$

Therefore, the period of the sequence is,

$$N = \frac{2\Pi}{\omega}$$

Some examples of discrete-time periodic signals are shown





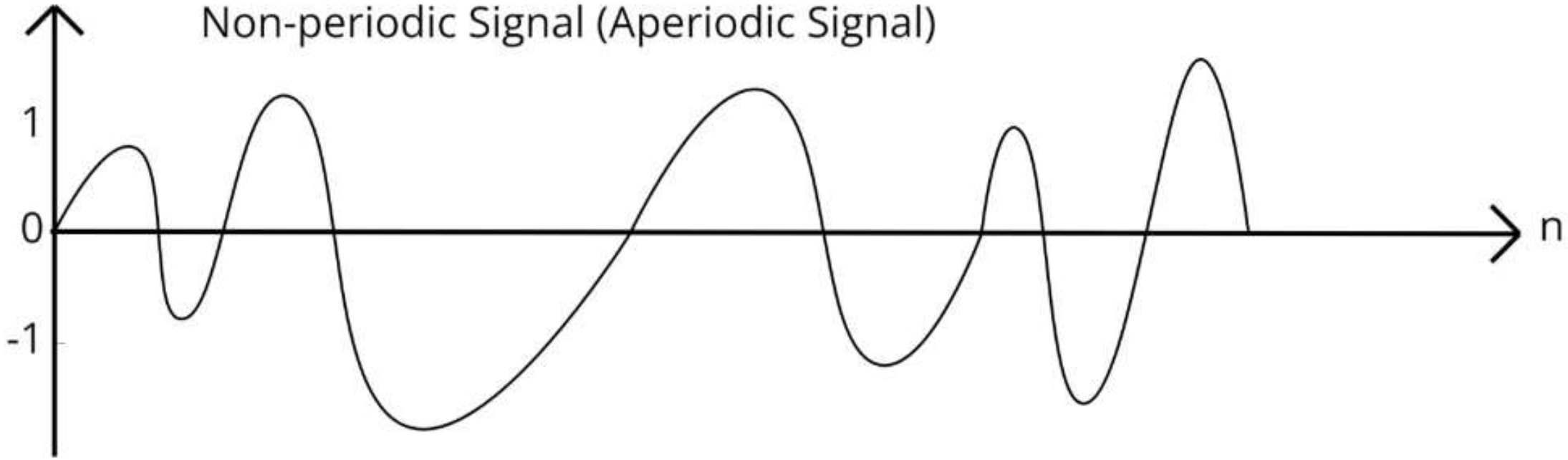
Aperiodic Signal / Non-periodic Signal

- A non-periodic signal is a signal that does not repeat itself at regular intervals. This means that it does not have a fundamental frequency, and its frequency spectrum is generally not composed of discrete frequencies.

- Non-periodic signals can be continuous or discrete, and they can be either deterministic or random.

- Examples of non-periodic signals include white noise, impulses, and arbitrary waveforms. Non-periodic signals are often encountered in real-world applications, such as speech, music, and environmental noise.

Aperiodic Signal

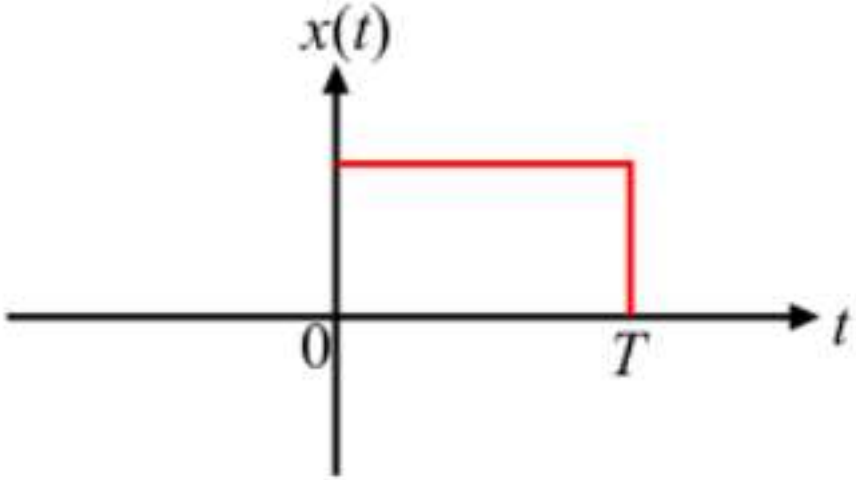


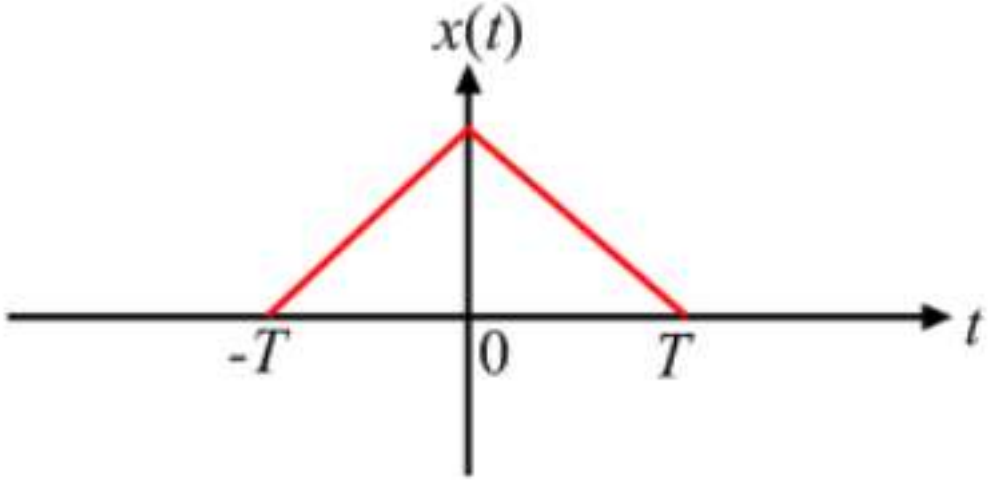
Continuous time Aperiodic Signal

If a continuous time signal does not have a definite pattern and does not repeat at regular intervals of time is known as continuous time aperiodic signal or nonperiodic signal.

In other words, a signal x(t) for which no value of time t satisfies the condition of periodicity, is known as aperiodic or non-periodic signal.

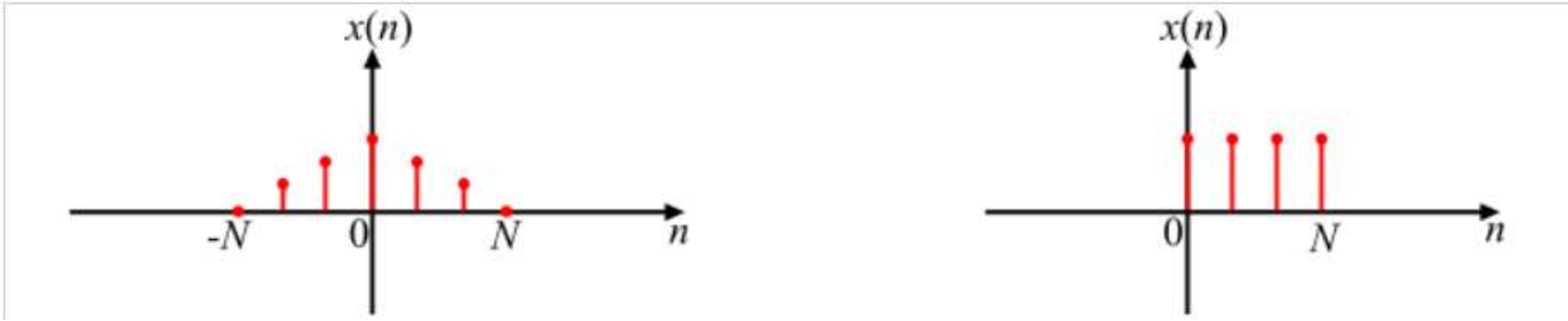
Some examples of continuous-time aperiodic signals is shown





Discrete time Aperiodic Signal

If the condition of periodicity is not satisfied even for one value of n for a discrete-time signal x(n), then the discrete-time signal is aperiodic or nonperiodic.



Difference between Periodic and Non-Periodic Signal:

PARAMETER	PERIODIC SIGNAL	NON-PERIODIC SIGNAL
Explanation	A periodic signal is a type of signal that repeats at regular intervals. Periodic signals are characterized by a constant frequency and may have a constant or varying amplitude.	Non-periodic signals, also known as aperiodic signals, do not repeat at regular intervals. Aperiodic signals may have varying frequencies and amplitudes, and may have complex or irregular waveforms.
Repetition	Periodic signals repeat at regular intervals.	Non-Periodic signals do not repeat.
Frequency	Periodic signals have a constant frequency.	Non-Periodic signals may have varying frequencies.
Amplitude	Periodic signals may have a constant or varying amplitude.	Non-Periodic signals may have a constant or varying amplitude.
Waveform	Periodic signals have a repeating waveform.	Non-Periodic signals may have a complex or irregular waveform.
Spectrum	Periodic signals have a discrete set of frequencies in their spectrum.	Non-Periodic signals may have a continuous spectrum.

Difference between Periodic and Non-Periodic Signal:

PARAMETER	PERIODIC SIGNAL	NON-PERIODIC SIGNAL
Energy	Periodic signals have energy concentrated at specific frequencies.	Non-Periodic signals may have energy distributed over a range of frequencies.
Time Domain Representation	In the time domain, periodic signals appear as repeating waves.	Non-Periodic signals may appear as random or chaotic.
Sampling	Periodic signals can be accurately represented by a finite number of samples.	Non-Periodic signals may require an infinite number of samples to be accurately represented.
Application	Periodic signals are commonly used in communication and control systems.	Non-Periodic signals are used in audio and image processing, and in the study of chaotic systems.
Example	Examples of periodic signals include sine waves, square waves, and triangle waves.	Examples of Non-Periodic signals include noise, music, and speech. Non-periodic signals are commonly used in audio and image processing, and in the study of chaotic systems.

Numerical Example

- $x(t) = \sin 10\pi t =$

- $x(t) = \sin \pi t \; u(t) =$

Numerical Example

Given signal is,

- $x(t) = \sin 10\pi t$; $\sin \omega t$

Since the signal x(t) is a sinusoidal signal. PERIODIC SIGNAL

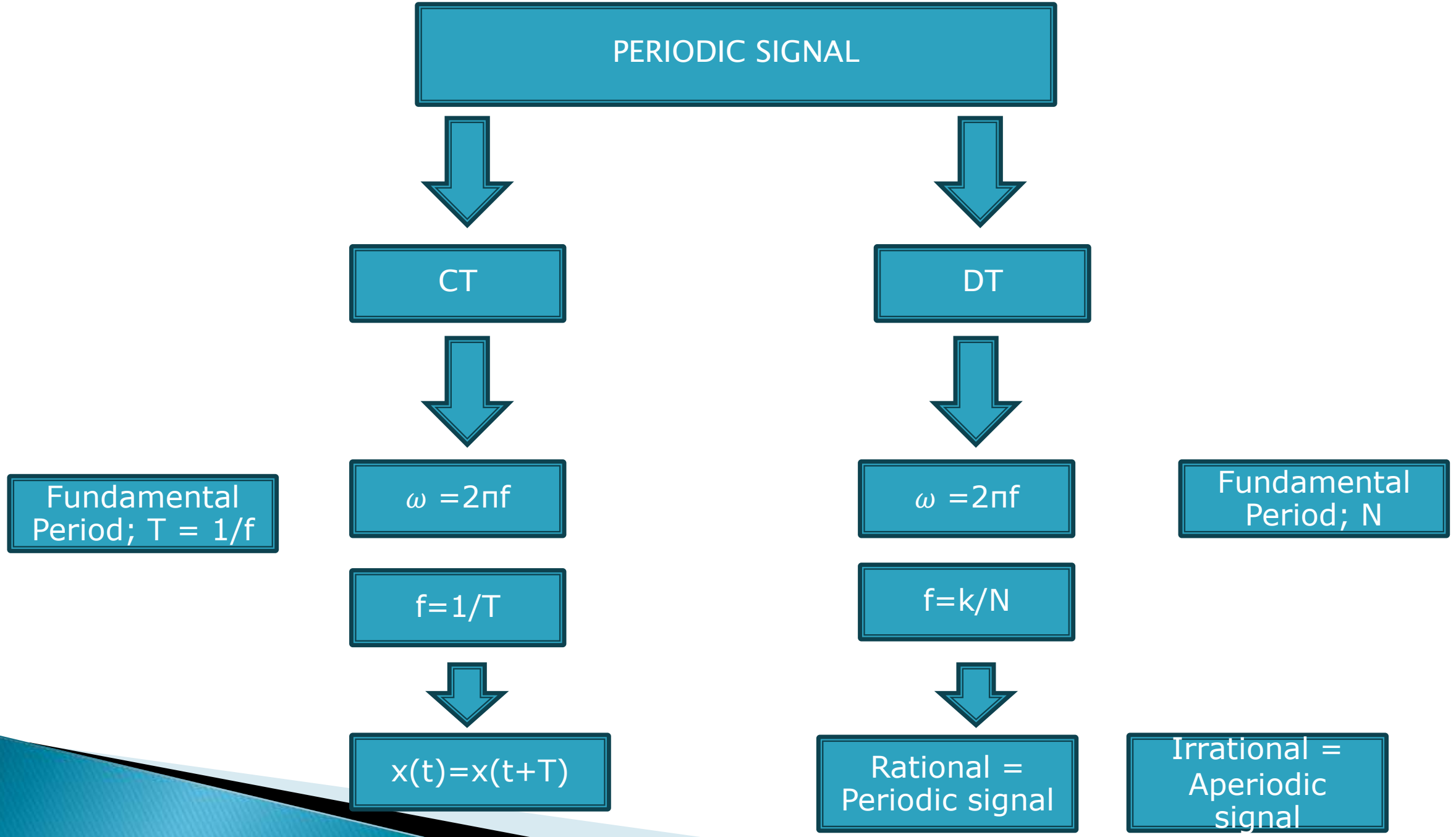
$$\omega = 10\pi$$

$$\text{Fundamental Period, } T = \frac{2\Pi}{\omega} = \frac{2\Pi}{10\Pi} = \frac{1}{5} \text{ sec}$$

- $x(t) = \sin \pi t \; u(t)$ = Aperiodic Signal
  - $x(t) = \sin \pi t$ - Periodic signal
  - $x(t)$ = u(t) - Aperiodic signal

Aperiodic signal x Periodic signal = Aperiodic Signal

Aperiodic signal + Periodic signal = Aperiodic Signal


PERIODIC SIGNAL
CT
DT
Fundamental Period; T = 1/f
ω =2πf
ω =2πf
Fundamental Period; N
f=1/T
f=k/N
x(t)=x(t+T)
Rational = Periodic signal
Irrational = Aperiodic signal

1. $\sin \frac{\pi}{4}$ (n)

2. $\sin^2 \frac{\pi}{4}$ (n)

3. Sin 3(n)

4. $\sin\frac{2\pi}{3}(n) + \cos\frac{\pi}{2}(n)$

Discrete Fourier Transform (DFT)

Discrete Fourier Transform (DFT)

It's a mathematical technique to transform a sequence of values into components of different frequencies. It converts a signal from its original domain (often time) into the frequency domain.

This is particularly useful in signal processing, image analysis, and many other fields.

The DFT of X(n) is a sequence consisting of N samples of X(k)

$$DFT\{x(n)\} = X(k) = \sum_{n=0}^{N} x(n) \cdot e^{-j2\pi kn/N}$$

N= number of samples/points
K= kth term
n= nth term for summation

1. Given the sequence x(n)= {1, -1, 2, -2}. Find the 4- point DFT of the given signal.

2. Compute the 3-point sequence of x(n)= {2, 1, 4} using DFT.

3. Using the same sequence X(n) = {1, 4, 3}. Compute the 6-point DFT.